LEARNIQ TRACK: AN AI-BASED LEARNING GAME AND DIGITAL WORK IMMERSION MONITORING SYSTEM FOR SENIOR HIGH SCHOOL STUDENTS OF PAETE SCIENCE AND BUSINESS COLLEGE INC.

A Capstone Project
Presented to the Faculty of
College of Computer Studies
Laguna State Polytechnic University

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LAGUNA STATE POLYTECHNIC UNIVERSITY
COLLEGE OF COMPUTER STUDIES
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ABSTRACT

This study presents LearnIQ Track, an AI-Based Learning Game and Digital Work Immersion Monitoring System designed for Senior High School students of Paete Science and Business College Inc. The study addresses the challenges associated with traditional learning activities and manual work immersion documentation, particularly the preparation of learning materials, paper-based attendance, handwritten journals, and the difficulty of monitoring student progress. The study aims to design, develop, test, and evaluate a web-based system that provides AI-generated quizzes, activities, and reviewers from teacher-uploaded learning materials, interactive quizzes with scores and leaderboards, digital work immersion attendance, daily journal submission, and teacher and administrator dashboards. The system was developed using HTML, CSS, and JavaScript for the front-end, Python with FastAPI for the back-end, Google Gemini AI for AI services, JSON for data exchange, Supabase for database and authentication services, and Cloudflare for web infrastructure. The study employs a developmental approach using Agile methodology and includes functional and black-box testing, non-functional testing, security testing, and user acceptance testing. System evaluation uses selected characteristics of ISO/IEC 25010 and constructs of the Technology Acceptance Model (TAM) to assess system quality, usability, perceived usefulness, and user acceptance. The study is significant in providing the school with a centralized digital platform that supports learning activities and organizes work immersion records. The developed system is intended to provide a more organized and accessible approach to academic learning and work immersion monitoring.

Keywords: *Artificial Intelligence, Learning Game, Digital Work Immersion, Web-Based System, Senior High School Students, Educational Technology, Student Monitoring*

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DEFINITION OF TERMS

The researchers explain the terminologies utilized in the development of the system, which can be beneficial for thoroughly understanding the key terms used in the study.

OPERATIONAL TERMS

AI-Generated Activities	Digital learning tasks automatically produced by the system from uploaded lesson materials for student practice and review.
AI-Generated Quizzes	Sets of assessment questions automatically generated by the system from teacher-uploaded files such as lesson documents and presentation materials.
Attendance Monitoring	The digital recording, storage, and viewing of students' work-immersion time-in and time-out logs through the system.
Daily Journal	A student's digital reflection or daily record of tasks, experiences, and learning during work immersion.
Dashboard	The role-based interface used by teachers, coordinators, and administrators to monitor attendance,

	journals, scores, and learner activity.
Leaderboard	A ranking feature that displays points, scores, or performance standings of students based on quiz and game outcomes.
Learning Materials	Teacher-uploaded lesson files used by the system as source content for AI-generated quizzes, activities, and reviewers.
Reviewer	A summarized study aid automatically generated by the system from uploaded lesson content.
Work Immersion Monitoring	The digital supervision of student attendance, journal submissions, and activity progress during work immersion.
Overall Acceptability	The respondents overall assessment of LearnIQTrack based on their experience with its functions, usability, and suitability for its learning and work immersion monitoring purposes.

TECHNICAL TERMS

Artificial Intelligence	A set of computational approaches used to automate or augment tasks such as tutoring, assessment, prediction, and personalization in education.
Authentication	The process of verifying a user's identity before granting access to protected system resources.
Gamification	The integration of game design elements such as points, badges, feedback, and leaderboards into non-game contexts like learning systems.
ISO/IEC 25010	An international software-quality model that defines product quality characteristics such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.
Supabase	A Postgres-based development platform that provides database, authentication, storage, APIs, and related backend services.
Technology Acceptance Model	A user-acceptance framework in which perceived usefulness and perceived ease of use influence attitude, intention, and actual

	system use.
Perceived Ease of Use	The extent to which users perceive that LearnIQ Track is easy to learn, understand, and use with minimal effort.
Perceived Usefulness	The extent to which users perceive that LearnIQ Track can help them perform their learning or work immersion monitoring tasks.
Attitude Toward Using	The users' overall positive or negative feeling toward using LearnIQ Track for its intended learning and work immersion monitoring functions.
Behavioral Intention to Use	The users' willingness or intention to use LearnIQ Track when the system is available for its intended purposes.
User Acceptance Testing	Acceptance testing carried out by future users in a simulated or real operational environment with emphasis on user needs and requirements.
Web-Based System	A system accessed through a web browser over a network rather than being installed only as a standalone desktop application.

CHAPTER I

INTRODUCTION

Education is increasingly shaped by digital platforms, analytics, and artificial intelligence. International guidance now recognizes that AI can improve teaching, learning, and assessment, but it also emphasizes that educational AI must remain human-centered, equitable, and pedagogically validated. In the Philippine context, this issue has become especially timely because the Department of Education issued foundational AI guidelines for basic education in 2026, confirming that AI may legitimately support teaching, learning, assessment, and school operations when used responsibly.

At the same time, online and web-based learning environments remain important because they extend access to lessons, activities, and feedback beyond the classroom. Dhawan (2020) argued that online learning became indispensable during major educational disruptions and highlighted its flexibility, accessibility, and support for self-paced study. Research on gamification also shows that learning systems can benefit from game elements such as points, badges, leaderboards, and feedback, although these elements must be selected carefully and tied to real learning goals rather than novelty alone.

For Senior High School in the Philippines, work immersion remains a formal curricular requirement. DepEd's work-immersion guidance establishes it as part of the SHS program, while recent Philippine studies show that immersion contributes to technical skill, responsibility, communication, performance, and workplace readiness when it is properly supervised and documented. At the same time, related studies on digital monitoring systems indicate that digitized attendance and reporting systems can

support the organization, accessibility, and monitoring of student records. These strands of evidence create a strong rationale for a single platform that supports both learning engagement and work immersion monitoring.

It is within this context that LearnIQ Track is proposed. The system is designed to support Senior High School students of Paete Science and Business College Inc. by combining AI-assisted study support, gamified learning interaction, and digital work-immersion monitoring in one integrated environment. Instead of treating learning review and immersion documentation as separate tasks handled by different tools or paper-based processes, the system is intended to unify them into one role-based platform for students, teachers, coordinators, and administrators.

1.1 Project Context

The project arises from two educational concerns relevant to the study setting. First, traditional study practices, such as reading printed materials and answering conventional exercises, may provide limited opportunities for interactive practice, immediate feedback, and repeated self-testing. Although these approaches can still support learning, digital learning environments can provide additional opportunities for students to engage with lesson materials through interactive activities and feedback. Literature on Artificial Intelligence and gamification supports the use of these technologies in educational settings, particularly for assessment support, personalized learning, feedback, and interactive learning activities.

Second, the school currently does not have a dedicated digital system for work immersion monitoring. Based on an interview with the school principal, only Grade 12 students participate in work immersion, with a total Grade 12 population of 514 students. At present, work immersion attendance is recorded using paper sheets, while

students submit their daily journals through handwritten or printed documents. These records are handled and checked by the teacher assigned to work immersion. Although no major problems with the existing process were specifically reported, the absence of a dedicated digital monitoring platform presents an opportunity to centralize attendance and journal records and provide teachers with a more accessible means of monitoring student immersion activities. The current processes therefore involve Grade 12 students as participants in work immersion and teachers as the personnel responsible for checking attendance and journal records, while the school administration has an oversight role in the program.

The interview with the school principal also indicated that the school had previously intended to develop an AI-based learning-game system, but the planned project was not continued after the school's IT team left. The principal expressed support for the development of LearnIQ Track because it addresses this previously identified need. The proposed digital work immersion monitoring component was also viewed positively as an additional feature that could complement the learning-game system. Thus, LearnIQ Track is proposed as an integrated platform that combines AI-assisted learning activities, gamification, and digital work immersion monitoring within one school-oriented system.

Table 1.1 Current Process and Proposed Process of LearnIQ Track.

Current Process	Gap / Concern	LearnIQ Track Solution
Paper-based attendance	No dedicated digital system for work immersion attendance records	Digital time-in and time-out

continuation

Handwritten or printed journals	No dedicated digital platform for journal submission and monitoring	Digital daily journal submission
Manual preparation of learning activities	Requires teachers to prepare learning activities from lesson materials	AI-assisted generation of quizzes, activities, and reviewers
Conventional lesson review	Provides fewer interactive game-based learning elements	Gamified quizzes with scores and leaderboard
Separate learning and work immersion records	Learning and immersion functions are not integrated into one platform	Centralized teacher and admin dashboards

The table highlights the technology gap addressed by LearnIQ Track. Existing learning platforms generally provide functions for managing learning materials, coursework, quizzes, and student activities, while work immersion monitoring systems primarily focus on attendance, reporting, and supervision. LearnIQ Track brings these functions together in a single school-oriented platform. It allows teachers to upload learning materials, generate AI-assisted quizzes, activities, and reviewers, provide students with interactive and gamified learning activities, and monitor work immersion attendance and daily journals through role-based dashboards.

The integration of these functions is intended to provide students, teachers, and school administrators with a centralized platform for both academic lesson review and work immersion monitoring. Rather than requiring these activities to be managed through separate processes, LearnIQ Track places the learning and monitoring functions within one system while maintaining teacher oversight of AI-generated content and student records.

The necessity of LearnIQ Track is also strengthened by current AI governance guidance. UNESCO notes that the rapid adoption of generative AI in education requires appropriate safeguards, including attention to validation and user privacy. This is relevant to a school platform that handles both academic content and student immersion records. Therefore, the project considers teacher oversight, privacy, and educational appropriateness in the design and use of its AI-assisted features.

1.2 Research Objective

1.2.1 General Objective

The main objective of this study is to design, develop, test, and evaluate LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The system is intended to provide an interactive platform for lesson review through AI-generated quizzes, activities, and reviewers while supporting the digital recording and monitoring of work immersion attendance and daily journals.

1.2.2 Specific Objectives

Specifically, this study aims to:

1. To design and develop an AI-based learning game and digital work immersion monitoring system using a web-based platform that provides teachers with the ability to upload learning materials and validate AI-generated quizzes, activities, and reviewers; provides students with interactive quizzes, scores, and leaderboard features for lesson review; and supports digital work immersion time-in and time-out recording and daily journal submission.

2. To test the functionalities of the developed system using functional and black-box testing in terms of:

2.1. User Account Management – registration, login, logout, user authentication, profile management, and role-based access for students, teachers, and admin.

2.2. Learning Material Management – uploading, storing, viewing, and managing PowerPoint presentations and document files uploaded by teachers.

2.3. AI-Generated Learning Content – generation of quizzes, activities, and reviewers based on uploaded learning materials.

2.4. Learning Game / Interactive Quiz System – student access to AI-generated quizzes and activities through an interactive quiz interface.

2.5. Score Computation and Leaderboard System – automatic recording of quiz scores, accumulated points, rankings, and leaderboard updates.

2.6. Digital Work Immersion Attendance – recording of students' time-in and time-out during work immersion.

2.7. Daily Journal Submission – submission of students' daily journals, reflections, and work immersion activity reports.

2.8. Teacher Dashboard – monitoring of students' learning progress, quiz performance, attendance records, journal submissions, and work immersion status.

2.9. Admin Dashboard – management of user accounts, viewing of system records, monitoring of reports, and supervision of the overall system by the school principal.

2.10. Reports and Data Tracking – generation of organized reports for student performance, attendance, journal submissions, and system records.

2.11. Security, Data Protection, and System Stability – protection of user accounts and student records, proper access control, and stable system performance during testing.

3. To evaluate the developed system based on selected characteristics of ISO/IEC 25010 and constructs of the Technology Acceptance Model (TAM) in terms of:

a. Functional Suitability

b. Usability

c. Performance Efficiency

d. Reliability

e. Security

f. Perceived Ease of Use

g. Perceived Usefulness

h. Attitude Toward Using

i. Behavioral Intention to Use

j. Overall Acceptability

1.3 Conceptual Framework

1.3.1 Components of the Conceptual Framework

The conceptual framework illustrates the development process of LearnIQ Track, beginning with the identification of problems and user needs, followed by system planning and design, development and implementation, testing and evaluation, and feedback and improvement. It presents the major activities undertaken by the researchers in developing an integrated platform for AI-assisted learning and digital work immersion monitoring. The framework covers the integration of AI-generated quizzes, activities, and reviewers, interactive learning games, score tracking, leaderboards, digital attendance, daily journal submission, and monitoring dashboards. The system is evaluated to determine whether its functions meet the needs of its intended users.

1.3.2 Conceptual Framework Diagram



Figure 1.1 Conceptual Framework of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc.

Figure 1.1 presents the conceptual framework of LearnIQ Track using an Input–Process–Output–Evaluation–Improvement structure. The inputs consist of the learning materials, user requirements, work immersion requirements, and user roles that serve as the basis for developing the system. The process includes requirements analysis, system design, AI processing, generation of quizzes and activities, gamification, attendance monitoring, journal management, dashboard and reporting, teacher validation, and system development and integration. The system outputs include AI-generated quizzes, activities, and reviewers; interactive quiz scores and leaderboard results; attendance records; daily journals; student monitoring records; and teacher and administrator reports. The developed system is then evaluated through functional and black-box testing, selected ISO/IEC 25010 characteristics, Technology Acceptance Model constructs, and overall acceptability. The evaluation results and user feedback provide the basis for identifying issues and making necessary system revisions. The improvement stage forms an iterative feedback loop that allows the system to return to the development process for further refinement, consistent with the Agile approach used in the study.

1.4 Project Purpose

The main purpose of this study is to design, develop, and evaluate LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The

proposed system provides a centralized digital platform that integrates AI-assisted learning activities, gamified quizzes, and digital work immersion monitoring.

The system aims to address the school's existing reliance on paper-based attendance records and handwritten or printed daily journals for work immersion, as well as the absence of a dedicated platform for centralizing these records. Through its features, LearnIQ Track intends to support lesson review, organize student learning records, and provide digital tools for recording attendance, submitting daily journals, and monitoring work immersion activities.

1.4.1 Intended Beneficiaries

LearnIQ Track is intended to benefit the following individuals and groups: Senior High School students, teachers and work immersion coordinators, the school principal or administrator, Paete Science and Business College Inc., and future researchers.

1.4.2 Expected Benefits

The following are the expected benefits of the developed system:

Students - The system provides students with AI-generated quizzes, activities, and reviewers based on teacher-uploaded learning materials. Its interactive quizzes, scores, and leaderboard support game-based lesson review. For work immersion, students can record their time-in and time-out and submit their daily journals digitally, providing an organized way to document their activities.

Teachers and Work Immersion Coordinators - LearnIQ Track allows teachers to upload learning materials, review and validate AI-generated learning content, and monitor students' academic performance and work immersion

records. Through the teacher dashboard, they can access quiz results, attendance records, and daily journal submissions in one platform.

School Principal / Administrator - The system provides access to student accounts, learning activities, quiz performance, attendance records, journals, and work immersion reports through the admin dashboard. These centralized records can support the principal in overseeing system activities and monitoring relevant student information.

Paete Science and Business College Inc. - The developed system provides the school with a centralized digital platform for selected academic learning and work immersion monitoring processes. It can help organize learning records, attendance information, journal submissions, and monitoring reports within one system.

Future Researchers - This study may serve as a reference for future researchers interested in developing or evaluating AI-based educational systems, gamified learning platforms, and digital work immersion monitoring systems. Its system design, development process, testing procedures, and evaluation approach may provide useful information for related studies.

1.5 Scope and Limitation of the Study

The study entitled "LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc." focuses on three major points: 1) the design and development of the system, 2) the testing of its functionality and features, and 3) the evaluation of its selected software quality characteristics, technology acceptance, and overall acceptability among students, teachers, and the admin. The scope of the study

is limited to the Senior High School department of Paete Science and Business College Inc.

1.5.1 Scope of the Study

The following are the scope of the study:

- The study is primarily focused on the development of an AI-based learning game and digital work immersion monitoring system for Senior High School students of Paete Science and Business College Inc.
- The system encompasses the design, development, and implementation of features such as user authentication, learning material upload, AI-generated quizzes, activities, and reviewers.
- The study covers the inclusion of an interactive quiz or learning game system with score computation and leaderboard ranking to provide students with an interactive approach to lesson review and encourage participation in learning activities.
- The project also includes digital work immersion monitoring features such as time-in and time-out recording, daily journal submission, and teacher monitoring of student attendance and immersion activities.
- The system provides a teacher dashboard where teachers can upload learning materials, validate AI-generated content, monitor student learning progress, and manage work immersion records. In this study, teachers also serve as the work immersion coordinators.
- The system provides an admin dashboard for the principal of the school to manage user accounts, view reports, monitor records, and supervise the overall use of the system.

- The research evaluates the functionality, usability, reliability, and acceptability of the system by gathering feedback from selected Grade 11 students, Grade 12 students, and teachers of Paete Science and Business College Inc.

1.5.2 Limitations of the Study

Yet, the study still has its own limitations:

- The study is limited to Senior High School students, teachers, and the school principal of Paete Science and Business College Inc., with Grade 11 and Grade 12 students and teachers serving as the selected respondents.
- The system has only three user roles: students, teachers, and admin. Teachers will also serve as the work immersion coordinators, while the admin role will be assigned to the school principal.
- The system will be tested and evaluated only within Paete Science and Business College Inc. and will not cover full implementation in other schools. Therefore, the findings may not be directly generalizable to schools with different populations, policies, or work immersion procedures.
- The testing and evaluation will involve only the selected respondents within the study population and will not represent all Senior High School students, teachers, or administrators.
- The quality and accuracy of AI-generated quizzes, activities, and reviewers depend on the clarity, completeness, and quality of the uploaded learning materials. AI-generated content may contain inaccurate, incomplete, irrelevant, or inappropriate information, including information that is not supported by the uploaded learning materials. Therefore, teacher validation is

required to check the accuracy, relevance, completeness, and appropriateness of generated content before it is used by students.

- The system requires an internet connection for login, file uploading, AI content generation, data storage, attendance recording, journal submission, and dashboard monitoring.
- The system depends on the availability and compatibility of the user's device and web browser. Differences in device specifications, screen sizes, browser versions, or internet conditions may affect the user's experience with the system.
- The system will initially support selected file formats only, such as PowerPoint presentations and document files.
- The study focuses on the development, testing, and user evaluation of LearnIQ Track and does not include a longitudinal measurement of students' learning outcomes over an extended period.
- The study does not provide direct evidence of actual improvement in academic achievement, grades, or long-term learning performance. The evaluation focuses on system functionality, selected software quality characteristics, technology acceptance, and overall acceptability.

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter presents the relevant literature and studies that the researcher considered in strengthening the claim and importance of the present study.

2.1 Studies and Literature

2.1.1 Selection of Related Literature

The literature and studies reviewed in this chapter were selected based on their relevance to the objectives of LearnIQ Track, their applicability to its proposed features and technologies, and the credibility of their sources. Priority was given to studies and publications discussing artificial intelligence in education, automated question generation, gamification, digital learning systems, work immersion monitoring, technology acceptance, and software quality. These references provide the concepts, methods, and technological foundations supporting the development and evaluation of LearnIQ Track.

2.1.2 Review and Analysis of Studies

The reviewed studies were examined and analyzed to identify their contributions, findings, and relevance to the development of LearnIQ Track. The discussion focuses on existing research related to artificial intelligence in education, automated question generation, gamification, digital learning systems, and work immersion monitoring. Their methods and findings were considered to determine how they support the proposed system and provide insights into its features and development.

Artificial Intelligence in Education

Artificial Intelligence has become an increasingly important area in educational technology because of its ability to support learning, assessment, feedback, personalization, and other educational processes. AI-based educational systems can process information and use different computational techniques to assist students and teachers in activities that traditionally require substantial manual effort. In the context of LearnIQ Track, Artificial Intelligence is particularly relevant because the system is designed to analyze learning materials uploaded by teachers and use them as the basis for generating quizzes, activities, and reviewers. The current manuscript specifically identifies AI-generated learning content as one of the main system functions, while also requiring teachers to validate the generated materials before they are provided to students.

Zawacki-Richter, Marín, Bond, and Gouverneur (2019) conducted a systematic review of research on Artificial Intelligence applications in higher education to determine how AI was being used and what areas of education were receiving the most attention. From 2,656 initially identified publications covering research from 2007 to 2018, 146 articles were included in the final synthesis after applying their inclusion and exclusion criteria. The review identified major applications of AI involving profiling and prediction, assessment and evaluation, adaptive systems and personalization, and intelligent tutoring systems. The researchers also noted that much of the research came from Computer Science and STEM disciplines and that the perspectives of educators were less visible in the literature. This finding is important to the development of LearnIQ Track because it demonstrates that AI has already been

applied to several educational functions, but it also emphasizes that technological capabilities should be considered together with actual teaching practices and the needs of educators. LearnIQ Track follows this principle by placing teachers in the process of using AI-generated content rather than allowing the system to operate without human involvement. Teachers will upload the learning materials, review the generated quizzes, activities, and reviewers, and determine whether the materials are suitable for students. In this way, the study by Zawacki-Richter et al. supports the use of AI as an educational support tool while also reinforcing the importance of teacher participation, which is directly reflected in the design of LearnIQ Track.

Holmes and Tuomi (2022) examined the development and implications of Artificial Intelligence in Education and discussed how AI can contribute to educational practices through intelligent tutoring, assessment, learning support, and other technology-assisted approaches. Their discussion is relevant because AI in education is not limited to replacing a conventional classroom activity with an automated version; instead, AI can provide additional forms of assistance that may support teachers and students. The use of AI becomes more meaningful when it is connected to an educational purpose, such as helping learners receive feedback, supporting assessment, or providing learning resources. This perspective is applicable to LearnIQ Track because the purpose of its AI component is not simply to demonstrate that AI can generate text. The AI component has a specific instructional purpose: it analyzes teacher-provided learning materials and generates quizzes, activities, and reviewers that students can use for lesson review. The generated content is then incorporated into the interactive learning component of the system, where students can answer

quizzes and receive scores. Therefore, the literature on AI in education supports the basic direction of LearnIQ Track, while the system applies the concept specifically to Senior High School lesson review and teacher-supported learning.

Ogunleye et al. (2024) conducted a systematic review concerning the use of generative Artificial Intelligence for teaching and learning practice. The study is particularly relevant to the present project because generative AI represents a newer development in educational technology that can produce text and other learning-related outputs from user-provided information. The researchers examined the growing use of generative AI in educational settings and discussed its possible applications as well as the challenges that accompany its use. Their work emphasizes that generative AI can provide opportunities for teaching and learning, but responsible implementation remains necessary because generated outputs may require evaluation and educational judgment. This is closely related to LearnIQ Track because the system does not intend to treat AI-generated quizzes, activities, and reviewers as automatically correct instructional materials. Instead, the teacher remains responsible for reviewing and validating the content before students use it. This distinction is important because LearnIQ Track is intended for Senior High School students, meaning that generated materials need to remain aligned with the lessons provided by the school and appropriate for the learners' educational level. Thus, the literature on generative AI provides support for incorporating AI into LearnIQ Track while also supporting the system's decision to retain teacher validation as part of the process.

AI-Generated Learning Materials and Teacher Support

AI-generated learning materials are particularly relevant to LearnIQ Track because one of the system's main purposes is to reduce some of the manual work involved in preparing quizzes, activities, and reviewers. Teachers traditionally need to examine their lesson materials, identify important concepts, formulate questions, prepare answer choices, and organize review activities before these materials can be used by students. When teachers handle several classes or subjects, preparing repeated learning activities may consume considerable time. AI can assist with this process by producing draft materials that teachers can examine, revise, and approve. The current LearnIQ Track manuscript specifically states that teachers will upload PowerPoint presentations and document files and that these materials will be analyzed by AI to generate quizzes, activities, and reviewers.

Gao et al. (2024) conducted a systematic review of automatic assessment of text-based responses in post-secondary education and examined how artificial intelligence and natural language processing techniques have been applied to assessment. Their review is relevant because it demonstrates how computational methods can assist with educational assessment tasks that can otherwise require significant instructor effort. The study also illustrates that automated educational processes should not simply be judged by whether a computer can produce an output; the quality and educational usefulness of that output are equally important. This is relevant to LearnIQ Track because the system's AI component is intended to assist teachers rather than remove their role in evaluating learning materials. The AI can generate possible questions and activities from uploaded content, but the teacher remains responsible for

deciding whether the generated output is appropriate. This approach allows LearnIQ Track to use automation to reduce repetitive preparation while maintaining academic supervision. Therefore, the study supports the concept of AI-assisted educational assessment and provides a basis for treating the system's generated materials as teacher-reviewed resources rather than completely autonomous instructional content. This process represents a human-in-the-loop approach, in which AI assists with content generation while the teacher remains involved in reviewing, validating, and approving the generated materials before they are used by students.

Chomphooyod, Suchato, Tuaycharoen, and Punyabukkana (2023) developed an Artificial Intelligence-assisted system for generating English grammar multiple-choice questions using a Text-to-Text Transfer Transformer (T5) model. Their system used controllable text generation techniques, including keywords and part-of-speech templates, to control the content and grammar topics of the questions. The researchers evaluated the generated multiple-choice questions through human experts and reported an 86% acceptance rate, while the controllability of content and grammar topics reached 96.86% and 98.57%, respectively. The findings demonstrate that AI-based question generation can produce educational assessment materials with useful levels of control, but the use of human evaluation in the study is also significant because it shows that generated questions still need to be checked for educational quality. This is directly related to LearnIQ Track because its AI component is also intended to generate quizzes from teacher-uploaded educational materials. Although the proposed system will not be limited to English grammar and will generate materials from the school's own lessons, the

underlying concept is similar: AI is used to reduce the manual effort required to produce assessment materials, while human users remain involved in checking the results. The study therefore supports the technical feasibility of AI-assisted question generation and reinforces LearnIQ Track's teacher-validation feature.

Ling and Afzaal (2024) examined automatic question-answer pair generation using pre-trained large language models in higher education. Their study addressed the problem that manually producing question-and-answer pairs for assessments can be time-consuming and energy-intensive for teachers. The researchers examined several approaches for generating question-answer pairs and evaluated these approaches using datasets associated with different courses and educational assessment considerations. An important part of the study is its emphasis on evaluating generated questions in an educational context rather than relying only on automated measures. This is highly relevant to LearnIQ Track because the system is intended to generate quizzes and learning activities from actual educational materials uploaded by teachers. The study demonstrates the potential of large language models to assist with assessment-material preparation while also highlighting the need to examine the quality and usefulness of generated questions. In LearnIQ Track, this concern is addressed through the teacher dashboard, where teachers can review and validate the AI-generated content before making it available to students. The study therefore provides recent support for using large language models to assist educational assessment while reinforcing the importance of human review in the proposed system.

AI Hallucination and Factual Accuracy

AI-generated educational content requires careful evaluation because generative AI systems may produce information that appears plausible but is inaccurate, incomplete, or not adequately supported by the source material. This issue is commonly referred to as AI hallucination, wherein a natural language generation system produces unintended or factually unsupported content. Ji et al. (2023) explained that hallucination is a significant problem in natural language generation because generated outputs may contain information that is inconsistent with the intended source or factual information. In educational settings, such inaccuracies are particularly important because generated content may affect students' understanding of lessons or result in inappropriate assessment materials. Therefore, AI-generated content should be evaluated not only based on whether the system can produce an output, but also on its accuracy, relevance, clarity, and alignment with the intended learning material.

For LearnIQ Track, this concern is directly relevant because the system will use AI to generate quizzes, activities, and reviewers from teacher-uploaded learning materials. Although the system is designed to use the uploaded materials as the basis for generation, the resulting content may still require human checking. UNESCO (2023) emphasizes that teachers and learners should critically assess generative AI outputs because such systems can produce incorrect statements, and educational institutions should establish appropriate validation mechanisms for the use of generative AI. Therefore, teacher validation is included in LearnIQ Track to allow generated questions and learning materials to be reviewed for factual accuracy, relevance to the lesson, appropriateness, and completeness before they are presented to students. This

human-centered approach allows AI to function as an educational support tool while teachers retain responsibility for reviewing and approving the final instructional content.

Automated Question Generation

Automated question generation is one of the most directly related areas of literature to LearnIQ Track because the proposed system will transform teacher-uploaded learning materials into quizzes and activities. The purpose of automated question generation is not simply to create random questions but to produce questions from source information that can be used for assessment, practice, or review. This is important for LearnIQ Track because the generated quizzes are expected to remain connected to the learning materials provided by the teacher. The system therefore uses AI as a means of converting existing educational content into additional opportunities for students to practice and review their lessons.

Kurdi et al. (2020) conducted a systematic review of automatic question generation for educational purposes and examined research on techniques used to automatically create questions from educational materials. The researchers explained that exam-style questions are fundamental educational tools but that manually constructing questions is a complex process requiring training, experience, and resources. This creates a practical limitation when teachers need a large and continuous supply of questions for practice activities or assessment. Their review examined different approaches to automatic question generation, the types of questions produced, and the methods used to evaluate generated questions. The study is especially relevant to LearnIQ Track because the

proposed system is designed around the same general problem: teachers should be able to provide educational material and obtain a set of questions that can be used for student review without having to manually construct every question from the beginning. At the same time, the review makes clear that question generation involves quality and evaluation concerns, meaning that automated output should not automatically be treated as perfect. This supports LearnIQ Track's design in which teachers validate the generated quizzes and activities before students use them. In relation to the proposed study, automatic question generation therefore serves both a practical purpose—reducing repetitive preparation—and an educational purpose—providing students with additional opportunities to practice the content of their lessons.

Chomphooyod et al. (2023) further demonstrated the practical application of automatic multiple-choice question generation by developing a T5-based system for English grammar questions. Their approach allowed the generated questions to be controlled according to specific content and grammar topics and used human experts to assess the quality of the generated questions. The reported acceptance rate of 86% indicates that the generated questions were considered usable by evaluators, while the high controllability results indicate that the model could generate questions according to the specified content requirements. This is relevant to LearnIQ Track because the proposed system will also depend on the relationship between the source learning material and the generated quiz content. The LearnIQ system is intended to process materials selected and uploaded by teachers, meaning that the resulting questions should correspond to the lesson rather than being unrelated questions generated from general knowledge. While Chomphooyod et al. focused specifically on English

grammar, their findings provide technical support for the broader idea of using language-generation models to produce multiple-choice educational questions. LearnIQ Track extends this idea to Senior High School learning materials and incorporates the generated questions into a gamified learning environment.

Ling and Afzaal (2024) investigated the generation of question-answer pairs using pre-trained large language models and evaluated different approaches using educational course datasets. Their work is important because it moves beyond simply demonstrating that a language model can generate questions and instead examines how generated assessment materials perform in an actual educational context. The study recognizes that question generation can reduce the burden associated with manually preparing assessment materials, but it also considers the need to evaluate the generated questions according to educational criteria. This is particularly applicable to LearnIQ Track because the proposed system will generate learning materials from teacher-provided files, but the final decision regarding whether a question is appropriate remains with the teacher. The relationship is therefore both technical and practical: the study demonstrates that large language models can support question generation, while LearnIQ Track applies this capability to the specific problem of preparing review materials for Senior High School students. The generated questions will then be used in the system's interactive quiz feature, making the AI-generated content part of a broader learning process rather than an isolated text-generation function.

Gamification and Game-Based Learning

Gamification is the incorporation of game-related elements into activities that are not traditionally games. In education, these elements may include points, scores, rankings, leaderboards, challenges, feedback, and other mechanisms intended to encourage participation. Gamification differs from game-based learning because gamification adds selected game elements to an existing learning activity, whereas game-based learning uses an actual game as the primary environment or method for learning. In LearnIQ Track, gamification is applied to academic lesson review through points, scores, leaderboards, and interactive quizzes rather than requiring students to learn through a separate standalone game. LearnIQ Track incorporates gamification through its interactive quiz system, automatic score computation, accumulated points, and leaderboard ranking. The purpose is to make lesson review more interactive and encourage students to participate repeatedly in academic activities rather than simply reading learning materials. The current manuscript specifically identifies these features as part of the proposed learning game.

Bai, Hew, and Huang (2020) conducted a meta-analysis and synthesis of qualitative research to examine whether gamification improves student learning outcomes. Their quantitative analysis included 30 independent interventions involving 3,202 participants from 24 quantitative studies, while their qualitative synthesis considered 32 studies examining learners' experiences with gamification. The meta-analysis found a statistically significant medium overall effect in favor of gamification compared with non-gamified learning, with a Hedges' g of 0.504. Their qualitative findings identified enthusiasm, performance feedback, recognition, and goal setting as major reasons learners

enjoyed gamification, while lack of additional utility and experiences such as anxiety or jealousy were identified as reasons some learners disliked it. These findings are important for LearnIQ Track because they support the use of game elements while also showing that gamification should not be treated as automatically beneficial in every situation. The proposed system uses scores and leaderboards because these features can provide students with immediate performance information and encourage them to improve their results, but the primary activity remains answering questions based on their lessons. Therefore, the study supports the use of gamification in LearnIQ Track while also suggesting that the game elements should remain connected to meaningful academic tasks rather than becoming competition without educational purpose.

Li, Hew, and Du (2024) conducted a meta-analysis and systematic review examining the effects of gamification on students' intrinsic motivation and psychological experiences. Their research synthesized 35 independent interventions involving approximately 2,500 participants and examined outcomes related to intrinsic motivation, autonomy, relatedness, and competence. The study reported that gamification had a positive overall effect on intrinsic motivation and positive effects on students' perceptions of autonomy and relatedness, while its effect on competence was comparatively minimal. These findings are relevant to LearnIQ Track because the proposed system uses game elements to encourage students to voluntarily participate in repeated lesson-review activities. Points and leaderboard rankings can provide students with immediate feedback about their quiz performance and progress, allowing them to recognize their current standing and identify opportunities for improvement. However, the effectiveness of these elements depends on how

they are integrated into the learning activity. LearnIQ Track therefore applies gamification to academic content rather than creating a separate game unrelated to school lessons. The study supports this approach by showing that gamification can influence motivational experiences, while also indicating that game elements alone should not be assumed to improve every aspect of learning.

The use of gamification in LearnIQ Track is therefore based on the combination of evidence rather than on the assumption that students will automatically learn better because a system contains game elements. The literature shows that gamification can support learning performance and motivation, but it should be designed around meaningful learning objectives. In LearnIQ Track, students gain points by answering AI-generated quizzes and participating in learning activities, while leaderboard rankings provide an additional form of feedback. This means that the game elements are directly tied to the academic purpose of the system. The approach is consistent with the current manuscript, which describes the learning game as a way to encourage students to answer quizzes, review lessons, and improve their learning performance.

Web-Based Learning Systems

Web-based learning systems allow students to access learning materials and activities through internet-connected devices rather than depending entirely on physical classroom resources. This approach is relevant to LearnIQ Track because the proposed system is intended to provide students with access to AI-generated learning materials and digital work immersion monitoring functions through a web-based platform. The system uses HTML, CSS, and JavaScript

for the front-end, Python with FastAPI for the back-end, Google Gemini AI for generating AI-assisted learning content, JSON for data exchange, and Supabase for database and authentication services. These technologies work together to provide different user roles with access to the learning, monitoring, and administrative functions appropriate to their responsibilities.

Dhawan (2020) examined online learning during the COVID-19 crisis and discussed how educational institutions were forced to transition from traditional face-to-face instruction toward online modes of teaching and learning. The study highlighted the role of online learning in maintaining educational activities during a period when physical classroom interaction was severely restricted. It also discussed both the opportunities and challenges associated with online education, including flexibility and accessibility as well as limitations involving technological infrastructure and learner engagement. This study is relevant to LearnIQ Track because it demonstrates the importance of digital learning environments that allow students to access educational activities beyond a strictly physical classroom setting. However, LearnIQ Track is not intended to reproduce emergency remote teaching; instead, it applies the flexibility of digital learning to regular Senior High School lesson review and work immersion monitoring. Students can use the system to access generated learning materials and complete quizzes, while work immersion information can be recorded and monitored digitally. Thus, the study supports the broader use of online learning environments while LearnIQ Track applies the concept to a more specific school-based purpose.

El-Sabagh (2021) developed an adaptive e-learning environment based on students' learning styles and examined its impact on student engagement. The

study used a development approach to create the adaptive environment and a quasi-experimental design to compare the proposed approach with conventional e-learning. The study is relevant because it demonstrates that digital learning environments can be designed not only to deliver content but also to influence how learners interact with that content. The importance of engagement is particularly relevant to LearnIQ Track because the proposed system is designed to address passive and repetitive review practices through interactive quizzes and game elements. Although LearnIQ Track is not being developed as a learning-style adaptive system, the study supports the general principle that digital learning platforms should consider learner interaction and engagement rather than simply transferring printed materials into an online format. LearnIQ Track applies this principle through AI-generated activities, interactive quizzes, scores, and leaderboard features, allowing the digital platform to provide a more active learning experience.

Learning Management Systems and Student Monitoring

Learning Management Systems provide digital environments for organizing learning materials, assignments, quizzes, grades, announcements, and other academic activities. Platforms such as Moodle, Google Classroom, and Schoology demonstrate how schools can centralize learning resources and communication between teachers and students. These platforms are useful references for LearnIQ Track because they show the value of having a centralized digital learning environment. However, the current manuscript identifies a specific difference between general-purpose learning platforms and the proposed system: LearnIQ Track is intended to combine AI-generated

learning content and gamified quizzes with work immersion attendance and daily journal monitoring.

The existence of established learning management platforms demonstrates that digital management of learning activities is already a practical approach in education. However, general-purpose LMS platforms are normally designed to accommodate broad educational use rather than the specific work immersion requirements of one Senior High School institution. LearnIQ Track is therefore not proposed simply as another LMS. Its purpose is to provide a more focused platform for the needs identified in the study setting, particularly the combination of lesson review and work immersion documentation. The system allows teachers to upload learning materials, generate AI-supported quizzes and reviewers, monitor student performance, and manage work immersion records within the same environment. This distinction is important to the technology gap of the study because the proposed system does not claim that digital learning platforms are new; instead, it identifies the lack of integration between specific academic and immersion-related functions within the intended school context.

Work Immersion Programs in Senior High School

Work immersion is an important component of the Philippine Senior High School curriculum because it provides learners with opportunities to experience actual workplace environments and apply knowledge and competencies in practical settings. DepEd Order No. 30, s. 2017 provides the official Guidelines for Work Immersion and establishes work immersion as a key feature of the Senior High School curriculum. The guidelines provide the

basis for implementing work immersion and describe its role in exposing learners to work-related environments and experiences. This is directly related to LearnIQ Track because work immersion is one of the two major areas supported by the proposed system, alongside academic lesson review.

The implementation of work immersion also requires students to comply with the activities, attendance, documentation, and supervision requirements established by the school and applicable DepEd guidelines. These requirements provide a basis for monitoring student participation and progress throughout the immersion period. Attendance records and daily journals are particularly relevant because they provide documentation of students' participation and activities during their work immersion. For LearnIQ Track, these requirements are reflected in the digital time-in and time-out feature, daily journal submission, and teacher monitoring functions of the proposed system.

Monis (2024) examined the work immersion experiences and challenges encountered by Technical-Vocational Senior High School students through a qualitative case-analysis approach. The researcher conducted interviews with Tech-Voc Senior High School graduates from selected schools to identify their first-hand experiences and challenges during work immersion. The study reported positive experiences involving interaction with co-workers and superiors, constructive feedback, and the development of work ethics, while also seeking to identify challenges that could be addressed through a contextualized work immersion program. This study is relevant to LearnIQ Track because it demonstrates that work immersion is not simply a requirement for completing Senior High School but an experience through which students develop workplace-related competencies and encounter practical challenges. Because

students are expected to document and participate in these experiences, schools need effective ways of supervising and organizing immersion information. LearnIQ Track addresses this need through digital time-in and time-out recording and daily journal submission, allowing students to document their participation while teachers can monitor their immersion records. The study therefore supports the importance of work immersion in the target population and provides context for why the proposed monitoring functions are relevant to Senior High School students.

Ador et al. (2023) explored the work immersion experiences of Grade 12 STEM students from a local science-integrated high school through narrative research. The researchers examined students' experiences before, during, and after work immersion, focusing on how students experienced their exposure to actual work environments. The study is relevant to LearnIQ Track because it demonstrates the importance of understanding work immersion from the students' perspective rather than treating the program only as an administrative requirement. Students' experiences during immersion involve activities, responsibilities, workplace interactions, and learning that can be documented and reflected upon. This supports the inclusion of daily journals in LearnIQ Track because the journal feature is intended to give students a digital space to record their activities and reflections while participating in immersion. It also supports teacher monitoring because the information submitted by students can help coordinators follow their progress even when students are completing immersion activities outside the school.

Digital Attendance Monitoring Systems

Attendance monitoring is particularly important in work immersion because students must complete their required immersion activities and hours. Manual attendance sheets can provide a straightforward way of recording presence, but they can also require teachers or coordinators to collect, organize, and verify multiple physical records. The current LearnIQ Track manuscript identifies manual attendance and journal recording as existing processes that require physical documentation and teacher checking, creating an opportunity for a more centralized digital approach to work immersion monitoring.

Hege and Igon (2025) developed a web-based monitoring system for State Vocational High School 7 Kupang that addressed the monitoring of students at distant placement sites, attendance reporting, and activity journals. Their system allowed students to take attendance and fill out activity journals online while supervising teachers could monitor student activities. The system was designed to facilitate real-time monitoring, improve transparency, support accurate reporting, and maintain digital records of attendance and journals. This study has a particularly strong connection to LearnIQ Track because the proposed systems have closely related monitoring functions. LearnIQ Track will also allow students to record time-in and time-out during work immersion and submit daily journals, while teachers will monitor these records through the teacher dashboard. The difference is that LearnIQ Track places these functions within a broader system that also includes AI-generated learning materials and gamified lesson review. Therefore, the study by Hege and Igon provides direct support for the feasibility and usefulness of digitizing work immersion

attendance and journal monitoring while also helping establish the specific technology gap addressed by LearnIQ Track.

Kesuma, Tanjung, and Suhardi (2025) developed a web-based application for monitoring participants' attendance and activities. Their study identified problems associated with manual monitoring, including delayed reports and inaccurate data, and developed a system using PHP, MySQL, and HTML with functions for attendance recording, activity reporting, documentation uploads, certificate management, and real-time assessment. The system was developed using the Waterfall method and tested using Black Box testing, with the researchers reporting that the system's major functions operated correctly and that the digital approach improved accuracy, transparency, and monitoring efficiency. This study is relevant to LearnIQ Track because it provides additional evidence that digital attendance and activity monitoring can support the organization and management of student documentation. LearnIQ Track similarly aims to digitize the recording of student time-in and time-out and provide teachers with a centralized view of attendance and immersion activities. However, LearnIQ Track extends the concept by integrating attendance monitoring with daily journals, teacher dashboards, AI-supported academic review, and gamified learning activities.

Digital Journal and Reflection Submission

Daily journals are important in work immersion because they provide students with a way to document what they did, what they experienced, and what they learned during their workplace exposure. In addition to serving as documentation, reflection can help students connect practical workplace

experiences with the knowledge and competencies developed in school. However, handwritten journals can be difficult for teachers to organize and monitor when many students are participating in work immersion at different locations. The current LearnIQ Track manuscript identifies this issue and proposes digital journal submission so that students can submit their reflections through the system and teachers can review and monitor them digitally.

Hege and Igon (2025) provide direct support for this approach because their monitoring system included online activity journal submission alongside attendance recording. Their study identified challenges related to activity journal monitoring and developed a system that allowed students to enter their activities digitally while supervising teachers monitored those submissions. The system therefore demonstrates how digital journals can become part of a broader monitoring process rather than remaining as isolated student documents. This is directly relevant to LearnIQ Track because its daily journal feature is intended to provide the same basic advantage: students can document their daily immersion activities through the system, while teachers can access and monitor submissions without having to collect physical notebooks or papers. LearnIQ Track also connects the journal records with attendance and student monitoring, allowing the teacher dashboard to provide a more centralized view of each student's immersion status.

Digital Work Immersion Monitoring Systems

Digital work immersion monitoring systems demonstrate how technology can be used to organize information that is traditionally handled through paper-based attendance sheets, journals, and reports. These systems can

provide centralized records, improve accessibility, and allow teachers or supervisors to monitor students who are completing workplace activities outside the school. This area of literature is particularly important to LearnIQ Track because work immersion monitoring is one of its two major purposes, and the proposed system is intended to reduce manual monitoring processes.

Cinense et al. (2025) developed and evaluated Workify, a Senior High School Work Immersion Management System with data analytics designed for Philippine secondary education. The researchers identified administrative challenges involving inefficient manual processes and student-placement matching and developed a system that integrated automated matching, real-time progress tracking, and dashboard analytics. Workify was developed using Scrum and underwent requirements gathering, user-centered design, iterative development, testing, and phased deployment. The evaluation involved 50 participants, including IT professionals and end-users, and the system received an overall ISO/IEC 25010 score of 3.96 out of 5, while user acceptance using TAM produced a mean of 3.88 out of 5. The study also reported reductions in placement mismatches and processing time. This study is highly relevant to LearnIQ Track because it is specifically situated in Philippine Senior High School work immersion and demonstrates the usefulness of a digital system for managing immersion-related processes. However, Workify focuses on work immersion management, matching, progress tracking, and analytics, while LearnIQ Track adds a separate academic learning component involving AI-generated quizzes, activities, reviewers, interactive quizzes, scores, and leaderboards. This distinction strengthens the technology gap of the proposed study because it shows that digital work immersion management already exists,

but the combination of work immersion monitoring with AI-supported and gamified academic learning remains a distinct focus of LearnIQ Track.

Hege and Igon (2025) likewise demonstrate the usefulness of integrating attendance and journal submission into an online monitoring system. Their system was designed for vocational high school students and allowed supervising teachers to monitor student activities while students recorded attendance and submitted activity journals online. The study is especially relevant to LearnIQ Track because the target users are also Senior High School students and the monitoring requirements are similar. However, LearnIQ Track is designed around the specific context of Paete Science and Business College Inc. and combines work immersion monitoring with an AI-based learning game. This means that findings from digital student monitoring studies can provide a foundation for the monitoring component, while the AI and gamification literature provides the foundation for the learning component. The integration of these two areas is what distinguishes the proposed system from systems that focus only on work immersion monitoring.

User Acceptance of Technology

User acceptance is important because the success of a system depends not only on whether its technical functions work but also on whether its intended users perceive the system as useful and manageable. LearnIQ Track will be used by students, teachers, and the school administrator, and each group will interact with different features. Students will use the learning game, quizzes, reviewers, scores, attendance, and journals; teachers will upload learning materials, validate AI-generated content, and monitor students; and the admin will manage

accounts and oversee records. The current manuscript uses selected characteristics of ISO/IEC 25010 to evaluate the quality of the developed system and selected constructs of the Technology Acceptance Model to assess user acceptance, specifically perceived ease of use, perceived usefulness, attitude toward using, and behavioral intention to use.

Davis (1989) introduced the Technology Acceptance Model to explain user acceptance of information technology and developed measures for two major beliefs: perceived usefulness and perceived ease of use. Perceived usefulness refers to the extent to which a person believes that using a particular system will improve their job performance, while perceived ease of use concerns the extent to which a person believes that using the system will require little effort. Davis's work provided an important foundation for later research on technology adoption because it demonstrated that users' beliefs about usefulness and ease of use are important factors in technology acceptance. This is directly relevant to LearnIQ Track because the proposed system contains many features that students and teachers must learn to use, and its success depends on whether they believe those features actually help them. Students need to perceive the AI-generated reviewers, quizzes, scores, and work immersion tools as useful, while teachers need to perceive the upload, validation, monitoring, and reporting functions as useful and manageable. Therefore, TAM provides an appropriate theoretical basis for determining whether LearnIQ Track is not only technically functional but also acceptable to its intended users.

Cinense et al. (2025) provide a particularly relevant application of TAM because their Workify system was also designed for Senior High School work immersion management. Their evaluation reported strong user acceptance, with

an overall TAM mean of 3.88 out of 5 and perceived usefulness receiving a mean of 4.01. This is relevant to LearnIQ Track because it demonstrates how a Philippine work immersion information system can be evaluated according to users' perceptions of its usefulness and acceptance. LearnIQ Track follows a similar evaluation principle but applies it to a broader system that includes both academic and work immersion functions. The study also provides a useful example of evaluating a developed school information system rather than relying only on technical testing. For LearnIQ Track, this means that user feedback from Grade 11 students, Grade 12 students, and teachers can help determine whether the integrated system is perceived as useful, understandable, and appropriate for actual school use. TAM in this study is used to assess users' perceptions and acceptance of LearnIQ Track rather than to measure actual improvement in students' academic learning outcomes.

Software Quality and ISO/IEC 25010

Software quality is important because a system may contain the required features but still fail to meet user needs if it is unreliable, difficult to use, inefficient, insecure, or incompatible with its intended environment. LearnIQ Track contains multiple interconnected functions, including user authentication, file uploading, AI-generated content, quizzes, score computation, leaderboards, attendance, journals, dashboards, and reports. Therefore, evaluating only whether each feature exists would not be enough to determine whether the system is an acceptable software product.

ISO/IEC 25010:2023 provides a recognized product-quality model for ICT and software products. The standard defines nine quality characteristics and

corresponding subcharacteristics that can be used as a reference for specifying, measuring, and evaluating software quality throughout its lifecycle. It is applicable to different stakeholders, including developers, quality assurance personnel, acquirers, and evaluators. For this study, five quality characteristics are selected for the evaluation of LearnIQ Track: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. These characteristics were selected because they directly relate to the system's primary functions, user interaction, performance, operational stability, and protection of user and student records. The other quality characteristics, including Compatibility, Maintainability, and Portability, are not included as separate evaluation criteria because they are outside the specific evaluation scope of the study. Since LearnIQ Track is a web-based system accessed through web browsers, compatibility and portability are addressed through system testing within the intended device and browser environment rather than through separate user-evaluation criteria. Maintainability is likewise outside the scope of the user evaluation, which focuses on the developed system's functionality, selected software quality characteristics, technology acceptance, and overall acceptability.

Although ISO/IEC 25010:2023 provides nine quality characteristics, the study selects Functional Suitability, Usability, Performance Efficiency, Reliability, and Security for the evaluation of LearnIQ Track. These characteristics were selected because they directly correspond to the system's primary functions, user interaction, system performance, operational stability, and protection of user and student records. The other characteristics, including Compatibility, Maintainability, and Portability, are not included as separate

evaluation criteria because they are outside the specific evaluation scope of the study. Since LearnIQ Track is a web-based system accessed through standard web browsers, compatibility and portability are addressed primarily through system testing on the intended devices and browser environment rather than through separate user-evaluation criteria. Maintainability is likewise outside the scope of the user acceptance evaluation because the study focuses on the developed system's functionality, quality, and acceptance by its intended users.

Cinense et al. (2025) also demonstrate the practical use of ISO/IEC 25010 in evaluating a Philippine Senior High School work immersion system. Their Workify system received an overall ISO/IEC 25010 score of 3.96 out of 5, with particularly strong ratings in security and usability. This is relevant to LearnIQ Track because it demonstrates that software quality evaluation can be applied to school-based work immersion systems and can identify areas of strength in the developed platform. The study is especially useful as a related system because Workify and LearnIQ Track both involve work immersion management, although LearnIQ Track includes additional AI and gamification components. The use of ISO/IEC 25010 in the related study supports the decision to use a recognized software-quality framework in evaluating LearnIQ Track, helping the researchers determine whether the proposed system meets acceptable quality standards beyond simple functional operation.

Data Privacy and Security in Educational Systems

Data privacy and security are important considerations in LearnIQ Track because the system will collect and store student information, account details, quiz scores, attendance records, daily journals, and other school-related

information. The system will also use role-based access to distinguish between students, teachers, and administrators. This means that information should only be accessible to users who are authorized to view or manage it. The current manuscript recognizes this requirement by including user authentication, role-based access, database protection, and system security among the system requirements.

The Data Privacy Act of 2012 (Republic Act No. 10173) provides the Philippine legal framework for protecting personal information and establishes requirements for organizations that collect and process personal data. This is directly relevant to LearnIQ Track because the proposed platform will process information belonging to Senior High School students and school personnel. The system therefore needs to collect only information necessary for its functions, protect stored information, restrict access according to user roles, and implement appropriate safeguards against unauthorized use. For LearnIQ Track, this means that students should primarily access their own learning and immersion records, teachers should access the information necessary for their monitoring responsibilities, and the administrator should have broader authorized access for system supervision. Data privacy is therefore not simply a technical feature of the system; it is part of responsible implementation because the platform will combine academic information with work immersion records.

UNESCO's guidance concerning generative AI in education also emphasizes the importance of responsible and human-centered use of AI. The guidance recognizes the potential of generative AI in education while emphasizing concerns involving privacy, safety, appropriate use, and human

oversight. This is relevant to LearnIQ Track because the system combines AI with student information and educational materials. The AI component should therefore operate within an environment where teachers remain involved in validating generated content and where user information is protected. This supports the system's decision to treat AI as an educational assistance tool rather than an autonomous replacement for teacher judgment. It also reinforces the need for authentication, role-based permissions, and responsible handling of uploaded learning materials and student records.

2.1.3 Thematic or Chronological Organization

The related literature and studies are organized thematically according to the major concepts and features of LearnIQ Track. The discussion begins with artificial intelligence in education and automated question generation, followed by gamification and digital learning systems. It then covers work immersion monitoring, technology acceptance, software quality, and data privacy and security. This organization provides a clear presentation of the concepts and findings related to the proposed system.

2.1.4 Relevance to the Present Project

The reviewed literature and studies provide a foundation for the development of LearnIQ Track by presenting relevant concepts, technologies, and findings related to its proposed features. These references support the integration of AI-generated learning materials, interactive quizzes, gamification, and digital work immersion monitoring. They also provide guidance for selecting appropriate software quality characteristics and technology acceptance factors for system evaluation. Thus, the reviewed studies

help establish the basis for the design, development, testing, and evaluation of the proposed system.

2.1.5 Citation and Referencing

The researchers ensured that all literature and studies included in this chapter were properly cited and referenced to acknowledge their original authors and sources. Relevant information was presented using appropriate in-text citations, while complete bibliographic details were provided in the reference list. This practice maintains the credibility of the study and recognizes the contributions of previous researchers.

2.2 Related Systems and Technology Gap Analysis

This section presents existing systems related to the proposed LearnIQ Track and examines their features, functions, and technologies. It also identifies similarities, differences, and limitations of these systems to determine the technology gaps that the proposed project aims to address.

2.2.1 Identification of Related Systems

The following systems were identified based on their relevant similarities to LearnIQ Track in terms of AI-assisted learning, automated quiz generation, gamified learning activities, student performance monitoring, and digital school management. These systems were examined to understand their existing functions and determine their relevance to the proposed project. Each system is briefly described below, together with the reason for its inclusion in the analysis.

1. Google Classroom.

Google Classroom is a web-based and mobile-supported learning platform that allows teachers to create classes, distribute learning materials, manage assignments, and provide feedback to students. It also enables teachers to monitor student submissions, grades, and missing activities through its grading features. It is included because its classroom management and student monitoring functions are related to the learning material management and teacher dashboard features of LearnIQ Track.

2. Moodle Learning Management System (LMS).

Moodle is an open-source learning management system that allows educators to create online courses, provide learning resources, conduct quizzes, and manage student activities. It includes features for tracking course progress, managing assignments, providing feedback, and awarding badges to learners. It is included because its assessment, progress tracking, and learning management functions are comparable to the educational features of LearnIQ Track.

3. Open edX.

Open edX is an open-source online learning platform that provides course creation, learner management, assessments, and learning analytics. It allows instructors to develop course materials, create different types of assessments, and monitor learner performance through reports and dashboards. It is included because its course management and assessment features are relevant to the learning material and student performance monitoring components of LearnIQ Track.

4. Canvas Learning Management System.

Canvas LMS is an educational platform that provides tools for managing courses, assignments, quizzes, grades, and student communication. It also includes analytics features that allow instructors and administrators to monitor course activities and student performance. It is included because its assessment management, grading, and analytics functions are related to the monitoring and reporting features of LearnIQ Track.

5. Kahoot!

Kahoot! is a game-based learning platform that allows teachers to create interactive quizzes and learning games for students. It provides live and self-paced activities, while its reports allow teachers to review participant scores, question results, and leaderboard rankings. It is included because its interactive quiz and gamification features are closely related to the learning game, score computation, and leaderboard components of LearnIQ Track.

6. Wayground (formerly Quizizz).

Wayground is an AI-supported learning platform that enables teachers to create interactive assessments, quizzes, lessons, and other learning resources. It provides live and self-paced activities, different question types, and performance reports that show student accuracy, completion rates, and scores. It is included because its AI-supported assessment creation and student performance tracking features are relevant to the AI-generated learning content and quiz monitoring functions of LearnIQ Track.

7. Quizlet.

Quizlet is a digital learning platform that provides flashcards, study activities, and practice tests. Its AI-powered tools allow users to generate

practice tests and study guides from uploaded notes, slides, and other learning materials. It is included because its ability to transform learning materials into study resources is similar to the AI-assisted generation of quizzes, activities, and reviewers in LearnIQ Track.

8. Nearpod.

Nearpod is an interactive learning platform that allows teachers to create lessons using presentations, videos, quizzes, polls, and other classroom activities. It also provides real-time dashboards and reports that help teachers monitor student responses and learning progress. It is included because its interactive learning activities and teacher monitoring features are related to the gamified learning and dashboard components of LearnIQ Track.

9. eSkooly School Management System.

eSkooly is a school management system that provides administrative and academic management features, including student information, attendance, examinations, homework, and reports. It allows teachers and administrators to manage student records and monitor academic activities through designated system modules. It is included because its attendance management, student records, and administrative reporting functions are relevant to the digital monitoring and admin dashboard features of LearnIQ Track.

10. Microsoft Teams for Education.

Microsoft Teams for Education is a digital learning and collaboration platform that provides class teams, assignment management, grading, and communication tools. Its Education Insights feature allows teachers to monitor student engagement, assignment submissions, and meeting attendance through

visual reports. It is included because its student progress monitoring, digital submissions, and teacher reporting features are related to the monitoring functions of LearnIQ Track.

2.2.2 System Feature Comparison

The following table presents a comparison of the features of the identified related systems and the proposed LearnIQ Track. The comparison focuses on AI-assisted learning material generation, interactive quizzes and gamification, student performance monitoring, and digital work immersion documentation. It highlights the similarities and differences between existing systems and the proposed project.

Table 2.1. Comparison of Features of Related Systems and LearnIQ Track

Related Systems	AI-Assisted Learning Material Generation	Quizzes and Interactive Activities	Gamification and Leaderboard	Student Progress Monitoring and Reports	Digital Attendance Monitoring	Daily Journal and Work Immersion Monitoring
Google Classroom	—	✓	—	✓	—	—
Moodle LMS	—	✓	✓	✓	—	—
Open edX	—	✓	—	✓	—	—
Canvas LMS	—	✓	—	✓	—	—
Kahoot!	—	✓	✓	✓	—	—
Wayground (Quizizz)	✓	✓	✓	✓	—	—
Quizlet	✓	✓	—	—	—	—
Nearpod	—	✓	—	✓	—	—

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eSkooly School Management System	—	✓	—	✓	✓	—
Microsoft Teams for Education	—	✓	—	✓	—	—
LearnIQ Track	✓	✓	✓	✓	✓	✓

The comparison shows that existing systems provide various features for online learning, assessment, gamification, and student monitoring. However, the reviewed systems generally focus on specific aspects of educational management and do not document the complete combination of learning activities and work immersion monitoring proposed in LearnIQ Track. The proposed system integrates AI-generated learning materials, interactive quizzes, a leaderboard, student performance monitoring, digital attendance recording, and daily journal submission in one platform. This integration distinguishes LearnIQ Track from the related systems and addresses the identified need for a centralized platform supporting both academic learning and work immersion monitoring.

2.2.3 Analysis of Existing Systems

The analysis of the existing systems shows that each platform offers features that support different aspects of education and student management. Google Classroom, Moodle LMS, Open edX, and Canvas LMS provide tools for managing learning materials, assignments, assessments, and student progress. Kahoot!, Wayground, and Quizlet support interactive learning through

quizzes, game-based activities, and AI-assisted study tools. Meanwhile, Nearpod provides interactive lessons and assessment reports, while eSkooly and Microsoft Teams for Education offer features for student management, attendance-related records, and monitoring. Although these systems provide useful educational functions, the reviewed features do not demonstrate the same integrated combination of AI-generated learning activities, gamified assessments, and digital work immersion documentation proposed in LearnIQ Track. Therefore, the analysis of these systems serves as a basis for developing a centralized platform that combines academic learning activities with work immersion attendance and daily journal monitoring.

2.2.4 Technology Gap

Based on the analysis of the related systems, existing platforms provide different features for online learning, assessment, gamification, and student management. However, the reviewed systems do not document the same integrated combination of AI-generated learning materials, interactive quizzes, gamified activities, and digital work immersion monitoring. Most focus on academic learning or student management rather than combining both functions in a single platform. To address this gap, LearnIQ Track is proposed as an integrated system that allows teachers to upload learning materials and generate AI-assisted quizzes, activities, and reviewers. It also provides interactive learning activities, score tracking, and a leaderboard, together with digital work immersion attendance recording and daily journal submission. Through these features, the system aims to provide teachers and administrators with a centralized platform for monitoring students' academic activities and work immersion progress.

2.2.5 Proposed Response to the Gap

To address the identified technology gap, the researchers propose LearnIQ Track, an AI-Based Learning Game and Digital Work Immersion Monitoring System designed for senior high school students of Paete Science and Business College Inc. The system integrates AI-assisted generation of quizzes, activities, and reviewers from teacher-uploaded learning materials, allowing students to participate in interactive learning activities with score tracking and a leaderboard. It also provides digital work immersion monitoring through time-in and time-out recording, daily journal submission, and progress tracking. Teachers and administrators can use the system to monitor students' academic performance and work immersion activities through centralized dashboards. By combining these functions, LearnIQ Track aims to provide a unified platform that supports both classroom learning and work immersion monitoring.

2.3 Synthesis

2.3.1 Integration of Findings

The reviewed literature and studies on Artificial Intelligence in Education established the importance of AI in supporting teaching and learning processes. Holmes and Tuomi (2022) and Wang et al. (2024) discussed how AI technologies can assist in assessment, adaptive learning, and educational support. UNESCO (2023) further emphasized the importance of responsible AI use and teacher supervision when integrating generative AI into educational activities. These findings support the development of LearnIQ Track,

particularly its use of AI to generate quizzes, activities, and reviewers from teacher-uploaded learning materials while maintaining teacher validation of the generated content.

Studies on AI-generated learning materials and automated assessment also highlighted the potential of AI to assist teachers in preparing educational resources and assessment activities. Gao et al. (2023) discussed automated assessment approaches and their potential to reduce repetitive tasks, while Mustafa et al. (2024) emphasized the importance of considering educational needs and context when implementing AI-based systems. These findings provide a basis for LearnIQ Track's AI-assisted learning features, which are intended to support teachers in preparing learning activities and provide students with additional materials for reviewing their lessons.

The reviewed literature on gamification and digital learning also demonstrated the value of interactive and technology-supported learning environments. Studies on gamification discussed the use of game elements such as points, rewards, and leaderboards to encourage student participation. Meanwhile, literature on web-based learning emphasized the accessibility and flexibility provided by digital platforms. These findings support the integration of interactive quizzes, score tracking, and leaderboard features in LearnIQ Track to provide students with a more engaging way of reviewing their lessons.

Furthermore, the literature and studies related to work immersion monitoring emphasized the importance of proper documentation, supervision, and monitoring of students' workplace activities. Digital monitoring approaches provide ways to organize attendance records, activity journals, and student progress information. These findings support LearnIQ Track's digital work

immersion features, particularly time-in and time-out recording, daily journal submission, and teacher monitoring through the system's dashboard.

The analysis of related systems also revealed that existing educational platforms offer useful features for learning management, assessment, gamification, and student monitoring. However, the reviewed systems focus on different educational and administrative functions. This indicates an opportunity to integrate AI-assisted learning, gamified assessments, and digital work immersion documentation into one platform. The reviewed concepts, technologies, and findings collectively provide the foundation for the design and development of LearnIQ Track for Senior High School students of Paete Science and Business College Inc.

2.3.2 Identified Gap

Based on the reviewed literature, studies, and existing systems, AI-assisted learning, automated quiz generation, gamification, and digital work immersion monitoring have been explored as separate areas of educational technology. Existing learning platforms provide features for managing educational materials, conducting assessments, and monitoring student performance, while work immersion systems focus on attendance recording, journal documentation, and student supervision. However, the reviewed studies and systems do not demonstrate the same integration of these functions in a single platform designed for Senior High School students. This identified gap provides the basis for developing LearnIQ Track, which combines AI-generated learning materials, interactive quizzes, scores and leaderboards, digital work immersion attendance, daily journal submission, and centralized monitoring dashboards. The proposed system aims to address the need for an integrated

platform that supports both academic learning activities and work immersion monitoring at Paete Science and Business College Inc.

CHAPTER III

RESEARCH METHODOLOGY

This chapter presents the methods and procedures that the researchers will employ in designing, developing, testing, and evaluating the proposed system. It includes the research design, participants of the study, data gathering procedures, project design, system architecture, software development methodology, system testing and evaluation, questionnaire validation, and statistical treatment.

3.1 Materials and Methods

This section presents an overview of how LearnIQ Track was researched, developed, and evaluated. It identifies the research and fact-finding activities, system development activities, and resources and technologies used by the researchers in gathering information, designing the system, developing its features, and evaluating the completed product. Each of these areas is discussed in detail in the corresponding sections of this chapter.

3.1.1 Research and Fact-Finding Activities

The researchers gathered information about the existing learning and work immersion monitoring practices of Paete Science and Business College Inc. through preliminary interviews, observation, and online research. These activities helped identify the needs of students and teachers, particularly in relation to learning materials, academic activities, attendance recording, and daily journal submission. A review of related systems was also conducted to identify existing features and technology gaps. A structured survey questionnaire was prepared to gather user evaluations of the developed system.

These activities are discussed further under the corresponding sections of this chapter.

3.1.2 System Development Activities

LearnIQ Track was planned, designed, developed, tested, and improved following the Agile Software Development Model. The development process consisted of Requirements Planning, System Design, Development, Testing and Evaluation, and Feedback and Improvement. These phases guided the researchers in identifying user needs, designing the system structure and interfaces, developing and integrating its features, testing its functionality, and applying necessary improvements based on testing results and user feedback. The activities performed during each phase are discussed in detail under Section 3.9, Software Development Methodology.

3.1.3 Resources and Technologies

LearnIQ Track is a web-based application developed using HTML, CSS, and JavaScript for its front-end interface, while Python with FastAPI handles the backend processes and system operations. Google Gemini AI is integrated to generate quizzes, activities, and reviewers based on teacher-uploaded learning materials. JSON is used for data exchange between system components, while Supabase provides database management and backend services for storing and managing system records. Cloudflare is used as the platform for hosting and deployment.

The researchers also used Visual Studio Code as the primary development environment for writing, editing, and debugging the system's source code. These technologies work together to support the system's academic

learning features, AI-assisted content generation, digital work immersion monitoring, and administrative functions.

3.2 Research Design

The study will employ a Developmental Research Design and a Descriptive-Evaluative Research Design. The Developmental Research Design will guide the systematic planning, designing, development, testing, and improvement of LearnIQ Track, a web-based system that integrates AI-assisted learning activities, gamified quizzes, and digital work immersion monitoring.

The Descriptive-Evaluative Research Design will be used to assess the developed system based on the experiences and evaluations of its intended users. The descriptive aspect will present the respondents' assessments, while the evaluative aspect will determine the system's quality and acceptability using selected ISO/IEC 25010 quality characteristics and Technology Acceptance Model (TAM) constructs. The evaluation will cover Functional Suitability, Usability, Performance Efficiency, Reliability, Security, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, and Overall Acceptability.

These research designs will guide the development and evaluation of LearnIQ Track, which includes user account management, learning material uploading, AI-generated quizzes, activities and reviewers, interactive quizzes, leaderboard, digital time-in and time-out, daily journal submission, teacher dashboard, and admin dashboard. The system will be evaluated by selected Grade 11 students, Grade 12 students, and teachers according to their respective roles and experiences with the system.

3.2.1 Rationale for the Selected Research Design

The Developmental Research Design is appropriate for this study because its primary objective is to develop a functional information system that addresses the learning and work immersion monitoring needs of Paete Science and Business College Inc. It provides a structured approach for identifying system requirements, designing the system, developing its features, testing its functions, and making necessary improvements based on identified issues and feedback.

Meanwhile, the Descriptive-Evaluative Research Design is suitable because the study also aims to determine how the intended users assess the completed system. Through this design, the researchers will gather and analyze respondents' evaluations of the system's selected software quality characteristics and technology acceptance factors. The results will provide information about the system's perceived usability, usefulness, functionality, and overall acceptability.

The combination of these research designs will allow the researchers to address both the development and evaluation objectives of the study. It will provide a systematic basis for producing LearnIQ Track and determining whether its implemented features meet the needs and expectations of its intended users.

3.3 Population of the Study

The population of the study will consist of stakeholders from Paete Science and Business College Inc. who are directly involved in or are intended to use LearnIQ Track. The study will be conducted at the institution located in Paete, Laguna,

Philippines. The school was selected as the locale because the proposed system is specifically designed to support its Senior High School students, teaching faculty, and school administrator in academic learning activities and work immersion monitoring. The target population will include Grade 11 students, Grade 12 students, and teaching faculty. Based on the population data provided for the study, there are 409 Grade 11 students, 514 Grade 12 students, and 48 teaching faculty, resulting in a total population of 971.

Grade 11 and Grade 12 students are included because they are the primary student users of LearnIQ Track. They will use the system to access learning materials, answer AI-generated quizzes and activities, review lessons, participate in interactive quizzes, and view their scores and leaderboard rankings. Grade 12 students are also included because they participate in work immersion and will use the digital attendance and daily journal features.

The teaching faculty are included because they will use the system to upload learning materials, review and validate AI-generated content, monitor student learning progress, and manage work immersion attendance and journal records. The school principal will serve as the system administrator and a key informant, providing information about the school's requirements and overseeing user accounts, reports, and system records. However, the principal will not be included in the statistical survey population because the sample-size computation covers only Grade 11 students, Grade 12 students, and teaching faculty.

The identified population is appropriate for the study because these groups represent the primary users and stakeholders who will interact with or oversee the functions of LearnIQ Track. Their participation will provide relevant assessments of the system based on their respective roles and experiences.

3.4 System Testers

The developed LearnIQ Track system will undergo technical testing with the participation of ten (10) IT experts. These testers were selected based on their educational backgrounds, technical qualifications, professional positions, and relevant experience in information technology and related fields. Their involvement is intended to provide technical assessment of the system's functionality, usability, reliability, performance, and security.

The system testers will examine the major functions of LearnIQ Track, including user authentication and role-based access, learning material management, AI-generated content, interactive quizzes, score computation, leaderboard ranking, digital attendance, daily journal submission, and the teacher and administrator dashboards. Their observations and feedback will help identify technical issues and areas that may require improvement.

3.4.1 Profile of System Testers

The researchers identified ten (10) IT experts as system testers. Their educational backgrounds, professional credentials, current positions, and relevant experience are presented in the following table.

Table 3.1 Profile of System Testers

System Tester	Name	Educational Background / Credentials and Professional Experience
1	Robin Paolo Bona	BSIT Graduate (2018); Computer System Servicing NC II; Administrative Aide – Information Technology, Municipal Government of Santa Cruz, Laguna (2019–Present).

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2	Mharjun V. Tobias	BS Information Technology Graduate (2013); Civil Service Professional (2022); Graphic Artist / Technical Support, MGO Santa Cruz, Laguna (2020–2022).
3	Kevin Adlaon	BS Computer Science Graduate (2012); Google Cloud Certified Professional Data Engineer; Analyst Programmer with more than 10 years of experience.
4	Justine Jhay Lopez	BSIT (2022–2026); Full-Stack Developer; MIS Programmer, Management Information Systems Office (2025–Present).
5	Jonas T. Santos	BSIT Graduate (2022); Civil Service Subprofessional Eligibility; MIS IT Staff, Municipal Government of Santa Cruz, Laguna (2025–Present).
6	John Marvin A. Tabo	BSIT Graduate (2023); Civil Service Examination Professional Passer; Computer Troubleshooting and Hardware Configuration.
7	Wilmar T. Morota	Computer Programming undergraduate, AMA Computer College; Career Service Examination Professional Passer; Job-Order personnel, Management Information Systems Office (2025–Present).
8	Vic Marco Orbista	Mega Cat Studios; Spec Ops.
9	Mike Angelo Matrido	Director of Technical Services, One Depot, Makati, Philippines.
10	Ferdinand T. Celis	Website Developer / Designer.

3.5 Respondents for User Evaluation

The respondents for the user evaluation of LearnIQ Track will consist of selected Grade 11 students, Grade 12 students, and teaching faculty of Paete Science and Business College Inc. These groups were selected because they represent the

primary users of the proposed system and are directly involved in the academic learning and work immersion activities supported by its features.

Grade 11 and Grade 12 students will participate in the evaluation because they will use the system to access learning materials, answer AI-generated quizzes and activities, review lessons, participate in interactive quizzes, and view their scores and leaderboard rankings. Grade 12 students will also evaluate the work immersion features, including digital time-in and time-out recording and daily journal submission.

The teaching faculty will be included because they will use LearnIQ Track to upload learning materials, review and validate AI-generated content, monitor student learning progress, and manage work immersion attendance and journal records. Their evaluation will provide feedback on the system's usefulness and suitability for their teaching and monitoring responsibilities.

The total population of the three respondent groups is 971, consisting of 409 Grade 11 students, 514 Grade 12 students, and 48 teaching faculty. From this population, a sample of 283 respondents will be selected for the user evaluation. The school principal will serve as the system administrator and key informant but will not be included in the statistical survey sample. The ten (10) IT experts discussed in Section 3.4 will participate separately in the technical testing of the system.

3.5.1 Distribution of Respondents

The 283 respondents will be distributed proportionally among the three identified groups based on their respective population sizes. This distribution will ensure that each group is represented in the user evaluation according to its proportion of the total population.

Table 3.2 Distribution of Respondents for User Evaluation

Respondent Group	Total Population	Sample Size	Sampling Design
Grade 11 Students	409	119	Stratified Random Sampling
Grade 12 Students	514	150	Stratified Random Sampling
Teaching Faculty	48	14	Stratified Random Sampling
Total	971	283	

As presented in Table 3.2, the study has a total population of 971, from which 283 respondents will be selected. The sample consists of 119 Grade 11 students, 150 Grade 12 students, and 14 teaching faculty. The allocation is based on the proportional representation of each group in the total population.

The selected respondents will evaluate LearnIQ Track after interacting with the system and using the features relevant to their respective roles. Their responses will serve as the basis for assessing the system's selected software quality characteristics, technology acceptance factors, and overall acceptability.

3.6 Sampling Design

This study will employ stratified random sampling, a probability sampling technique in which the population is divided into distinct groups or strata before respondents are randomly selected. This technique is appropriate for the study because the intended users of LearnIQ Track belong to three different groups: Grade 11 students, Grade 12 students, and teaching faculty of Paete Science and Business College Inc. It

ensures that each group is represented in the user evaluation according to its proportion of the total population.

The total population of 971 will be divided into three strata: 409 Grade 11 students, 514 Grade 12 students, and 48 teaching faculty. The required sample size of 283 respondents will be allocated proportionally among these groups, resulting in 119 Grade 11 students, 150 Grade 12 students, and 14 teaching faculty.

The sample size was determined using Slovin's formula:

$$n = \frac{N}{1 + N(e)^2}$$

Where:

- n = required sample size
- N = total population
- e = allowable margin of error

Using a total population of 971 and a 5% margin of error:

$$n = \frac{971}{1 + 971(0.05)^2}$$

$$n = \frac{971}{1 + 971(0.0025)}$$

$$n = \frac{971}{3.4275}$$

$$n = 283.269 \approx 283$$

Thus, the required sample size is 283 respondents, following the sample-size computation provided by the study's statistician.

After determining the sample size for each stratum, the researchers will prepare separate lists of eligible Grade 11 students, Grade 12 students, and teaching faculty. Respondents will then be selected randomly from their respective lists using an appropriate random-selection procedure. This process will provide representation from all three groups while maintaining the probability-based nature of the sampling method.

The selected respondents must be officially enrolled Grade 11 or Grade 12 students, or teaching faculty of the Senior High School department who are involved in the use, monitoring, or evaluation of LearnIQ Track. They must also be willing to participate, provide the required consent, interact with the developed system, and complete the evaluation questionnaire. Individuals who do not meet these criteria will be excluded from the statistical survey.

The school principal will not be included in the statistical sample. Instead, the principal will serve as the system administrator and key informant, providing institutional context and assisting in the authorization and implementation of the study.

3.7 Data Collection Instrument

The study will use a structured survey questionnaire, interview guide, and observation checklist to gather the information needed for the development and evaluation of LearnIQ Track. The survey questionnaire will serve as the primary instrument for assessing the system's software quality, user acceptance, and overall acceptability.

3.7.1 Instrument Development

The researchers will prepare a structured questionnaire based on the study's objectives and evaluation criteria. It will cover five selected ISO/IEC 25010 characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security, as well as four Technology Acceptance Model (TAM) constructs: Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. A five-point Likert scale will be used to measure the respondents' evaluations.

An interview guide and observation checklist will also be used to gather information about the school's existing learning and work immersion monitoring practices.

3.7.2 Instrument Validation

The survey questionnaire will be reviewed by the thesis adviser to ensure that its statements are clear, relevant, and aligned with the research objectives and evaluation criteria. Necessary revisions will be made based on the adviser's comments before the questionnaire is administered.

3.7.3 Administration of the Instrument

After securing permission from the school, the researchers will conduct interviews and observations to gather initial information. The selected respondents will then be oriented on the system and asked to use its features. Afterward, they will answer the survey questionnaire. The completed questionnaires will be collected, organized, and prepared for statistical analysis.

Participation will be voluntary, and the researchers will observe confidentiality and applicable data privacy requirements.

3.8 Project Design

The researchers designed LearnIQ Track as an integrated web-based system that combines AI-assisted academic learning activities with digital work immersion monitoring for Senior High School students of Paete Science and Business College Inc. The project design was developed based on the identified needs of students, teachers, and the school administrator, particularly in providing learning materials, preparing

quizzes and reviewers, monitoring student performance, and managing work immersion records.

The system consists of several major features, including user account management, learning material upload, AI-generated quizzes and activities, interactive quizzes, score computation, leaderboard ranking, digital attendance monitoring, daily journal submission, and monitoring dashboards. These features are organized according to the responsibilities of the three main user roles: Student, Teacher, and Admin. This arrangement allows the system to support both academic learning activities and work immersion monitoring within one platform.

The development of LearnIQ Track follows an iterative process in which the identified features are designed, developed, tested, and improved. The system requirements serve as the basis for determining the functions of each module, while the system design establishes how users interact with the platform and how information is processed and stored. Through this project design, the researchers aim to provide an organized platform that supports learning activities and assists teachers and administrators in monitoring student progress and work immersion records.

3.8.1 System Requirements

The functional requirements of LearnIQ Track focus on providing the necessary features for academic learning and work immersion monitoring. The system must support three main user roles, each with specific functions based on their responsibilities.

For students, the system must provide access to uploaded learning materials, AI-generated quizzes, activities, and reviewers. Students must also be able to participate in interactive quizzes, view their scores and leaderboard rankings, record their work immersion time-in and time-out, and submit their

daily journals. These functions allow students to access learning activities and maintain their work immersion records through the platform.

For teachers, the system must allow the uploading of learning materials that will serve as the basis for AI-generated learning content. Teachers must also be able to review and validate the generated quizzes, activities, and reviewers before they are made available to students. In addition, the system must provide access to student performance records, quiz results, attendance information, and submitted daily journals to support monitoring and supervision.

For the Admin, represented by the school principal, the system must provide functions for managing user accounts, monitoring system records, viewing reports, and supervising the overall use of the platform. These functions support the administrative management of student learning and work immersion information.

The non-functional requirements focus on the quality and performance of the system. LearnIQ Track will be evaluated using five selected software quality characteristics from ISO/IEC 25010: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. These characteristics will serve as the basis for assessing whether the system provides the required functions, can be used effectively by its intended users, performs efficiently, operates consistently, and protects system information and user access. The requirements will be considered in the system's development and evaluation.

3.8.2 System Design

The system design of LearnIQ Track organizes the major processes, user roles, and interactions required to support its academic learning and work immersion monitoring functions. The design is based on the identified system

requirements and the responsibilities of the Student, Teacher, and Admin accounts.

The Student module focuses on accessing learning materials, answering AI-generated quizzes and activities, participating in the learning game, viewing scores and leaderboard rankings, recording work immersion attendance, and submitting daily journals. Information submitted by students will be available for authorized teachers and administrators to monitor.

The Teacher module focuses on managing learning materials and supervising student activities. Teachers can upload learning resources, review and validate AI-generated content, and monitor student quiz performance, attendance records, and daily journal submissions. The Admin module provides the school principal with access to user account management, system records, and reports for overall monitoring and supervision.

The system also incorporates the interaction of its main components, particularly the frontend interface, backend processes, database, and Google Gemini AI service. Learning materials uploaded by teachers serve as the basis for AI-generated learning content, while student activities and work immersion records are processed and stored for monitoring. The design ensures that the functions available to each user are organized according to their assigned roles and responsibilities.

The system design may be represented through appropriate diagrams, such as a use-case diagram and data-flow diagram, to illustrate the relationship between users, system processes, and information flow. These diagrams should correspond to the actual implemented functions of LearnIQ Track.

3.8.3 System Architecture

LearnIQ Track follows a web-based architecture that connects its user interface, backend services, database, and external AI service. Students, teachers, and administrators access the platform through their respective accounts and interact with the functions assigned to their roles.

The frontend is developed using HTML, CSS, and JavaScript, while Python with FastAPI handles the backend operations. Supabase is used for database and authentication services, and Google Gemini AI supports the generation of quizzes, activities, and reviewers from uploaded learning materials. JSON is used for data exchange between the frontend and backend, while Cloudflare serves as the hosting and deployment platform.

The interaction of these components allows the system to provide its academic learning and work immersion monitoring features through a centralized web-based platform. A detailed discussion of the system architecture, including its hardware and software requirements, system components, database structure, user interface, and communication mechanisms, is presented in Section 3.10.

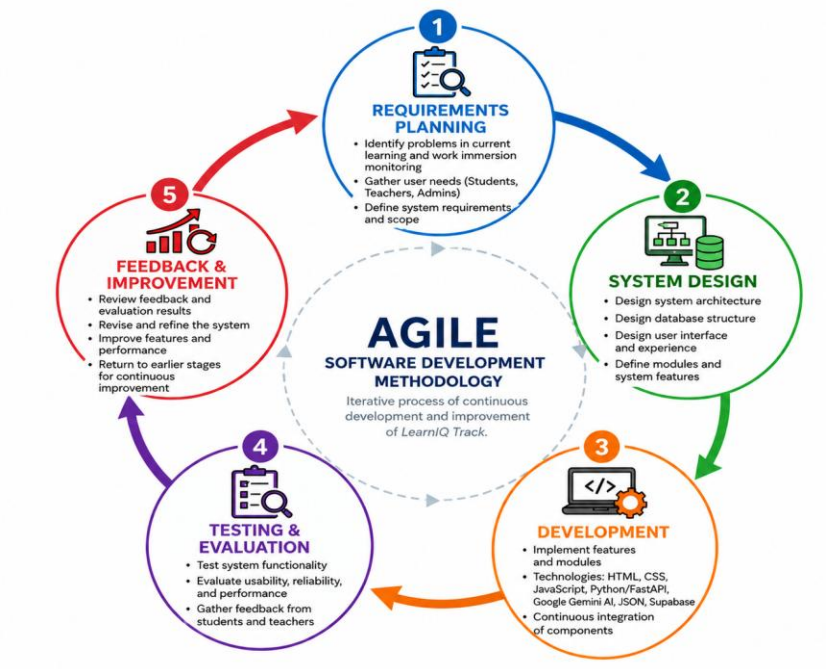
3.9 Software Development Model

The researchers will use the Agile Software Development Model in developing LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. Agile is an iterative approach that allows the system to be developed through a series of connected stages, including requirements planning, system design, development, testing and evaluation, and feedback and improvement. This model is appropriate for

the project because LearnIQ Track consists of several interconnected features that need to be developed, checked, and refined to ensure that they function according to the requirements of the intended users.

The Agile approach allows the researchers to evaluate the progress of the system throughout development rather than waiting until all features have been completed. Issues identified during testing or feedback from users can be used as a basis for revising the affected features. This process helps ensure that the system remains aligned with the needs of students, teachers, and the school administrator.

Figure 3.1 Agile Software Development Model



The first stage is Requirements Planning, where the researchers identify the existing problems and determine the needs of the intended users. This includes examining the current practices involving learning activities, quiz and review preparation, work immersion attendance, daily journal submission, and student monitoring. Information gathered from interviews, observations, and other research activities will be used to determine the necessary functions and features of LearnIQ

Track. The output of this stage is the list of identified system requirements, which will serve as the basis for the succeeding design and development activities.

The second stage is System Design, where the researchers plan the structure and overall organization of LearnIQ Track. This stage includes identifying the system's user roles, modules, database, interfaces, and technical components. The researchers will organize the functions for students, teachers, and administrators and determine how the different modules will interact to support academic learning and work immersion monitoring. The output of this stage includes the system design specifications, diagrams, and planned structure of the system.

The third stage is Development, where the researchers build and integrate the planned features using HTML, CSS, JavaScript, Python with FastAPI, Google Gemini AI, JSON, and Supabase. The development activities include implementing user account management, learning material upload, AI-generated quizzes and reviewers, interactive quizzes, score computation, leaderboard ranking, digital time-in and time-out, daily journal submission, and teacher and admin dashboards. These features will be integrated to provide the intended learning and monitoring functions within one platform. The output of this stage is the working version of LearnIQ Track with its major modules and features.

The fourth stage is Testing and Evaluation, where the researchers check whether the developed system operates according to its identified requirements. The major functions, including user account management, learning material upload, AI-generated content, quizzes, leaderboard, attendance, journal submission, dashboards, and report generation, will be tested. The system will also undergo evaluation involving selected Grade 11 students, Grade 12 students, and teachers to assess its software

quality and user acceptance. The output of this stage includes documented testing results, evaluation responses, and identified issues that may require improvement.

The fifth stage is Feedback and Improvement, where the researchers review the results of testing and evaluation to identify errors, limitations, and areas that need refinement. Necessary revisions will be made to the affected features based on the findings and feedback gathered during the process. Since Agile is iterative, improvements may require the researchers to return to earlier stages, such as planning, design, or development, to make the necessary adjustments. The output of this stage is a refined version of LearnIQ Track that addresses the identified issues and is prepared for its intended use.

Overall, the Agile Software Development Model is suitable for LearnIQ Track because it supports continuous development, testing, and refinement of the system. Its iterative process allows the researchers to address issues as they arise and make adjustments based on the requirements of the intended users. This approach supports the development of an integrated platform that combines AI-assisted learning activities and digital work immersion monitoring.

3.10 System Architecture

This section presents the overall structure and organization of the LearnIQ Track system, illustrating how its hardware, software, system components, database, user interfaces, and communication mechanisms work together. The architecture is designed to support the system's two main functions: AI-assisted academic learning and digital work immersion monitoring. It also establishes how students, teachers, and the admin access the system according to their assigned roles and responsibilities.

LearnIQ Track follows a web-based architecture that connects the frontend interface, backend application, database and authentication services, and external artificial intelligence service. Through this structure, students can access learning materials and activities, participate in interactive quizzes, and submit work immersion records. Teachers can upload learning resources, validate AI-generated content, and monitor student performance and work immersion activities. Meanwhile, the admin can manage user accounts, monitor system records, and view reports.

3.10.1 Hardware and Software Requirements

LearnIQ Track requires appropriate hardware and software resources for its development, operation, maintenance, and deployment. Since the system is web-based, users can access its features through a compatible desktop or mobile device with a modern web browser and an active internet connection. The system does not require users to install a separate native mobile application.

Table 3.3 Hardware and Software Requirements

Requirement Category	**Specification**
Development Device	Computer or laptop capable of running the development environment and supporting web application development
Client Device	Desktop computer, laptop, tablet, or smartphone
Operating System	Operating system compatible with a modern web browser and the required development tools
Frontend Technologies	HTML, CSS, and JavaScript
Backend Framework	Python with FastAPI

Continuation

Artificial Intelligence Service	Google Gemini AI for generating quizzes, activities, and reviewers from uploaded learning materials
Database and Authentication	Supabase for database and authentication services
Data Exchange Format	JSON for communication and data exchange between application components
Hosting and Deployment	Cloudflare for web infrastructure, hosting, and deployment
Development Environment	Visual Studio Code
Client Software	Modern web browser and an active internet connection

Table 3.3 presents the hardware and software requirements of LearnIQ Track. The development device is used by the researchers to create, configure, and maintain the application. Users access the system through desktop or mobile devices using a compatible web browser.

The frontend technologies, HTML, CSS, and JavaScript, are responsible for presenting the system's pages and interactive features. Python with FastAPI handles the backend processes and application requests. Supabase provides database and authentication services, while Google Gemini AI supports the generation of learning content based on materials uploaded by teachers. JSON serves as the data exchange format between the application components. Cloudflare provides the web infrastructure used for deployment and access to the system.

These resources allow LearnIQ Track to operate as an integrated web-based platform without requiring users to install a separate application on their devices.

3.10.2 System Components and Modules

LearnIQ Track is organized into major components and modules that work together to provide its academic learning and work immersion monitoring functions. The system is structured around three main user roles: Student, Teacher, and Admin. Each role is provided with functions appropriate to its responsibilities within the platform.

Presentation Layer

The presentation layer serves as the interface through which users access and interact with LearnIQ Track. It is developed using HTML, CSS, and JavaScript. This layer displays the available features, forms, learning activities, dashboards, and records based on the user's assigned role.

Students use this layer to access learning materials, answer quizzes, view scores and leaderboard rankings, record attendance, and submit daily journals. Teachers use it to upload learning materials, review AI-generated content, and monitor student activities. The admin uses the interface to manage user accounts, monitor records, and access reports.

Application and Business Logic Layer

The application and business logic layer is implemented using Python with FastAPI. It handles the system's main operations and processes requests received from the frontend. This includes managing user-related activities, processing learning materials, coordinating AI-generated content, handling quiz submissions and scores, and processing work immersion records.

This layer also connects the system's different modules to ensure that information entered by users is processed and made available to the appropriate functions.

Database and Authentication Layer

The database and authentication layer uses Supabase to provide data storage and authentication services. It supports the management of user accounts and the storage and retrieval of information needed by the system. These include learning materials, generated learning content, quiz results, leaderboard information, attendance records, and daily journals.

Authentication services also support user access according to the assigned account and role, allowing the system to provide appropriate functions to students, teachers, and the admin.

External Artificial Intelligence Layer

The external artificial intelligence layer uses Google Gemini AI to assist in generating quizzes, activities, and reviewers based on learning materials uploaded by teachers. The generated content is processed by the application and presented for teacher review and validation before being made available to students.

The teacher remains responsible for checking the generated learning content to ensure that it is appropriate and relevant to the uploaded materials.

Major System Modules

The major modules of LearnIQ Track are organized according to the functions available to its intended users.

System Module	Main Function
User Account Management	Handles user accounts and access according to the assigned roles of Student, Teacher, and Admin.
Learning Material Management	Allows teachers to upload learning materials that serve as the basis for AI-generated learning content.

Continuation

AI-Based Learning Content	Uses Google Gemini AI to generate quizzes, activities, and reviewers from uploaded learning materials.
Interactive Quiz and Learning Game	Allows students to answer quizzes and participate in learning activities.
Score and Leaderboard Management	Computes quiz scores and displays student rankings based on their recorded performance.
Work Immersion Attendance	Allows students to record their work immersion time-in and time-out.
Daily Journal Management	Allows students to submit daily journals for work immersion documentation.
Teacher Dashboard	Provides teachers with access to student performance, quiz results, attendance records, and journal submissions.
Admin Dashboard	Allows the admin to manage user accounts, monitor system records, and view reports.

These components and modules are integrated to provide a centralized platform for academic learning and work immersion monitoring. Each module contributes to the overall operation of LearnIQ Track and supports the responsibilities of its intended users.

3.10.3 Database Structure

The database structure of LearnIQ Track is designed to organize and manage the information required by its academic learning and work immersion monitoring features. Supabase is used as the system's database and authentication service, allowing the application to store and retrieve information needed by students, teachers, and the admin.

The information managed by the system includes user account details, uploaded learning materials, AI-generated learning content, quiz activities, student scores, leaderboard records, attendance information, and daily journal submissions. These records support the different processes performed within the platform.

The major information groups and their purposes are presented below.

Table 3.4 Main Information Groups in LearnIQ Track

Information Group	Description	Related System Functions
User Account Information	Contains information needed to identify users and support their assigned roles and access.	User Account Management and Authentication
Learning Material Information	Contains information about learning resources uploaded by teachers.	Learning Material Management
Generated Learning Content	Contains AI-generated quizzes, activities, and reviewers based on uploaded materials.	AI-Based Learning Content
Quiz and Activity Records	Contains information related to student quiz participation and submitted answers.	Interactive Quiz and Learning Game
Student Performance Records	Contains recorded quiz scores and information used for leaderboard rankings.	Score and Leaderboard Management
Attendance Records	Contains student work immersion time-in and time-out information.	Work Immersion Attendance
Daily Journal Records	Contains daily journal submissions made by students.	Daily Journal Management

Continuation

Monitoring and Report Information	Contains records used by teachers and the admin to monitor student activities and system information.	Teacher Dashboard and Admin Dashboard

The information groups are connected according to the processes performed by the system. User account information identifies the users who access the platform and determines the functions available to them. Teachers' uploaded learning materials serve as the source for AI-generated learning content. Students' quiz activities produce performance records that can be used for score computation and leaderboard ranking. Meanwhile, attendance and daily journal submissions provide information for work immersion monitoring.

Teachers can access the appropriate student records through the teacher dashboard, while the admin can monitor overall system information through the admin dashboard. Access to these records is provided according to the users' assigned roles.

3.10.4 User Interface Design

The user interface of LearnIQ Track is designed to provide an organized and accessible environment for students, teachers, and the admin. The interface presents the system's features according to the user's assigned role, allowing each user to access the functions needed to perform their respective activities.

Student Interface

The student interface provides access to academic learning activities and work immersion monitoring features. Students can view learning materials uploaded by teachers, access AI-generated quizzes, activities, and reviewers, participate in interactive quizzes, and view their scores and leaderboard

rankings. The interface also provides functions for recording work immersion time-in and time-out and submitting daily journals.

Teacher Interface

The teacher interface provides tools for managing learning resources and monitoring student activities. Teachers can upload learning materials that will serve as the basis for AI-generated content. They can also review and validate generated quizzes, activities, and reviewers before students access them. The dashboard allows teachers to monitor quiz performance, student scores, attendance records, and daily journal submissions.

Admin Interface

The admin interface is intended for the school principal, who oversees the overall use of LearnIQ Track. It provides functions for managing user accounts, monitoring system records, viewing reports, and supervising the activities recorded within the platform.

The three interfaces are organized to ensure that users can access the appropriate features based on their responsibilities. This role-based arrangement helps maintain an organized system environment and supports the management of academic and work immersion information.

3.10.5 System Communication and Interaction

The system communication and interaction of LearnIQ Track describe how users, application components, database services, and the external artificial intelligence service exchange information to perform the system's functions.

Users access LearnIQ Track through a web browser using their respective accounts. When a user performs an action, such as uploading a

learning material, answering a quiz, recording attendance, or submitting a daily journal, the frontend sends the necessary information to the backend application. The backend, implemented using Python with FastAPI, processes the request and communicates with the appropriate service.

For AI-assisted learning content generation, teachers first upload learning materials through the system. The backend processes the request and communicates with Google Gemini AI to generate quizzes, activities, and reviewers based on the provided materials. The generated content is then returned to the application for teacher review and validation. Once validated, the content can be made available to students.

For academic learning activities, students access the available quizzes and activities through the frontend. Their submitted answers and corresponding scores are processed by the backend and recorded through the database service. The stored performance information can then be used to display student scores and leaderboard rankings.

For work immersion monitoring, students submit their time-in and time-out records and daily journals through the system interface. These submissions are processed and stored through the application's database services. Teachers can retrieve the appropriate records through the teacher dashboard to monitor student attendance and journal submissions. The admin can also access authorized system records and reports for overall supervision.

JSON is used as the data exchange format between the frontend and backend components. Supabase provides the database and authentication services needed to support user access and information management. Through these interactions,

the system connects its learning activities, AI-assisted content generation, student performance records, and work immersion monitoring functions.

Overall, the communication structure allows LearnIQ Track to operate as an integrated web-based platform. The interaction among users, the frontend, backend, Google Gemini AI, and Supabase supports the delivery of learning activities and the recording, storage, retrieval, and monitoring of student information.

3.11 System Testing and Evaluation Procedure

The System Testing and Evaluation Procedure will be conducted to determine whether LearnIQ Track functions according to its documented requirements and intended objectives. The procedure will focus on verifying the major features of the system, identifying possible errors, and assessing whether the system operates properly for its intended users. Testing will cover the academic learning features, AI-assisted content generation, student performance monitoring, and digital work immersion monitoring.

The testing activities will include functional testing, non-functional testing, and black box testing. Functional testing will determine whether each system feature performs its intended task. Non-functional testing will examine important software quality characteristics, including usability, reliability, performance efficiency, and security. Black box testing will verify the system's outputs based on the inputs provided without examining its internal source code.

The system will be tested by ten (10) IT experts who will examine its major functions and features. Their testing will help identify errors, usability concerns, and other technical issues that may require correction. The system will also undergo user

evaluation involving selected Grade 11 students, Grade 12 students, and teachers of Paete Science and Business College Inc. to assess their experience and acceptance of the developed platform. The testing and evaluation activities will be aligned with the system requirements and the research objectives.

3.11.1 Testing Plan

The testing plan establishes the sequence of activities that will be followed in examining the functions and performance of LearnIQ Track. The plan will begin with reviewing the system requirements and preparing the test environment, followed by the execution of test cases, documentation of results, correction of identified issues, and retesting when necessary.

The testing coverage will focus on the major modules of LearnIQ Track, including user account management, learning material management, AI-generated learning content, interactive quizzes, score computation, leaderboard ranking, work immersion attendance, daily journal submission, teacher dashboard, and admin dashboard.

The testing activities will be organized into the following areas:

Functional Testing. This will determine whether the system features perform according to their intended functions. The testing will cover user registration and login, role-based access, learning material upload, AI-generated quizzes and reviewers, interactive quizzes, score computation, leaderboard ranking, attendance recording, daily journal submission, and dashboard functions. Each feature will be tested using appropriate inputs or user actions, and the actual output will be compared with the expected result.

Non-Functional Testing. This will examine important quality characteristics of the system aside from its individual functions. The assessment

will focus on usability, reliability, performance, security, and stability. The researchers will observe whether the system responds properly during use, operates consistently while different features are accessed, protects user information, and provides an appropriate experience for its intended users.

Black Box Testing. This will be applied to verify the system's functionality by providing valid and invalid inputs and observing the resulting outputs. The internal source code will not be examined during this process. For example, the login feature will be tested using correct and incorrect credentials, while the attendance feature will be tested by submitting time-in and time-out information and checking whether the records are properly saved.

The ten (10) IT experts will be involved in examining the system's technical functions and overall operation. After the necessary system testing activities, selected students and teachers will use the platform according to their assigned roles as part of the user evaluation. Feedback and identified issues will be considered in determining the necessary system corrections and improvements.

3.11.2 Test Cases and Test Scenarios

The test cases and scenarios will be developed based on the functional requirements and major features of LearnIQ Track. Each test case will identify a specific system function, the action or input required, and the expected result. The actual result and testing status will be recorded during the testing process to determine whether the system performs according to its intended requirements.

The planned functional and black box test cases are presented in Table 3.5.

Table 3.5 Planned Functional and Black Box Test Cases

Test Case	Test Action/Input	Expected Result
User Registration	Enter valid user information and submit the registration form.	A user account is successfully created.
User Login	Enter valid account credentials.	The user is successfully logged in and directed to the appropriate page based on the assigned role.
Role-Based Access	Access system functions using Student, Teacher, and Admin accounts.	Each user can access only the functions assigned to their role.
Learning Material Upload	The teacher uploads a supported learning material.	The learning material is successfully uploaded and made available to the intended users.
AI-Generated Quiz	Generate a quiz based on an uploaded learning material.	The system generates and displays quiz content based on the provided material.
AI-Generated Activities and Reviewers	Request the generation of activities and reviewers using uploaded learning resources.	The system generates and displays the requested learning content.
Teacher Content Validation	The teacher reviews and validates AI-generated learning content.	The teacher can review the generated content before it is made available to students.
Interactive Quiz	The student answers and submits an available quiz.	The student's answers are successfully submitted and processed.
Score Computation	Complete and submit a scored quiz or activity.	The system computes and displays the corresponding score.
Leaderboard Ranking	Access the leaderboard after student quiz results are recorded.	The leaderboard displays student rankings based on recorded performance.

Continuation

Work Immersion Attendance	The student records time-in and time-out information.	The attendance information is successfully recorded and stored.
Daily Journal Submission	The student completes and submits a daily work immersion journal.	The journal is successfully submitted and can be accessed by authorized teachers.
Teacher Dashboard	The teacher accesses student performance, attendance, and journal records.	The dashboard displays the appropriate student monitoring information.
Admin Dashboard	The admin accesses user management, system records, and reports.	The admin can access the appropriate administrative functions and information.

The test cases presented in Table 3.5 cover the major features of LearnIQ Track and are intended to verify whether each function produces the expected result. These test cases will serve as the basis for documenting the actual testing outcomes, identifying errors, and determining whether corrections are necessary.

3.11.3 Testing Data

The testing data will consist of sample and dummy information prepared to represent realistic conditions during the use of LearnIQ Track. These data will be used to examine the system's functions without relying on actual student personal information or unnecessary production records.

The test data will include sample user account information for Student, Teacher, and Admin roles; sample learning materials for AI-generated quizzes, activities, and reviewers; sample quiz questions and student answers; and sample work immersion attendance and daily journal entries. These data will

allow the researchers and system testers to examine the system's ability to process, store, and retrieve the information required by its different modules. For user account testing, valid and invalid credentials will be used to verify login and role-based access. For learning content generation, sample learning materials will be uploaded and used as the basis for generating quizzes, activities, and reviewers. For quiz testing, sample answers will be submitted to verify answer processing and score computation. For work immersion monitoring, sample time-in and time-out entries and daily journal submissions will be used to determine whether the records are properly saved and made available to authorized users.

The use of sample and dummy data will help demonstrate the system's expected behavior while minimizing the exposure of actual personal information. The test data will be selected according to the requirements of each test case and will be used consistently when comparing the expected and actual system results.

3.11.4 Testing Procedure

The testing procedure will be conducted systematically to ensure that the major functions of LearnIQ Track are examined and that the results are properly documented. The following steps will be followed:

1. **Review of System Requirements.** The researchers will review the documented functional and non-functional requirements of LearnIQ Track to identify the features and quality characteristics that need to be tested.
2. **Preparation of the Testing Environment.** The researchers will prepare the deployed system, verify access to the required services, and ensure that the necessary test accounts and sample data are available.

3. **Preparation of Test Cases.** The researchers will organize the test cases based on the system's major modules, including user account management, learning materials, AI-generated content, quizzes, leaderboard, attendance, journals, and dashboards.

4. **Execution of Test Cases.** The system testers will perform the specified actions using the prepared test data. Each feature will be accessed and tested according to its corresponding test scenario.

5. **Comparison of Expected and Actual Results.** The actual output produced by the system will be compared with the expected result stated in each test case. This will determine whether the tested function operates as intended.

6. **Documentation of Testing Results.** The researchers will record the actual results and testing status of each test case. Supporting evidence, such as screenshots and observed errors, may also be documented to provide a record of the testing activities.

7. **Correction and Retesting.** Any identified errors or unexpected behavior will be reviewed and addressed by the researchers. The affected test cases will be performed again when necessary to verify whether the corrections have resolved the identified issues.

8. **Review of Testing Results.** The documented testing results will be reviewed to determine whether the system's major functions satisfy their intended requirements. The findings will serve as the basis for the discussion of system testing in Chapter IV.

Through this procedure, LearnIQ Track will be examined in terms of its functionality and selected software quality characteristics. The documented

results will provide evidence of the system's performance during testing and will help determine whether further corrections or improvements are necessary before its final implementation.

3.12 System Testing

System testing will be conducted to determine whether LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. meets its specified functional and non-functional requirements. It will focus on verifying the proper operation of the system's major features, identifying errors, and determining whether the system produces the expected results when used under appropriate conditions.

The study will employ Functional Testing and Black Box Testing to assess the system's major modules. Functional testing will determine whether each feature performs its intended purpose, while Black Box Testing will examine the system's behavior by providing valid and invalid inputs and comparing the actual outputs with the expected results. Testing will cover user authentication, learning material management, AI-generated learning content, interactive quizzes, score computation, leaderboard ranking, work immersion attendance, daily journals, teacher monitoring, and administrative functions.

Non-functional testing will also be conducted to observe the system's usability, reliability, performance, security, and stability. These testing activities will help determine whether LearnIQ Track operates properly and handles the information and activities of its intended users.

3.12.1 Test Cases

The researchers will use test cases to verify whether the major features of LearnIQ Track function according to their intended requirements. Each test case will identify the feature being tested, the test condition or action, and the expected result. The actual result and test status will be recorded during the execution of each test.

The test cases will cover the system's major functions, including user registration and login, role-based access, learning material upload, AI-generated quizzes and activities, teacher validation, interactive quizzes, score computation, leaderboard ranking, attendance recording, daily journal submission, teacher dashboard, and admin dashboard.

Table 3.6 System Test Cases

Test Case	Test Condition / Action	Expected Result
User Registration	Enter valid user information and submit the registration form.	The system successfully creates a user account.
User Login	Enter valid and invalid account credentials.	The system allows authorized users to log in and rejects incorrect credentials.
Role-Based Access	Log in using Student, Teacher, and Admin accounts and access their respective pages.	Each user can access only the functions and information permitted for their assigned role.
Learning Material Upload	Upload a valid PowerPoint or document file using a teacher account.	The system accepts the file and makes the uploaded material available for its intended use.

Continuation

AI-Generated Learning Content	Submit an uploaded learning material for AI-generated quizzes, activities, or reviewers.	The system generates learning content based on the submitted material or displays an appropriate error if generation cannot be completed.
Teacher Content Validation	Review the AI-generated learning content using a teacher account.	The teacher can review the generated content before it is made available to students.
Interactive Quiz	Open a quiz, answer the questions, and submit the responses.	The system records the submitted answers and displays the corresponding quiz result.
Score Computation	Complete a quiz with known correct and incorrect answers.	The system computes and displays the score according to the quiz's correct answers.
Leaderboard	Complete quizzes and access the leaderboard.	The system displays student rankings based on the recorded scores or points.
Work Immersion Attendance	Record time-in and time-out using a student account.	The system records and saves the attendance information correctly.
Daily Journal Submission	Enter and submit a daily work immersion journal.	The system saves the journal and makes it accessible to authorized users.
Teacher Dashboard	Access the teacher dashboard and view student performance, attendance, and journal records.	The system displays the appropriate student monitoring information.
Student Monitoring and Work Immersion Records	View the available student progress and work immersion records using an authorized teacher account.	The system displays the corresponding records accurately.

Continuation

Admin Dashboard and User Management	Access the admin dashboard and perform authorized user management and report-related functions.	The system allows the admin to manage user accounts and access the appropriate system reports.

Table 3.6 presents the test cases that will be used to assess the major functions of LearnIQ Track. During actual testing, the researchers will document the observed output for each test case and compare it with the expected result. The actual result and corresponding status will be recorded in the System Testing Tool.

3.12.2 Testing Instrument

System testing will use the System Testing Tool for System Testers, a structured instrument prepared by the researchers to document and evaluate the performance of LearnIQ Track. The instrument is designed to guide the testers in executing the test cases, recording actual system behavior, identifying defects, and documenting the results of testing.

The instrument contains sections for testing information, functional and Black Box test cases, AI-generated content testing, security and access-control testing, reliability and data-integrity testing, performance and compatibility observations, defect and error logging, retesting, and the system testing summary. It also includes a final tester checklist and a certification section for documenting the completion of testing.

The testing information section records the tester's name, testing role, date, system version, device, operating system, browser, network, and testing

environment. Each test case provides fields for the actual result, test status, and remarks so that the observed behavior can be compared with the expected result.

The instrument also includes a defect and error log for documenting identified problems and their severity. When a defect is corrected, the corresponding test case may be executed again, and the retest result will be recorded. This process will help the researchers maintain a record of the issues identified and the actions taken during system testing.

3.12.3 Validation of Testing Instrument

The System Testing Tool for System Testers was prepared by the researchers based on the functional and non-functional requirements of LearnIQ Track. The instrument was organized to ensure that its test cases and evaluation areas correspond to the system's major features, including AI-generated learning content, teacher validation, student learning activities, work immersion monitoring, and administrative functions.

Before its formal use, the instrument will be submitted to the research adviser for review to determine whether its contents, instructions, and testing criteria are appropriate and aligned with the objectives of the study. Necessary revisions and recommendations will be incorporated to improve the clarity and completeness of the instrument.

The researchers will also ensure that the instrument provides sufficient space for recording actual results, test status, identified defects, corrective actions, and retesting outcomes. This will help ensure that the testing records are based on observable system behavior rather than assumptions.

3.12.4 Interpretation of Test Results

The results of system testing will be interpreted by comparing the actual output of each test case with its expected result. Each executed test case will be assigned an appropriate status based on the observed system behavior. The researchers will use the following classifications:

Table 3.7 Classification of System Testing Results

Classification	Description
Pass	The actual result meets the expected result, and no material defect is observed.
Fail	The actual result does not meet the expected result, or the function cannot be completed as specified.
Blocked	The test cannot be performed because of an environment, dependency, access, or prerequisite problem.

Aside from the test status, identified defects will be classified according to their severity to determine their possible effect on the system.

Table 3.8 Classification of Defect Severity

Severity	Description
Critical	A serious system failure, security or access-control issue, data loss or corruption, or defect that prevents a major system function from operating.
Major	An important system function fails or produces an incorrect result, although other parts of the system remain usable.

Continuation

Minor	A limited defect that affects a specific function but has an available workaround and does not prevent the main system operations.
Cosmetic	A presentation-related issue, such as text, spacing, alignment, or other visual inconsistencies that do not affect the main system functions.

Failed test cases will be documented in the defect and error log, together with the observed problem and its assigned severity. After the necessary correction, the affected test cases will be executed again to determine whether the identified issues have been resolved. Blocked test cases will also be documented, including the reason they could not be completed.

The overall testing results will be summarized according to the testing areas identified in the instrument: Functional/Black-Box Testing, AI Content, Security/Access, Reliability/Data Integrity, and Performance/Compatibility. The pass rate may be computed using the following formula:

Pass Rate (%)

$$\text{Pass Rate (\%)} = \frac{Number\ of\ Passed\ Test\ Cases}{Total\ Number\ of\ Executed\ Test\ Cases} \text{ x } 100$$

The pass rate will be computed only after the actual test cases have been executed and their results documented. The final testing summary will present the number of executed, passed, failed, blocked, and successfully retested cases. These results will serve as the basis for determining whether the system meets its documented requirements and identifying areas that require further correction or improvement.

3.13 System User Evaluation

System User Evaluation will be conducted to determine how the intended users perceive and assess the developed LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. This evaluation will focus on the system's software quality, usability, usefulness, user acceptance, and overall acceptability based on the actual experiences of the respondents while using the system.

The evaluation will involve selected Grade 11 students, Grade 12 students, and teaching faculty of Paete Science and Business College Inc. These respondents will evaluate the system according to their respective roles and the features they are expected to use. The school principal will serve as the system administrator and key informant but will not be included in the statistical survey sample.

A structured survey questionnaire using a five-point Likert scale will serve as the primary instrument for gathering the respondents' assessments. The evaluation will be based on five selected ISO/IEC 25010 software quality characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. The Technology Acceptance Model (TAM) will also be used through the constructs of Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. Overall Acceptability will also be assessed to determine the respondents' general assessment of LearnIQ Track.

3.13.1 User Evaluation Instrument

The researchers will use a structured survey questionnaire to gather the assessments of the selected Grade 11 students, Grade 12 students, and teachers regarding LearnIQ Track. The questionnaire will be administered after the

respondents have interacted with the developed system to ensure that their evaluations are based on their actual experiences.

The questionnaire will consist of four main parts. The first part will gather the respondents' basic profile and respondent group. The second part will evaluate the system based on the selected ISO/IEC 25010 software quality characteristics, namely Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. The third part will assess user acceptance based on the Technology Acceptance Model (TAM), specifically Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. The fourth part will determine the overall acceptability of LearnIQ Track and provide space for comments and suggestions.

The questionnaire will use a five-point Likert scale ranging from 5 (Strongly Agree) to 1 (Strongly Disagree). The statements will be formulated according to the system's functions and evaluation criteria. Student respondents will assess the features they directly use, such as learning materials, AI-generated quizzes, activities, reviewers, interactive quizzes, scores, leaderboard ranking, digital attendance, and daily journal submission. Teacher respondents will assess the functions related to learning material management, validation of AI-generated content, student progress monitoring, attendance, journals, and work immersion records.

3.13.2 Instrument Source and Validation

The user evaluation questionnaire will be researcher-made and developed based on the research objectives, selected ISO/IEC 25010 software quality characteristics, and Technology Acceptance Model (TAM) constructs.

The questionnaire items will be prepared to correspond to the intended functions of LearnIQ Track and the evaluation requirements of the study.

The questionnaire will undergo validation by the researchers' thesis adviser before it is administered to the selected respondents. The validation process will focus on the clarity, relevance, and appropriateness of the questionnaire statements, as well as their alignment with the research objectives and evaluation criteria. The adviser will also review whether the items appropriately represent the selected software quality characteristics and TAM constructs.

Based on the comments and recommendations of the thesis adviser, the researchers will revise the questionnaire as necessary before its final administration. This process will help ensure that the instrument is understandable to the intended respondents and appropriate for evaluating the developed system.

3.13.3 User Evaluation Procedure

The user evaluation will be conducted after the development and system testing of LearnIQ Track. Before the evaluation, the researchers will secure the necessary permission from the administration of Paete Science and Business College Inc. and coordinate with the concerned teachers regarding the schedule and participation of the selected respondents.

The researchers will first introduce LearnIQ Track to the respondents and explain its purpose, major features, and intended functions. The respondents will then be given an opportunity to interact with the system according to their respective roles. Students will explore the learning materials, AI-generated quizzes, interactive learning activities, scores, leaderboard, and applicable work

immersion features. Teachers will explore the learning material management, AI-generated content validation, student monitoring, attendance, and journal management functions.

After using the system, the respondents will be provided with the validated survey questionnaire. They will be instructed to answer the statements based on their actual experiences and perceptions while using LearnIQ Track. The researchers will collect the completed questionnaires and review them for completeness before organizing the responses for statistical analysis.

Participation will be voluntary, and the respondents will be informed about the purpose of the study and the confidentiality of the information they provide. For student respondents who are minors, the researchers will obtain the required parent or legal guardian consent and student assent in accordance with applicable school policies and ethical requirements. The collected responses will be coded and handled confidentially, and the information will be used only for the purposes of the study.

3.13.4 Evaluation Criteria

LearnIQ Track will be evaluated based on five selected ISO/IEC 25010 software quality characteristics, four Technology Acceptance Model (TAM) constructs, and Overall Acceptability. These criteria will provide a basis for determining how the intended users assess the quality and acceptance of the developed system.

The selected ISO/IEC 25010 characteristics are the following:

Functional Suitability determines whether LearnIQ Track provides the functions needed by its intended users and whether these functions produce the

expected results. This includes user authentication, learning material management, AI-generated learning content, interactive quizzes, score computation, leaderboard ranking, digital attendance, journal submission, and dashboard functions.

Usability determines whether the system is understandable, easy to learn, and convenient to use. This includes the organization of the interface, navigation, accessibility of features, and clarity of system instructions.

Performance Efficiency assesses the system's responsiveness and performance when users access its features. This includes the loading of pages, access to learning materials, quiz execution, and processing of system requests.

Reliability determines whether the system operates consistently and whether information remains available and properly recorded during normal use. This includes the saving and retrieval of quiz results, attendance records, daily journals, and other system information.

Security assesses whether LearnIQ Track protects user accounts and information through authentication, role-based access, and appropriate restrictions on student records and other system data.

The Technology Acceptance Model (TAM) will also be used to assess the respondents' acceptance of the system through the following constructs:

Perceived Ease of Use determines whether users believe that LearnIQ Track can be used with minimal difficulty and whether its features are easy to understand and navigate.

Perceived Usefulness determines whether users believe that the system provides benefits to their learning activities, teaching responsibilities, and work immersion monitoring tasks.

Attitude Toward Using assesses the users' general positive or negative feelings toward using LearnIQ Track as a platform for learning activities and work immersion monitoring.

Behavioral Intention to Use determines whether users intend or are willing to use the system in performing their related academic, teaching, or administrative activities.

Lastly, **Overall Acceptability** determines the general assessment of the respondents regarding the suitability and acceptability of LearnIQ Track based on their experience with the system.

The respondents' answers will be interpreted using the five-point Likert scale presented in Table 3.9

Table 3.9 Five-Point Likert Scale for Interpreting Weighted Means and Acceptability

Scale	Range	Verbal Interpretation
5	**4.21–5.00**	**Strongly Agree / Highly Acceptable**
4	**3.41–4.20**	**Agree / Acceptable**
3	**2.61–3.40**	**Neutral / Moderately Acceptable**
2	**1.81–2.60**	**Disagree / Less Acceptable**
1	**1.00–1.80**	**Strongly Disagree / Not Acceptable**

A higher weighted mean will indicate a more favorable assessment of LearnIQ Track, while a lower weighted mean will indicate a less favorable assessment. The results for each evaluation criterion will be summarized and

interpreted to determine the perceived software quality, user acceptance, and overall acceptability of the developed system.

3.16 Statistical Treatment

The data gathered from the survey questionnaires will be analyzed using descriptive statistical methods to determine the respondents' assessment of the developed LearnIQ Track system. The statistical tools will include frequency and percentage distribution, weighted mean, and standard deviation. These methods will be used to summarize the respondents' profiles and evaluate the system based on the selected ISO/IEC 25010 quality characteristics, Technology Acceptance Model (TAM) constructs, and overall acceptability.

3.16.1 Descriptive Statistics

Frequency and Percentage Distribution

Frequency and percentage distribution will be used to present the number and proportion of respondents according to their respective groups, specifically Grade 11 students, Grade 12 students, and teachers. Frequency will indicate the number of respondents in each group, while percentage will show their corresponding proportion in relation to the total number of respondents.

The percentage will be computed using the following formula:

$$P = \frac{f}{N} \text{ x } 100$$

Where:

- (P) = Percentage
- (f) = Frequency of respondents
- (N) = Total number of respondents

Weighted Mean

The weighted mean will be used to determine the average responses of the respondents to the statements in the user evaluation questionnaire. Since the questionnaire uses a five-point Likert scale, this statistical method will provide a numerical basis for interpreting the respondents' level of agreement and acceptance of LearnIQ Track.

The weighted mean will be used to evaluate the system in terms of Functional Suitability, Usability, Performance Efficiency, Reliability, Security, Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, Behavioral Intention to Use, and Overall Acceptability. It will also summarize the respondents' assessments of the system's features, including AI-generated quizzes and reviewers, interactive learning activities, leaderboard, digital work immersion attendance, daily journal submission, and the teacher and admin dashboards.

The weighted mean will be computed using the following formula:

$$\text{WM} = \frac{\Sigma fw}{N}$$

Where:

- (WM) = Weighted Mean
- (f) = Frequency of responses
- (w) = Weight assigned to each Likert-scale response
- (N) = Total number of valid responses

Standard Deviation

The standard deviation will be used to determine the variability or consistency of the respondents' answers. It will show how closely the individual

responses are distributed around the computed mean. A lower standard deviation indicates that the responses are relatively close to one another, suggesting greater consistency among the respondents. In contrast, a higher standard deviation indicates greater variation in their evaluations.

The standard deviation will supplement the weighted mean by showing the extent to which the respondents' assessments of LearnIQ Track differ from one another. It will be computed using the following formula:

$$s = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$$

Where:

- s = Sample standard deviation
- x_i = Individual response
- $\bar{x}$ = Mean of the responses
- n = Number of valid responses

The computed weighted means will be interpreted using the five-point Likert scale and corresponding interpretation ranges presented in the User Evaluation section of this chapter.

3.16.2 Qualitative Data Analysis

Qualitative information gathered through interviews, observations, and respondents' comments or suggestions will be examined descriptively. These responses will be used to identify relevant information about the existing learning and work immersion monitoring processes, system requirements, and possible improvements to LearnIQ Track.

The qualitative inputs will serve as supporting information in explaining the development and evaluation of the system. They will not be converted into numerical ratings or included in the computation of the weighted mean and standard deviation. Instead, relevant comments and observations will be summarized and considered in identifying areas for improvement.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the results and discussion of the development, functional testing, and evaluation of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The presentation follows the three research objectives: (1) the design and development of the system, (2) the testing of its functionalities, and (3) the evaluation of its selected software quality characteristics and user acceptability.

The chapter presents the developed system and its major components, followed by the results of functional and black-box testing, AI-generated content testing, security and role-based access testing, and reliability, data integrity, performance, and compatibility testing. It also presents the evaluation results gathered from 283 respondents using a five-point Likert scale. The evaluation covers five selected ISO/IEC 25010 software quality characteristics and four Technology Acceptance Model (TAM) constructs, together with the overall acceptability of the system.

4.1 General Presentation of Results

This chapter presents the results and discussion of the development, testing, and evaluation of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The findings are organized according to the objectives of the study, covering the development of the system, the results of system testing, and the evaluation of its selected software quality characteristics and overall acceptability.

The first part presents the developed system, including its major components, user roles, and functions. It describes how the system integrates learning material management, AI-assisted content generation, interactive quizzes, score computation, leaderboard, digital work immersion attendance, daily journal submission, and monitoring dashboards.

The second part discusses the results of functional and black-box testing, including the testing of AI-generated content, security and role-based access, reliability, data integrity, performance, and compatibility. The actual results are presented to determine whether the system's features operated according to their intended functions and expected outputs.

The final part presents the evaluation results gathered from 283 respondents using a five-point Likert scale. The evaluation covers five selected ISO/IEC 25010 software quality characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. It also examines the four Technology Acceptance Model (TAM) constructs: Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use, together with the overall acceptability of the system.

The results are presented through tables and corresponding discussions to provide a clear interpretation of the system's development, testing outcomes, and user evaluations.

4.2 System Architecture and Development Results

This section presents the actual results of the design and development of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc.

The developed system was implemented as a web-based platform that integrates AI-assisted academic learning activities and digital work immersion monitoring.

The development of LearnIQ Track covered user account management, learning material management, AI-generated quizzes, activities and reviewers, interactive quizzes, score computation, leaderboard display, work immersion attendance, daily journal submission, dashboards, and reports. These functions are organized according to the three user roles: Student, Teacher, and Admin.

The succeeding sections present the completed system, including its system overview, architecture, hardware and software requirements, database structure, user-interface design, and system modules.

4.2.1 System Overview

The developed LearnIQ Track is a web-based learning and work immersion monitoring system designed for senior high school students of Paete Science and Business College Inc. It provides a centralized platform for accessing learning materials, participating in AI-assisted learning activities, completing interactive quizzes, and recording work immersion requirements.

The system provides three types of user accounts: Student, Teacher, and Admin. Each account has specific functions based on the user's responsibilities. Students can access learning materials, answer quizzes, view their scores and leaderboard rankings, record work immersion attendance, and submit daily journals. Teachers can upload learning materials, review and validate AI-generated content, and monitor student quiz performance, attendance, and journal submissions. The Admin can manage user accounts, monitor system records, and access reports.

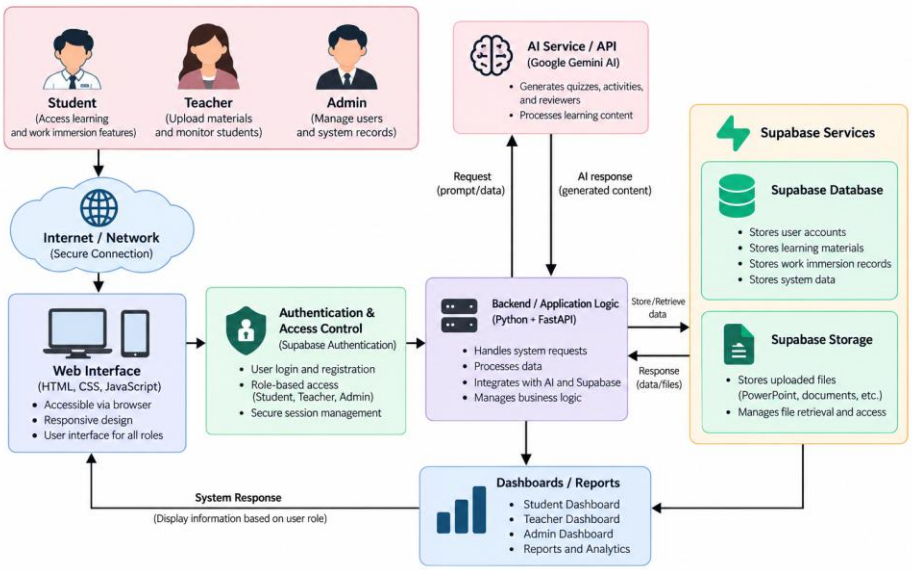
One of the main features of LearnIQ Track is its AI-assisted learning content generation. Teachers upload learning materials that serve as the basis for generating quizzes, activities, and reviewers through Google Gemini AI. The generated content can be reviewed and validated by teachers before it is made available to students.

The system also supports digital work immersion monitoring through time-in and time-out recording and daily journal submission. These records can be accessed by authorized teachers for monitoring purposes. Through these features, LearnIQ Track combines academic learning activities and work immersion documentation in one web-based platform.

4.2.2 System Architecture

LearnIQ Track follows a web-based architecture that connects the user interface, backend application, database and authentication services, and external artificial intelligence service. Students, teachers, and administrators access the platform through their respective accounts and interact with the functions assigned to their roles.

Figure 4.1. System Architecture of LearnIQ Track.



The system architecture consists of the following major layers:

Presentation Layer. The presentation layer serves as the interface through which users access and interact with LearnIQ Track. It is developed using HTML, CSS, and JavaScript. This layer displays the system pages, forms, learning activities, dashboards, and records according to the user's assigned role. Students use this layer to access learning materials, answer quizzes, view scores and leaderboard rankings, record attendance, and submit daily journals. Teachers use it to upload learning materials, review AI-generated content, and monitor student activities. The Admin uses the interface to manage user accounts, monitor records, and access reports.

Application and Business Logic Layer. The application and business logic layer is implemented using Python with FastAPI. It handles the system's main operations and processes requests received from the frontend. These include managing user-related activities, processing learning materials, coordinating AI-generated content, handling quiz submissions and scores, and processing work immersion records. This layer connects the different system modules and ensures that information entered by users is processed for the appropriate functions.

Database and Authentication Layer. The database and authentication layer uses Supabase to provide data storage and authentication services. It supports the management of user accounts and the storage and retrieval of information needed by the system, including learning materials, generated learning content, quiz results, leaderboard information, attendance records, and

daily journals. Its authentication services also support user access according to assigned accounts and roles.

External Artificial Intelligence Layer. The external artificial intelligence layer uses Google Gemini AI to support the generation of quizzes, activities, and reviewers based on learning materials uploaded by teachers. The backend coordinates the content-generation process and makes the generated content available for teacher review and validation before it is provided to students.

Communication and Deployment. JSON is used as the data exchange format between the frontend and backend components. Cloudflare serves as the web infrastructure and hosting and deployment platform used to make the system accessible through a web browser.

The interaction among these components allows LearnIQ Track to operate as an integrated web-based platform. The frontend provides the user interface, the backend processes application requests, Supabase provides database and authentication services, and Google Gemini AI supports the generation of learning content. These components work together to support the system's academic learning and work immersion monitoring functions.

4.2.3 Hardware and Software Requirements

The developed LearnIQ Track system requires appropriate hardware and software resources to support its development, operation, maintenance, and deployment. Since the system is web-based, users can access its features through a compatible desktop or mobile device using a modern web browser and an

active internet connection. Users are not required to install a separate native mobile application.

The software environment uses HTML, CSS, and JavaScript for frontend development and Python with FastAPI for backend operations. Supabase provides database and authentication services, while Google Gemini AI supports AI-assisted learning content generation. JSON serves as the data exchange format between application components, and Cloudflare provides the web infrastructure for deployment.

Table 4.1. Hardware and Software Requirements of LearnIQ Track

Requirement Category	Actual Requirement
Development Device	Computer or laptop capable of running the development environment and supporting web application development
Client Device	Desktop computer, laptop, tablet, or smartphone
Operating System	Operating system compatible with a modern web browser and the required development tools
Frontend Technologies	HTML, CSS, and JavaScript
Backend Framework	Python with FastAPI
Artificial Intelligence Service	Google Gemini AI
Database and Authentication	Supabase
Data Exchange Format	JSON

Continuation

Hosting and Deployment	Cloudflare
Development Environment	Visual Studio Code
Client Software	Modern web browser and an active internet connection

Table 4.1 presents the hardware and software requirements of LearnIQ Track. The development device is used by the researchers to create, configure, and maintain the application, while users access the system through desktop or mobile devices using a compatible web browser.

The frontend technologies are responsible for displaying the system's pages and interactive features, while Python with FastAPI handles backend processes and application requests. Supabase provides the database and authentication services required for user access and information management. Google Gemini AI supports the generation of learning content based on uploaded materials, while JSON enables data exchange between the frontend and backend components. Cloudflare provides the web infrastructure used for hosting and deployment.

These resources allow LearnIQ Track to operate as an integrated web-based platform without requiring users to install a separate application on their devices.

4.2.4 Database Structure

LearnIQ Track uses Supabase as its database and authentication service. The database supports the storage, retrieval, and management of information

required by the system's academic learning and digital work immersion monitoring functions.

The stored information supports user account management, learning materials, AI-generated learning content, quiz results, leaderboard information, work immersion attendance, and daily journal submissions. These data components allow the system to provide the necessary information to students, teachers, and administrators according to their assigned roles.

The database structure illustrates how the system organizes and manages the information required by its different modules. User account information supports authentication and role-based access, while learning-related records support the management of uploaded materials, generated content, quiz results, and student performance. Work immersion records support the storage and retrieval of attendance information and daily journal submissions.

The database structure enables the system to retrieve relevant information for student activities, teacher monitoring, and administrative functions.

4.2.5 User-Interface Design and System Modules

LearnIQ Track provides user interfaces and system modules designed according to the responsibilities of students, teachers, and administrators. The interfaces allow users to access the functions available to their assigned roles. The major modules of the system are described below:

Student Module. The Student Module allows students to access learning materials, answer AI-generated quizzes and activities, participate in the

learning game, view scores and leaderboard rankings, record work immersion attendance, and submit daily journals.

Teacher Module. The Teacher Module allows teachers to upload learning materials, generate AI-assisted learning content, review and validate generated quizzes, activities, and reviewers, and monitor student quiz performance, attendance records, and daily journal submissions.

Admin Module. The Admin Module provides administrative functions, including user account management, access to authorized system records, monitoring, and report viewing.

Learning Material Management Module. This module allows teachers to upload and manage learning resources that serve as the basis for AI-assisted learning content generation.

AI-Generated Learning Content Module. This module coordinates the generation of quizzes, activities, and reviewers using Google Gemini AI and the learning materials provided by teachers. Generated content is subject to teacher review and validation.

Interactive Quiz and Learning Game Module. This module allows students to answer available quizzes and activities, submit their responses, and view their corresponding scores.

Score Computation and Leaderboard Module. This module records quiz results, computes student scores, and displays leaderboard rankings based on recorded performance.

Digital Work Immersion Monitoring Module. This module allows students to record their work immersion time-in and time-out information and submit their daily journals. Authorized teachers can access these records for monitoring purposes.

Dashboard and Reports Module. This module presents relevant student performance, attendance, journal, and administrative information to authorized users.

The following figures present the actual user interfaces of the developed system. Each screenshot should demonstrate the corresponding module and its implemented functions.

- Student dashboard
- Teacher dashboard
- Admin dashboard
- Learning material upload and management interface
- AI-generated quizzes, activities, or reviewers interface
- Interactive quiz and score interface
- Leaderboard interface
- Work immersion attendance interface
- Daily journal submission and monitoring interface
- Reports interface, if implemented and accessible

The user interfaces and modules work together to provide the system's academic learning and work immersion monitoring functions. Their organization according to user roles allows students, teachers, and administrators to access the features and information relevant to their responsibilities.

4.3 System Testing Results

This section presents the results obtained from the functional and black-box testing of LearnIQ Track. The testing was conducted to determine whether the completed system performed according to its documented requirements and expected functions. The testing covered user account management, learning material management, AI-generated learning content, interactive quizzes, score computation, leaderboard functionality, work immersion attendance, daily journal submission, dashboards, reports, security, reliability, performance, and compatibility.

The testing results were documented using three statuses: PASS, FAIL, and BLOCKED. PASS indicates that the recorded actual result matched the expected behavior of the test case. FAIL indicates that the actual result did not meet the expected behavior. BLOCKED indicates that the test could not be completed under the available testing conditions.

4.3.1 Test Results by Requirement

The testing results are presented according to the major functional and non-functional requirements of LearnIQ Track. Each test case identifies the actual result observed during testing and its corresponding status.

Functional and Black-Box Testing

The functional and black-box testing examined the major operations of LearnIQ Track, including account registration, authentication, role-based access, learning material management, AI-generated content display, quiz processing, score computation, leaderboard ranking, attendance recording, journal submission, dashboards, and reports.

Table 4.2. Functional and Black-Box Testing Results.

Test ID	Actual Result	Status
FT-01	Account created successfully with the assigned role.	PASS
FT-02	Required fields identified when submitted empty.	PASS
FT-03	Duplicate account registration prevented.	PASS
FT-04	User logged in and redirected to the appropriate dashboard.	PASS
FT-05	Invalid credentials rejected with an appropriate message.	PASS
FT-06	Logout ended the session.	PASS
FT-07	Student access limited to authorized functions.	PASS
FT-08	Teacher functions accessible to the teacher account.	PASS
FT-09	Admin functions accessible to the admin account.	PASS
FT-10	Unauthorized access to restricted functions prevented.	PASS
FT-11	Uploaded learning material became available to intended users.	PASS
FT-12	Unsupported material upload rejected.	PASS
FT-13	Assigned learning material displayed correctly.	PASS
FT-14	Unauthorized learning material remained inaccessible.	PASS
FT-15	AI-generated quiz displayed for review.	PASS
FT-16	Learning activity generated and displayed.	PASS

Continuation

FT-17	AI-generated reviewer displayed for review.	PASS
FT-18	Teacher could review generated content before release.	PASS
FT-19	AI service failure could not be simulated in the test environment.	BLOCKED
FT-20	Quiz answers processed and submission completed.	PASS
FT-21	Quiz validation followed the configured rules.	PASS
FT-22	Score matched the defined scoring rule.	PASS
FT-23	Saved quiz score remained available.	PASS
FT-24	Leaderboard reflected the defined ranking rule.	PASS
FT-25	Time-in record saved with the appropriate information.	PASS
FT-26	Time-out record saved successfully.	PASS
FT-27	Invalid attendance sequence handled according to system rules.	PASS
FT-28	Daily journal saved successfully.	PASS
FT-29	Authorized user retrieved the saved journal.	PASS
FT-30	Teacher dashboard displayed relevant information.	PASS
FT-31	Student monitoring information displayed correctly.	PASS
FT-32	Authorized attendance records displayed.	PASS
FT-33	Journal submissions displayed for teacher monitoring.	PASS
FT-34	Work immersion records displayed for authorized users.	PASS

Continuation	dashboard displayed administrative functions.	PASS
FT-36	User-management changes were saved.	PASS
FT-37	Available reports displayed the expected records.	PASS
FT-38	Saved records remained available after refreshing or signing in again.	PASS
FT-39	Protected functions required authentication after logout.	PASS

The testing results showed that 38 of the 39 functional and black-box test cases were marked PASS, while one test case, FT-19, was marked BLOCKED. The passed cases covered the major system operations, including account management, learning activities, quiz processing, work immersion monitoring, and report access.

FT-19 was blocked because the AI service failure condition could not be simulated in the testing environment. Therefore, the system's behavior when the AI service becomes unavailable remains subject to further testing.

AI-Generated Content Testing

AI-generated content testing examined the relevance, accuracy, structure, clarity, and appropriateness of the learning content generated using Google Gemini AI. It also covered answer key consistency, question duplication, and the teacher review and validation workflow.

Table 4.3. AI-Generated Content Testing Results.

Test ID	Actual Result	Status

Continuation

AI-01	Generated questions corresponded to the supplied lesson.	PASS
AI-02	Generated content stayed relevant to the requested topic.	PASS
AI-03	No factual errors identified in the simulated review.	PASS
AI-04	Answer key matched the supplied source material.	PASS
AI-05	Requested question structure was produced.	PASS
AI-06	Generated questions showed no significant unnecessary duplication.	PASS
AI-07	Questions and choices were understandable.	PASS
AI-08	No inappropriate or clearly irrelevant content identified in the simulated review.	PASS
AI-09	Teacher review and validation workflow was available.	PASS
AI-10	AI service failure and retry behavior could not be tested.	BLOCKED

The results showed that nine of the ten test cases were marked PASS, while AI-10 was marked BLOCKED. The completed checks covered content relevance, question structure, answer keys, clarity, duplication, and teacher validation.

AI-10 remained blocked because AI service failure and retry behavior could not be tested. In addition, AI-03 and AI-08 in the original testing records explicitly refer to a simulated review. These entries should be identified

accurately and should not be presented as comprehensive verification of all AI-generated content.

Security and Role-Based Access Testing

Security and role-based access testing examined whether LearnIQ Track provided appropriate access according to user roles and protected restricted information. The tests covered authentication, invalid credentials, access to student records, role-specific permissions, logout protection, invalid input handling, and unauthorized record modification.

Table 4.4. Security and Role-Based Access Testing Results.

Test ID	Actual Result	Status
SEC-01	Valid credentials authenticated the authorized user.	PASS
SEC-02	Invalid credentials did not grant access.	PASS
SEC-03	Restricted records belonging to another student were not displayed.	PASS
SEC-04	Teacher account accessed authorized monitoring functions.	PASS
SEC-05	Student account could not access teacher/admin-only functions.	PASS
SEC-06	Admin account accessed authorized administrative functions.	PASS
SEC-07	Protected content was inaccessible after logout.	PASS
SEC-08	Invalid input was handled without exposing sensitive information.	PASS
SEC-09	Unauthorized record modification was prevented.	PASS

Continuation	screens displayed information appropriate to the assigned role.	PASS

All ten security and role-based access test cases were marked PASS. The results indicate that the listed security scenarios produced their expected outcomes, including restricting unauthorized access and displaying information appropriate to each assigned role.

However, these results are limited to the security scenarios included in the testing records. They do not establish that the system is free from all possible security vulnerabilities.

Reliability, Data Integrity, Performance, and Compatibility Testing

The testing also examined selected aspects of system reliability, data integrity, performance, and compatibility. These tests covered login and dashboard loading, interactive quiz loading, file upload, repeated operations, data persistence, record retrieval, information consistency, and compatibility across devices and browsers.

Table 4.5. Reliability, Data Integrity, Performance, and Compatibility Testing Results.

Test ID	Actual Result	Status
NFR-01	Login completed within the assumed acceptable response time.	PASS
NFR-02	Dashboard loaded without an abnormal delay in the example scenario.	PASS
NFR-03	Interactive quiz loaded with the expected elements.	PASS

Continuation

NFR-04	AI generation response could not be measured under a controlled failure scenario.	BLOCKED
NFR-05	Learning material upload completed without observed corruption in the example.	PASS
NFR-06	Repeated common operations completed without an unexpected crash in the example.	PASS
NFR-07	Saved information remained available after refreshing.	PASS
NFR-08	Saved records could be retrieved after signing in again.	PASS
NFR-09	Information remained consistent across the example's authorized views.	PASS
NFR-10	Compatibility across additional devices and browsers remained unverified.	BLOCKED

The results showed that eight of the ten test cases were marked PASS, while two were marked BLOCKED. NFR-04 was blocked because AI response behavior could not be measured under a controlled failure scenario. NFR-10 was blocked because compatibility across additional devices and browsers remained unverified.

The completed cases covered the recorded observations involving login, dashboard and quiz loading, file upload, repeated operations, data persistence, and information consistency. The blocked cases indicate that additional performance and compatibility testing is necessary.

4.3.2 Testing Summary

The overall testing results were consolidated to determine the number of test cases completed and their corresponding statuses. The summary includes functional and black-box testing, AI-generated content testing, security and role-based access testing, and reliability, data integrity, performance, and compatibility testing.

Table 4.6. Summary of System Testing Results.

Testing Category	Total Test Cases	Passed	Failed	Blocked
Functional and Black-Box Testing	39	38	0	1
AI-Generated Content Testing	10	9	0	1
Security and Role-Based Access Testing	10	10	0	0
Reliability, Data Integrity, Performance, and Compatibility	10	8	0	2
Total	**69**	**65**	**0**	**4**

Table 4.6 shows that 65 of the 69 listed test cases were marked PASS, none was marked FAIL, and four were marked BLOCKED. The results indicate that the majority of the documented test cases were completed successfully, while some scenarios could not be evaluated under the available testing conditions.

The pass rate was computed using the number of passed test cases divided by the number of executed test cases.

Formula:

Pass Rate = $(\frac{Number\ of\ Passed\ Test\ Cases}{Number\ of\ Executed\ Test\ Cases}) \times 100$

Substitution:

Pass Rate = $(\frac{65}{65}) \times 100$

Pass Rate = 100%

The computed 100% pass rate applies only to the 65 test cases recorded as executed and passed. It does not mean that all 69 listed test cases were completed, since four test cases remained blocked and require further testing.

4.3.3 Discussion of Testing Results

The testing results indicate that LearnIQ Track performed according to the expected outcomes in most of the documented functional and non-functional test cases. The completed functional tests demonstrated the operation of the system's major features, including user account management, learning material management, AI-generated content display, interactive quizzes, score computation, leaderboard ranking, work immersion attendance, daily journal submission, dashboards, and reports.

The AI-generated content testing showed that the recorded checks for relevance, question structure, answer keys, clarity, duplication, and teacher validation were marked PASS. The teacher review and validation workflow is an important part of the system because generated learning materials should be checked before being made available to students. However, the testing records also identify simulated review entries and an uncompleted AI service failure test. These limitations should be considered when interpreting the results.

The security and role-based access testing results showed that all ten listed test cases were marked PASS. These results demonstrate that the tested authentication and access-control scenarios produced their expected outcomes. Nevertheless, additional security assessment may still be conducted to examine threats and vulnerabilities beyond the scenarios included in the testing records.

For reliability, data integrity, performance, and compatibility, the completed test cases showed that selected operations and records behaved as expected during the documented testing. However, the blocked cases indicate that AI response behavior under a controlled failure scenario and compatibility across additional devices and browsers have not yet been verified.

The researchers also consulted IT experts regarding LearnIQ Track. The experts provided generally positive verbal feedback and offered few additional comments. However, they were unable to complete the formal evaluation survey because of their schedules. Therefore, their feedback is presented as supplementary qualitative input, and no numerical rating or weighted mean was computed from their comments.

Overall, the testing results provide evidence that the majority of the documented test cases produced their expected outcomes. The four blocked test cases identify specific areas requiring further verification. The results should therefore be interpreted according to the actual testing coverage and conditions rather than as proof that every possible system condition has been tested.

4.4 System User Evaluation Results

This section presents the results of the system user evaluation of LearnIQ Track. The evaluation was conducted to determine the respondents' perceptions of the system in terms of selected software quality characteristics and user acceptance. A five-point Likert scale was used to assess the system based on five selected ISO/IEC 25010 software quality characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. The evaluation also covered four Technology Acceptance Model (TAM) constructs: Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. Overall Acceptability was also evaluated to determine the respondents' general assessment of the system.

4.4.1 Respondent Profile

The system user evaluation included 283 respondents. The same number of responses was recorded for each item in the evaluation questionnaire, providing the basis for the computation of the weighted means presented in this section.

The available evaluation results in the manuscript do not provide a verified breakdown of the 283 respondents according to participant group or demographic characteristics. Therefore, a distribution table by student, teacher, administrator, age, or other demographic category is not presented to avoid reporting unsupported figures.

4.4.2 Evaluation Results by Criterion

The evaluation results were organized according to the selected ISO/IEC 25010 software quality characteristics, TAM constructs, and Overall Acceptability.

For the ISO/IEC 25010 characteristics, the evaluation covered Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. These criteria assessed the respondents' perceptions of the system's functions, ease of use, performance, consistency, and protection of information.

For user acceptance, the evaluation covered Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. Overall Acceptability was also included to summarize the respondents' general satisfaction and perception of the system's suitability for its intended purposes.

The summary of the evaluation results is presented below.

Table 4.7. Summary of Software Quality Characteristics.

Quality Characteristic	Overall Weighted Mean	Interpretation
Functional Suitability	4.59	Strongly Agree
Usability	4.58	Strongly Agree
Performance Efficiency	4.58	Strongly Agree
Reliability	4.55	Strongly Agree
Security	4.56	Strongly Agree

Continuation

Overall Mean	**4.57**	**Strongly Agree**

Table 4.7 shows that the five selected software quality characteristics obtained an overall mean of 4.57, interpreted as Strongly Agree. Functional Suitability received the highest category mean of 4.59, while Reliability received the lowest category mean of 4.55. The results indicate favorable respondent perceptions across the evaluated software quality characteristics.

Table 4.8. Summary of Technology Acceptance and Overall Acceptability Results.

Evaluation Construct	Overall Weighted Mean	Interpretation
Perceived Ease of Use	4.56	Strongly Agree
Perceived Usefulness	4.56	Strongly Agree
Attitude Toward Using	4.56	Strongly Agree
Behavioral Intention to Use	4.53	Strongly Agree
Overall Acceptability	4.56	Strongly Agree

Table 4.8 presents the results of the TAM constructs and Overall Acceptability. The weighted means ranged from 4.53 to 4.56, all interpreted as Strongly Agree. Behavioral Intention to Use obtained the lowest mean of 4.53, while Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Overall Acceptability each obtained a mean of 4.56.

4.4.3 Statistical Results

The weighted mean was used to summarize the respondents' ratings for each evaluation statement. The computation was based on the frequency of responses and the numerical value assigned to each rating.

The formula used was:

Weighted Mean = Σ(fx) / N

Where:

- **f** = frequency of responses for each rating
- **x** = numerical value assigned to the rating
- **N** = total number of responses for the item

The following interpretation scale was used to determine the verbal interpretation of the computed weighted means.

Table 4.9. Five-Point Likert Scale Interpretation.

Weighted Mean Range	Verbal Interpretation
4.21–5.00	Strongly Agree
3.41–4.20	Agree
2.61–3.40	Neutral
1.81–2.60	Disagree
1.00–1.80	Strongly Disagree

ISO/IEC 25010 Software Quality Characteristics

The following tables present the item-level weighted means for the five selected ISO/IEC 25010 software quality characteristics.

Table 4.10. Evaluation of Functional Suitability.

Item	Weighted Mean
1	4.61
2	4.53
3	4.64
4	4.62
5	4.58
6	4.55
Overall Weighted Mean	**4.59**
Interpretation	**Strongly Agree**

Functional Suitability obtained an overall weighted mean of 4.59, interpreted as Strongly Agree. The highest-rated item was the statement that the system gives appropriate results when users perform their assigned tasks, with a weighted mean of 4.64. The lowest-rated item concerned whether LearnIQ Track provides the expected learning or work immersion features for the user's role, with a weighted mean of 4.53.

Table 4.11. Evaluation of Usability.

Item	Weighted Mean
1	4.50
2	4.59

Continuation

3	4.62
4	4.65
5	4.59
6	4.59
7	4.50
Overall Weighted Mean	**4.58**
Interpretation	**Strongly Agree**

Usability obtained an overall weighted mean of 4.58, interpreted as Strongly Agree. The highest-rated item concerned the clarity of the layout and presentation of information, with a weighted mean of 4.65. The items concerning ease of learning and understanding or recovering from problems received the lowest weighted means, both at 4.50.

Table 4.12. Evaluation of Performance Efficiency.

Item	Weighted Mean
1	4.60
2	4.62
3	4.51
4	4.56
5	4.64

Continuation

6	4.56
Overall Weighted Mean	**4.58**
Interpretation	**Strongly Agree**

Performance Efficiency obtained an overall weighted mean of 4.58, interpreted as Strongly Agree. The highest-rated item concerned AI-generated content being produced within a reasonable time when the feature is available, with a weighted mean of 4.64. Accessing learning materials without unnecessary delay received the lowest weighted mean of 4.51.

Table 4.13. Evaluation of Reliability.

Item	Weighted Mean
1	4.58
2	4.58
3	4.51
4	4.49
5	4.56
6	4.59
Overall Weighted Mean	**4.55**
Interpretation	**Strongly Agree**

Reliability obtained an overall weighted mean of 4.55, interpreted as Strongly Agree. The highest-rated item concerned using LearnIQ Track without frequent unexpected failures, with a weighted mean of 4.59. The lowest-rated item concerned the continued availability of attendance and journal records, with a weighted mean of 4.49.

Table 4.14. Evaluation of Security.

Item	Weighted Mean
1	4.57
2	4.51
3	4.59
4	4.53
5	4.60
6	4.54
Overall Weighted Mean	**4.56**
Interpretation	**Strongly Agree**

Security obtained an overall weighted mean of 4.56, interpreted as Strongly Agree. The highest-rated item concerned appropriate authentication before protected features can be accessed, with a weighted mean of 4.60. The lowest-rated item concerned access according to the assigned user role, with a weighted mean of 4.51.

Technology Acceptance Model Constructs

The following tables present the results for the four TAM constructs and Overall Acceptability.

Table 4.15. Evaluation of Perceived Ease of Use.

Item	Weighted Mean
1	4.54
2	4.60
3	4.56
4	4.58
5	4.51
Overall Weighted Mean	**4.56**
Interpretation	**Strongly Agree**

Perceived Ease of Use obtained an overall weighted mean of 4.56, interpreted as Strongly Agree. The highest-rated item concerned learning to operate LearnIQ Track, with a weighted mean of 4.60. The statement that using the system does not require excessive effort received the lowest weighted mean of 4.51.

Table 4.16. Evaluation of Perceived Usefulness.

Item	Weighted Mean
1	4.55

Continuation	4.59
3	4.53
4	4.52
5	4.61
Overall Weighted Mean	**4.56**
Interpretation	**Strongly Agree**

Perceived Usefulness obtained an overall weighted mean of 4.56, interpreted as Strongly Agree. The highest-rated item concerned improving the way users perform relevant school tasks, with a weighted mean of 4.61. The statement that the system's features provide useful support for the user's role received the lowest weighted mean of 4.52.

Table 4.17. Evaluation of Attitude Toward Using.

Item	Weighted Mean
1	4.53
2	4.59
3	4.52
4	4.60
Overall Weighted Mean	**4.56**
Interpretation	**Strongly Agree**

Attitude Toward Using obtained an overall weighted mean of 4.56, interpreted as Strongly Agree. The highest-rated item concerned viewing the use of LearnIQ Track favorably in the school setting, with a weighted mean of 4.60. Comfort using the system for appropriate school-related activities received the lowest weighted mean of 4.52.

Table 4.18. Evaluation of Behavioral Intention to Use.

Item	Weighted Mean
1	4.52
2	4.50
3	4.56
4	4.53
Overall Weighted Mean	**4.53**
Interpretation	**Strongly Agree**

Behavioral Intention to Use obtained an overall weighted mean of 4.53, interpreted as Strongly Agree. The highest-rated item concerned recommending continued use of LearnIQ Track for appropriate school activities, with a weighted mean of 4.56. The statement about using the system again when its relevant features are needed received the lowest weighted mean of 4.50.

Table 4.19. Evaluation of Overall Acceptability.

Item	Weighted Mean
1	4.59
2	4.53
3	4.54
4	4.57
5	4.58
Overall Weighted Mean	**4.56**
Interpretation	**Strongly Agree**

Overall Acceptability obtained an overall weighted mean of 4.56, interpreted as Strongly Agree. The highest-rated item concerned satisfaction with the overall experience using LearnIQ Track, with a weighted mean of 4.59. The statement that the system is acceptable for its intended school use received the lowest weighted mean of 4.53.

In this study, overall acceptability refers to the respondents' reported satisfaction and perceptions of the system's suitability for its intended purposes.

4.4.4 Discussion of User Evaluation

The results of the system user evaluation indicate that the respondents generally assessed LearnIQ Track favorably. All five selected ISO/IEC 25010 software quality characteristics obtained interpretations of Strongly Agree,

with an overall mean of 4.57. Functional Suitability received the highest category mean of 4.59, suggesting that respondents favorably assessed the system's functions in relation to their intended tasks.

Usability obtained a weighted mean of 4.58, indicating that respondents generally perceived the system's interface and navigation as understandable and convenient. However, the lower item ratings for ease of learning and recovery from problems suggest areas where further interface improvements and user guidance may be beneficial.

Performance Efficiency also obtained a weighted mean of 4.58. This indicates favorable user perceptions of the system's loading behavior and response time. These findings represent respondents' perceptions and should not be interpreted as measured performance results under controlled testing conditions.

Reliability received a weighted mean of 4.55, while Security obtained 4.56. Both were interpreted as Strongly Agree. These results suggest that respondents generally perceived the system as consistent and considered its authentication and access-control features favorably. The findings should be considered alongside the system testing results, particularly the scenarios that remained blocked and require additional verification.

For the Technology Acceptance Model constructs, Perceived Ease of Use, Perceived Usefulness, and Attitude Toward Using each obtained a weighted mean of 4.56. These results indicate that respondents generally perceived LearnIQ Track as easy to operate, useful for its intended school-related functions, and favorable to use.

Behavioral Intention to Use obtained a weighted mean of 4.53, which was also interpreted as Strongly Agree. This suggests that respondents expressed a favorable intention to use or support the continued use of LearnIQ Track. However, actual continued use would need to be assessed through subsequent usage records or follow-up evaluation.

Overall Acceptability obtained a weighted mean of 4.56, interpreted as Strongly Agree. This indicates that respondents generally viewed LearnIQ Track favorably in terms of its suitability for its intended users and school-related purposes.

Overall, the evaluation results demonstrate favorable respondent perceptions of LearnIQ Track in terms of its selected software quality characteristics, user acceptance, and overall acceptability. However, these findings reflect user assessments and do not independently establish improvements in academic achievement, learning efficiency, or actual student performance.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This chapter presents the summary, conclusions, and recommendations derived from the design, development, testing, and evaluation of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. It consolidates the major findings of the study in relation to its specific objectives, discusses the conclusions supported by the results, and provides recommendations for the improvement, implementation, and future development of the system.

The summary presents the accomplishments of the study based on the development of the system, the results of functional and black-box testing, and the evaluation of its software quality and user acceptability. The conclusions are drawn from the findings presented in Chapter IV, while the recommendations identify possible improvements and further actions for the school, teachers, students, system developers, and future researchers.

5.1 Summary

This study aimed to design, develop, test, and evaluate LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The system was developed as a web-based platform that integrates AI-assisted learning activities, gamification, and digital work immersion monitoring. Specifically, the study was guided by three research objectives.

For the first objective, the study designed and developed LearnIQ Track as an integrated platform for learning activities and work immersion monitoring. The developed system incorporated user account management, learning material management, AI-generated quizzes, activities and reviewers, interactive quizzes, score computation, leaderboard display, digital work immersion time-in and time-out, daily journal submission, dashboards, and reports. It provided separate functions for students, teachers, and administrators according to their respective responsibilities. The system used HTML, CSS, and JavaScript for the front end; Python with FastAPI for the back end; Google Gemini AI for AI-assisted content generation; JSON for data exchange; Supabase for database and authentication services; and Cloudflare for web infrastructure.

For the second objective, the study tested the functionalities of LearnIQ Track using functional and black-box testing, AI-generated content testing, security and role-based access testing, and testing related to reliability, data integrity, performance, and compatibility. The supplied testing records contained 69 test cases, of which 65 were marked PASS, none were marked FAIL, and four remained BLOCKED. The recorded pass rate was 100% among the 65 test cases marked as executed and passed. The blocked cases involved AI service failure simulation, AI response measurement under a controlled failure scenario, and compatibility testing across additional devices and browsers. These cases require further verification.

For the third objective, the study evaluated LearnIQ Track using five selected ISO/IEC 25010 software quality characteristics: Functional Suitability, Usability, Performance Efficiency, Reliability, and Security. The evaluation also included four Technology Acceptance Model (TAM) constructs: Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use, together with

Overall Acceptability. The evaluation involved 283 respondents who rated the system using a five-point Likert scale.

The five selected ISO/IEC 25010 characteristics obtained an overall mean of approximately 4.57, interpreted as Strongly Agree. Functional Suitability obtained the highest category mean of 4.59, while Reliability obtained the lowest category mean of 4.55. The TAM results also showed favorable evaluations, with an Overall Acceptability mean of 4.56, interpreted as Strongly Agree. These findings indicate that the respondents generally viewed the system favorably in terms of its selected software quality characteristics, perceived usefulness, ease of use, and acceptability for its intended purposes.

Overall, the study achieved the design and development of a web-based platform integrating AI-assisted learning activities and digital work immersion monitoring. The testing results showed that the majority of the recorded test cases passed, although four cases remained blocked. The user evaluation results also reflected favorable perceptions of the system. However, the findings do not independently establish improvements in academic achievement, learning efficiency, or long-term student outcomes.

5.2 Conclusions

Based on the findings of the study, the following conclusions are drawn in relation to the three specific objectives.

First, LearnIQ Track was successfully designed and developed as a web-based platform that integrates AI-assisted learning activities, gamified quizzes, and digital work immersion monitoring. Its implemented functions include learning material management, AI-generated learning content, interactive quizzes, score computation,

leaderboard display, work immersion attendance recording, daily journal submission, dashboards, and reports. The system provides role-specific functions for students, teachers, and administrators, supporting the intended learning and monitoring activities of Paete Science and Business College Inc.

Second, the system testing results indicate that the majority of the recorded test cases performed according to their expected results under the conditions of the conducted testing. Of the 69 listed test cases, 65 were marked PASS, none were marked FAIL, and four remained BLOCKED. The results support the functional performance of the tested features within the scope of the completed test cases. However, the blocked cases indicate that further testing is necessary, particularly for AI service failure handling, controlled AI response measurement, and compatibility across additional devices and browsers.

Third, the user evaluation results indicate that LearnIQ Track received favorable ratings across the five selected ISO/IEC 25010 software quality characteristics and the evaluated TAM constructs. The overall mean of approximately 4.57 for the selected software quality characteristics and the Overall Acceptability mean of 4.56 were both interpreted as Strongly Agree. These results suggest that respondents generally perceived the system as suitable and acceptable for its intended educational and work immersion monitoring purposes.

Overall, LearnIQ Track provides an integrated digital platform intended to support learning activities and work immersion monitoring within the school. The findings support the system's intended purpose and demonstrate favorable user perceptions based on the conducted evaluation. Nevertheless, the study does not establish that the system directly improves academic achievement, student engagement,

or long-term learning outcomes. Further testing and evaluation are needed to examine these areas.

5.3 Recommendations

Based on the findings and conclusions of the study, the following recommendations are presented.

1. For the School Administration

The school administration may consider using LearnIQ Track as a supplementary platform for organizing learning activities and work immersion records. Before broader implementation, the administration should ensure that the system's functions are aligned with existing school procedures, user responsibilities, and applicable data privacy requirements. Proper orientation and user guidance should also be provided to students and teachers to support appropriate system use.

2. For Teachers

Teachers are encouraged to review and validate AI-generated quizzes, activities, and reviewers before making them available to students. They should verify the accuracy, relevance, clarity, and alignment of generated content with the uploaded learning materials. Teachers may also use the system's monitoring and reporting features to review students' quiz scores, attendance records, and daily journal submissions.

3. For Students

Students are encouraged to use LearnIQ Track responsibly when accessing learning materials, participating in quizzes, reviewing their scores, and completing work immersion requirements. They should ensure that their attendance records and

daily journal submissions are accurate and complete and follow the school's guidelines for the proper use of the platform.

4. For System Developers

Further testing should be conducted for the four blocked test cases. Developers should particularly examine the system's behavior during AI service interruptions, conduct controlled measurements of AI response times, and verify compatibility across additional devices and web browsers. Future system improvements may also include clearer error messages, enhanced recovery procedures, and additional compatibility checks.

5. For Data Privacy and Security

The developers and school administrators should continue reviewing the system's authentication, role-based access, data protection, and record-handling procedures. Access to student information should remain limited to authorized users. Regular security checks, appropriate data backup procedures, and responsible handling of personal information are also recommended to help maintain the confidentiality and integrity of school records.

6. For Future Researchers

Future researchers may conduct further studies to examine the system's effectiveness over a longer period of use. They may consider measurable learning outcomes, such as pretest and posttest scores, student activity records, or longitudinal performance data, provided that appropriate research methods and permissions are established. Additional evaluation may also include a wider range of devices and browsers, different network conditions, and further feedback from students, teachers, and IT experts. Any claims regarding improvements in academic performance, learning

engagement, or work immersion outcomes should be supported by appropriate evidence.

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APPENDICES

APPENDICES A – TECHNICAL BACKGROUND

INITIAL CLIENT INTERVIEW

Interview Date: March 11, 2026

Interview Location: Principal's Office, Paete Science and Business College Inc.

Interviewee: Ms. Ma. Gemma B. Calma, School Principal

Interviewers: De La Cruz, Aaron Karl U., De Leon, Francine Ann C., Yamomo, John Kaith L., Researchers

Purpose of the Interview

The initial interview was conducted to determine the school's willingness to serve as the client for the proposed project and to gather preliminary information regarding the academic needs of Senior High School students, the school's existing learning practices, and its openness to integrating technology into teaching and learning. The interview also aimed to identify the possible features and considerations that could guide the development of LearnIQ Track.

Interview Questions and Summary of Responses

1. How does the school currently support students in exam preparation and self-study?

Response: The school primarily supports students through the lessons provided by their teachers.

2. What digital tools or platforms are currently used in the school (Google Classroom, LMS, modules, etc.)?

Response: The school currently uses learning modules as part of its instructional materials.

3. What are the school's goals for integrating technology into teaching and learning?

Response: The interviewee expressed that the school was interested in having a system similar to the proposed LearnIQ Track. The school had previously intended to have a similar system, but its IT team was no longer available.

4. Do you think a game-based reviewer system would help motivate SHS students to study? Why or why not?

Response: The interviewee agreed that a game-based reviewer system was a great idea and expressed a positive view of the proposed concept.

5. What subjects or grade levels should be prioritized in the system?

Response: The system should prioritize Senior High School students, particularly Grade 11 and Grade 12.

6. Is the school open to using AI to automatically generate quizzes from uploaded lessons?

Response: Yes. The interviewee expressed openness to using artificial intelligence for quiz generation, particularly given the developments in the modern world.

7. Does the school have reliable internet and devices for students (computer labs, tablets, Wi-Fi)?

Response: Students have their own mobile phones that may be used to access the system. The availability and reliability of internet connectivity were not specifically established during the interview.

8. What recommendations can you give us to make this system more beneficial for SHS students and teachers?

Response: The interviewee recommended that the system should be engaging and user-friendly.

Summary of the Interview

The initial interview provided the researchers with preliminary information regarding the school's existing instructional practices, its interest in educational technology, and its expectations for the proposed system. The school currently uses teacher-provided lessons and learning modules to support students. The interviewee also expressed interest in a system similar to LearnIQ Track and positively received the idea of incorporating game-based learning and artificial intelligence to support quiz generation.

The interview further established that the proposed system should cater to Grade 11 and Grade 12 students. The interviewee emphasized the importance of making the system engaging and user-friendly. Students' access to personal mobile phones was also identified as a relevant consideration in developing the system.

Overall, the interview encouraged the researchers to proceed with the proposed project. The principal's positive response and willingness to assist the researchers with their study provided initial support for the development of the study.

Note on Interview Documentation

The initial interview was conducted on March 11, 2026, at the school's principal's office. The interview was not audio- or video-recorded because the interviewee expressed that she did not wish to be recorded. Therefore, the responses presented in this documentation are based on the information documented by the researchers. The responses are summarized and are not presented as verbatim statements.

Thematic Analysis

The following thematic analysis presents the major themes identified from the preliminary interview conducted with Ms. Ma. Gemma B. Calma, the School Principal of Paete Science and Business College Inc., on March 11, 2026. The interview aimed to understand the school's existing learning practices, available educational resources, and willingness to adopt a technology-based learning and monitoring system. The discussion also explored the school's perspective on game-based learning, artificial intelligence (AI)-generated quizzes, and the proposed system's intended users.

The researchers analyzed the information gathered during the interview to identify recurring ideas and important points relevant to the development of LearnIQ Track. Based on the summarized responses, the analysis established four main themes:

Theme 1: Existing Learning Practices and Available Educational Resources

Teacher-Led Instruction and Learning Materials

The interview revealed that the school currently supports students' learning and examination preparation primarily through teacher-provided lessons. The principal also identified modules as among the learning resources currently used by the school. This indicates that the existing learning process relies on instructional materials provided through teachers and available learning modules.

Availability of Personal Devices Among Students

The discussion also established that students have their own mobile phones. This provides a potential means for accessing a digital learning platform. However, the interview did not establish whether internet connectivity is consistently reliable among students.

Theme 2: Positive Perception of Game-Based Learning

Potential of Interactive and Game-Based Review Activities

The principal expressed a positive response to the idea of developing a game-based reviewer for students. This suggests that interactive learning activities may be considered a useful approach to making review sessions more engaging and encouraging students to participate in learning activities.

Importance of an Engaging and User-Friendly System

The principal emphasized that the proposed system should be engaging and easy to use. This highlights the importance of designing a platform that students can understand and navigate while participating in its learning activities.

Theme 3: Acceptance of Artificial Intelligence in Educational Activities

Interest in AI-Generated Quizzes

The interview revealed that the principal was open to the use of artificial intelligence to automatically generate quizzes. She expressed a positive view of incorporating AI into educational activities, particularly in the context of modern technological developments.

This response supports the consideration of AI-assisted quiz generation as one of the major features of LearnIQ Track. The system is intended to assist teachers in preparing learning activities based on uploaded instructional materials, subject to teacher review and validation.

Integration of Modern Technology in Education

The principal's positive response to AI integration indicates the school's openness to exploring modern technologies that may support teaching and learning

activities. This provides initial support for the proposed system's use of AI as an educational assistance tool.

Theme 4: School Readiness and Support for the Proposed System

Interest in a Technology-Based Learning Platform

During the interview, the principal expressed that the school had previously wanted something similar to the proposed system. However, the school's former IT team was no longer available. This response indicated an existing interest in having a technology-based solution that could support the school's educational activities.

Target Users and Potential System Accessibility

The principal identified Senior High School students, particularly Grade 11 and Grade 12, as the intended users of the proposed learning platform. The availability of personal mobile phones among students also provides a potential means of accessing the system.

Willingness to Support the Project

The principal responded positively to the researchers' proposed project and expressed willingness to assist them with the requirements needed to accomplish the study. Her response encouraged the researchers to proceed with the development of LearnIQ Track.

Overall, the thematic analysis indicates that Paete Science and Business College Inc. currently relies on teacher-provided lessons and learning modules to support student learning. The interview also revealed the principal's positive perception of game-based learning and AI-generated quizzes, as well as the school's initial interest in adopting a technology-based learning platform. The identified themes support the proposed development of LearnIQ Track as an integrated system that combines AI-

assisted quiz generation, interactive learning activities, and student monitoring features. These findings served as initial considerations in identifying the system's requirements and refining its proposed features.

STORY BOARD

Below is the storyboard of the developed LearnIQ Track web application. It presents the step-by-step processes and major functions of the system based on the different user roles, namely students, teachers, and administrators. The screenshots illustrate how each user accesses and navigates the system's features.

STUDENT JOURNEY

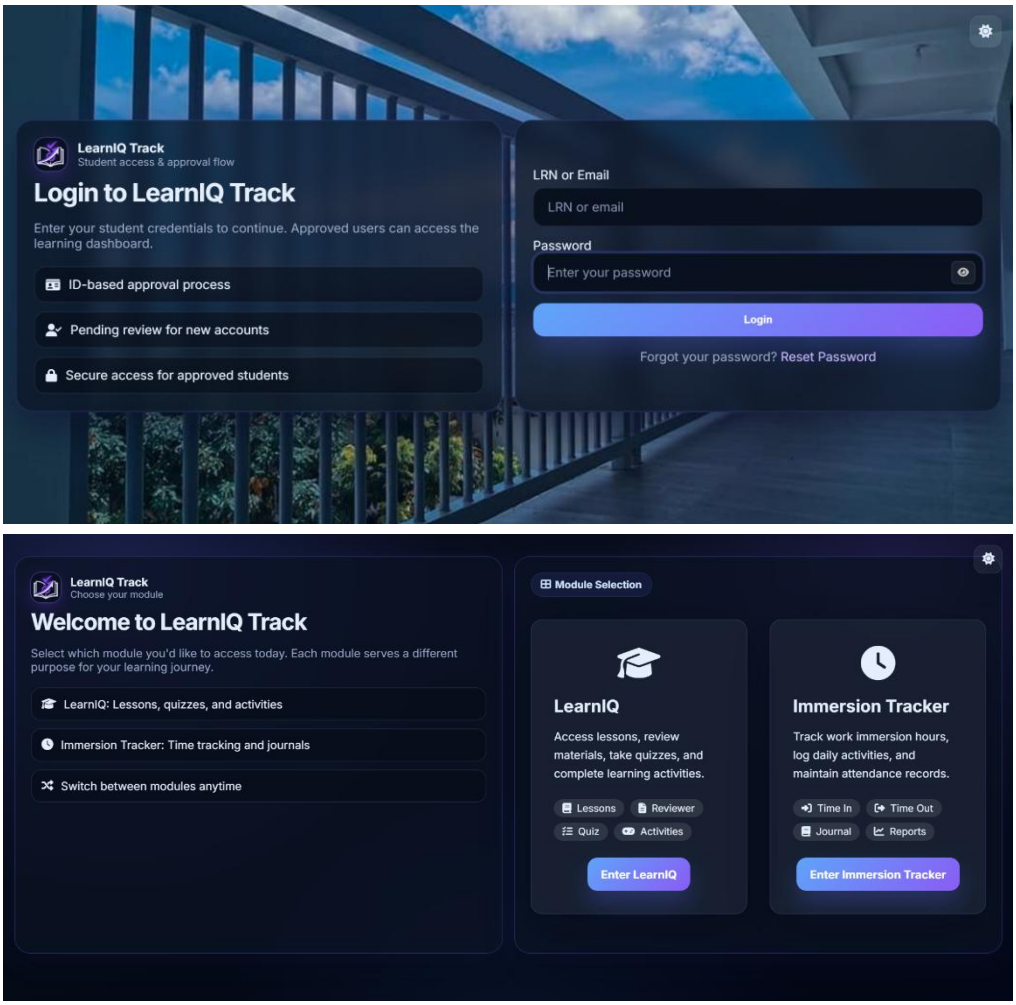
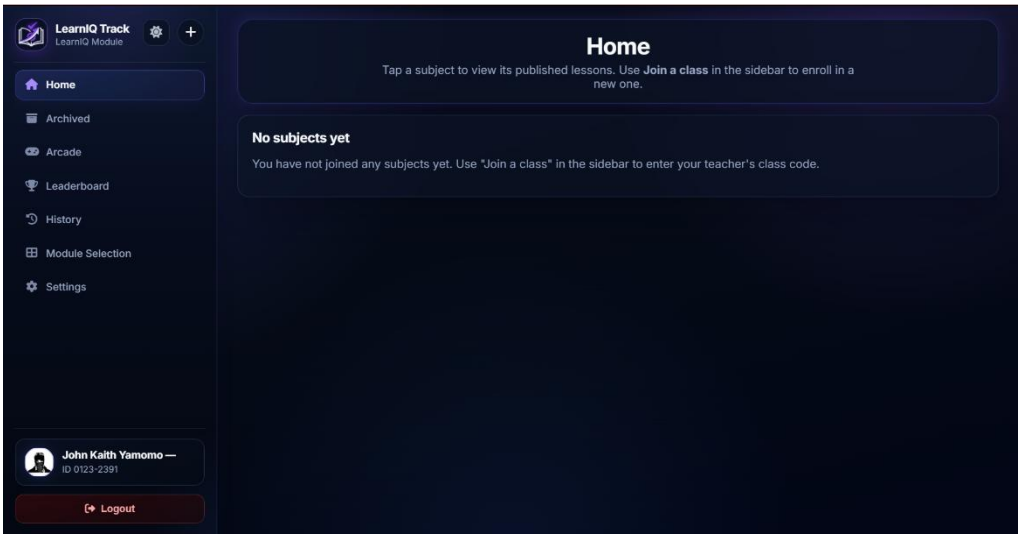


Figure A.1. Student account login, module selection, and access to the LearnIQ and Immersion Tracker modules.



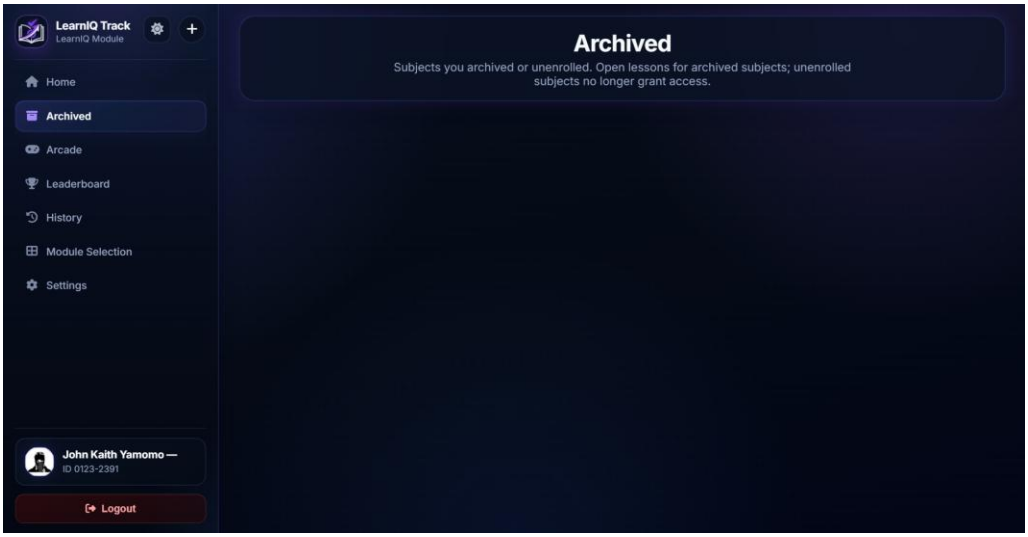


Figure A.2. Student Home and Archive page

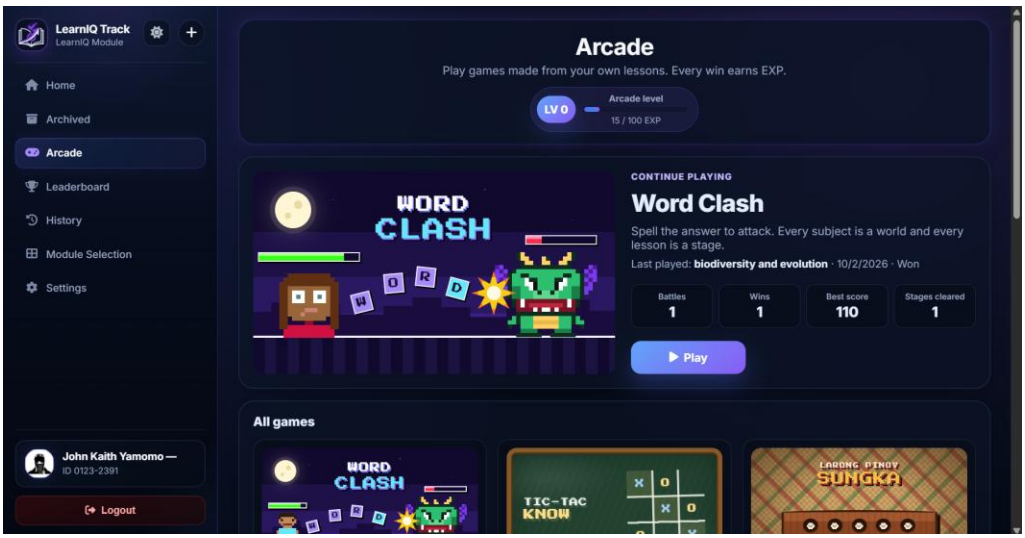
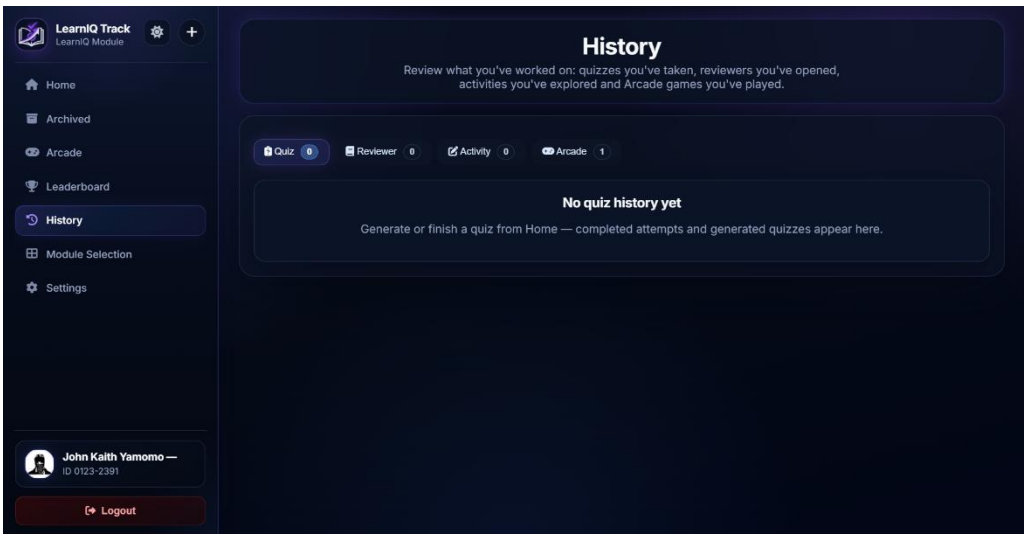
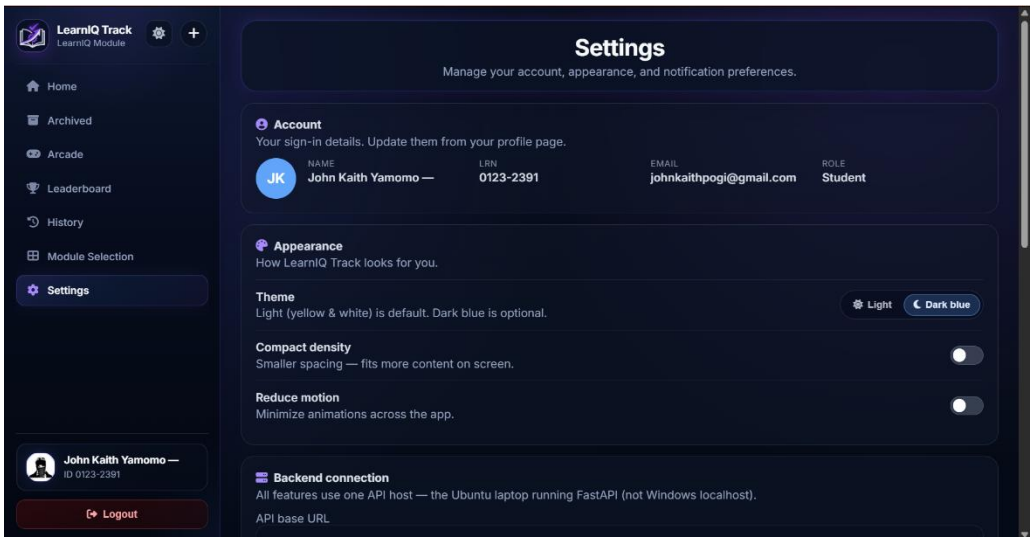




Figure A.3. Accessing educational games and viewing student rankings through the LearnIQ module.





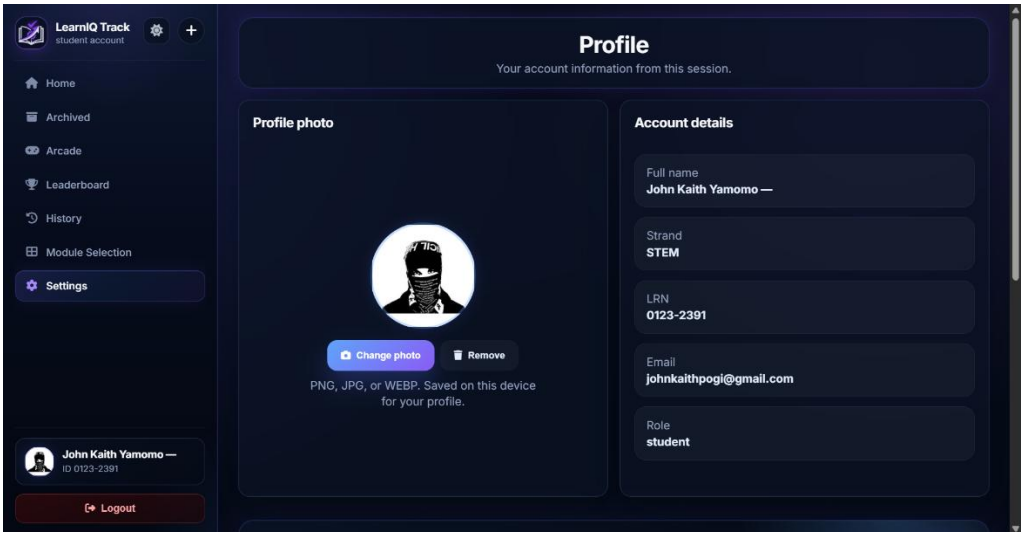


Figure A.4. Student History, Settings, and Profile pages.

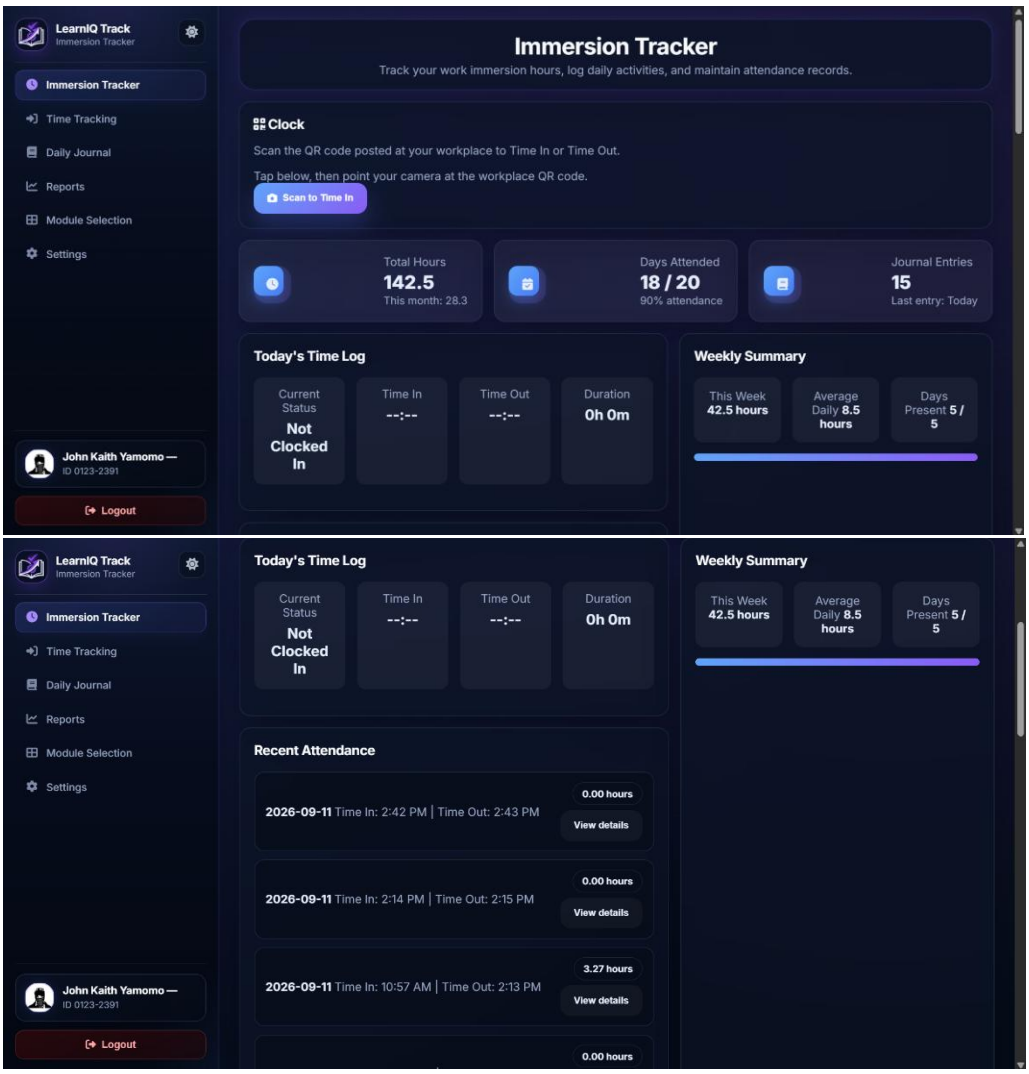


Figure A.5. Monitoring work immersion attendance, recording time-in and time-out, and viewing accumulated hours through the Immersion Tracker.

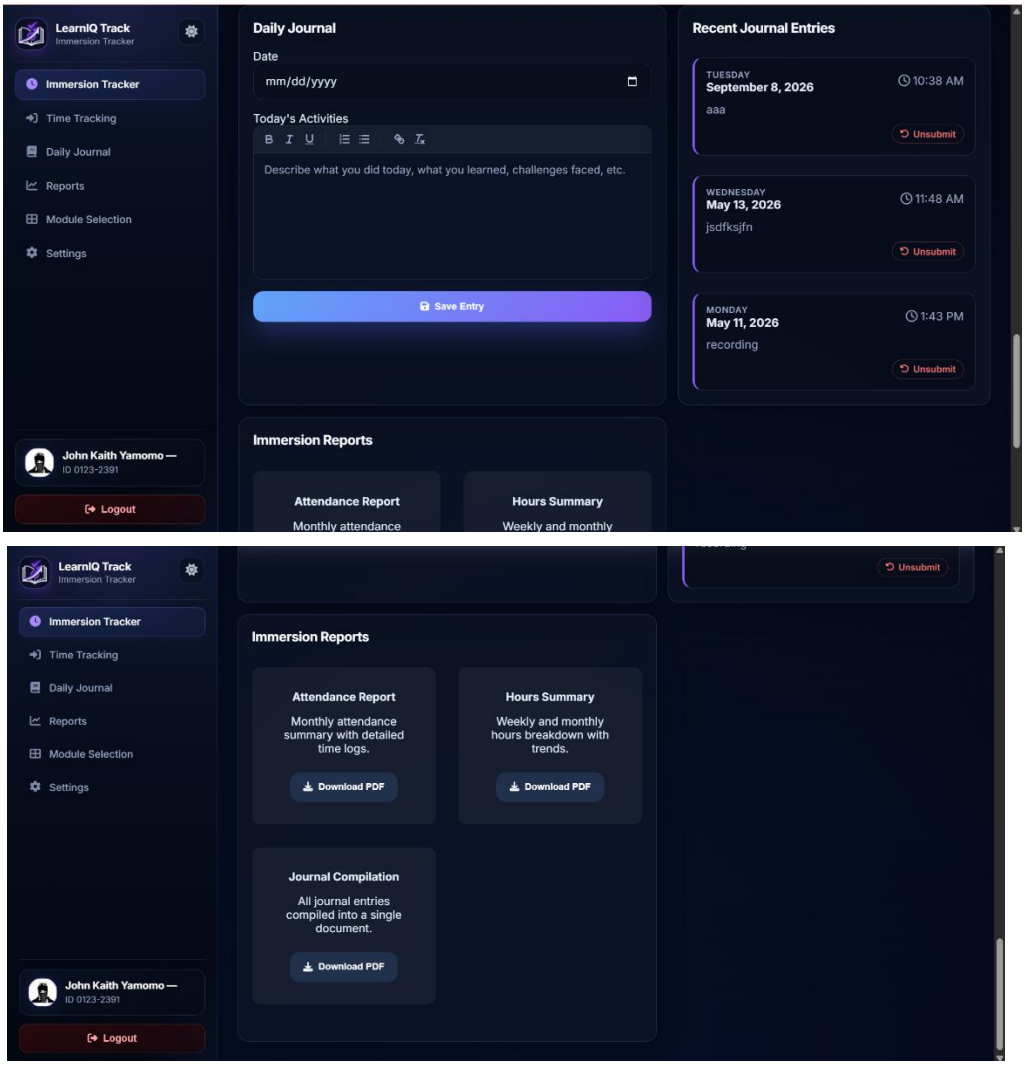
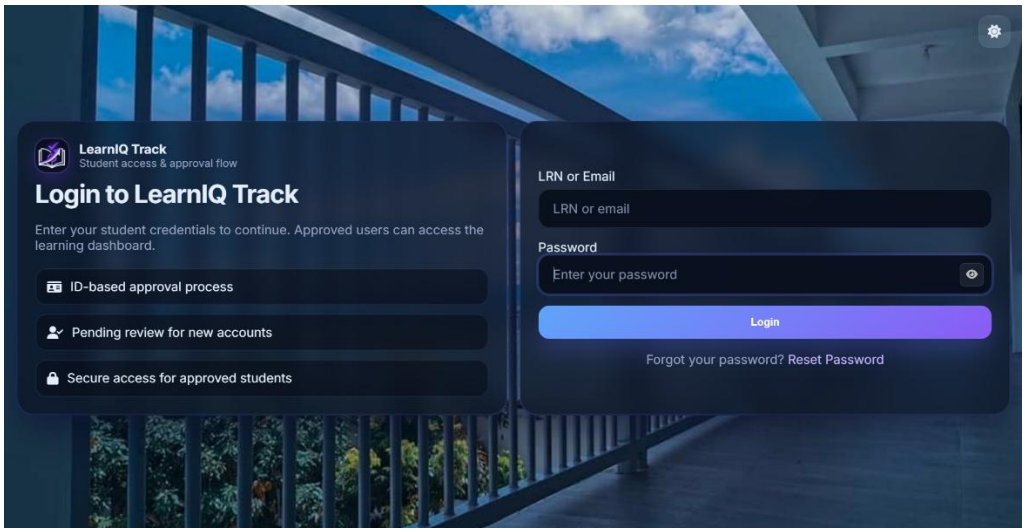


Figure A.6. Creating daily journal entries and accessing work immersion reports.

TEACHER JOURNEY



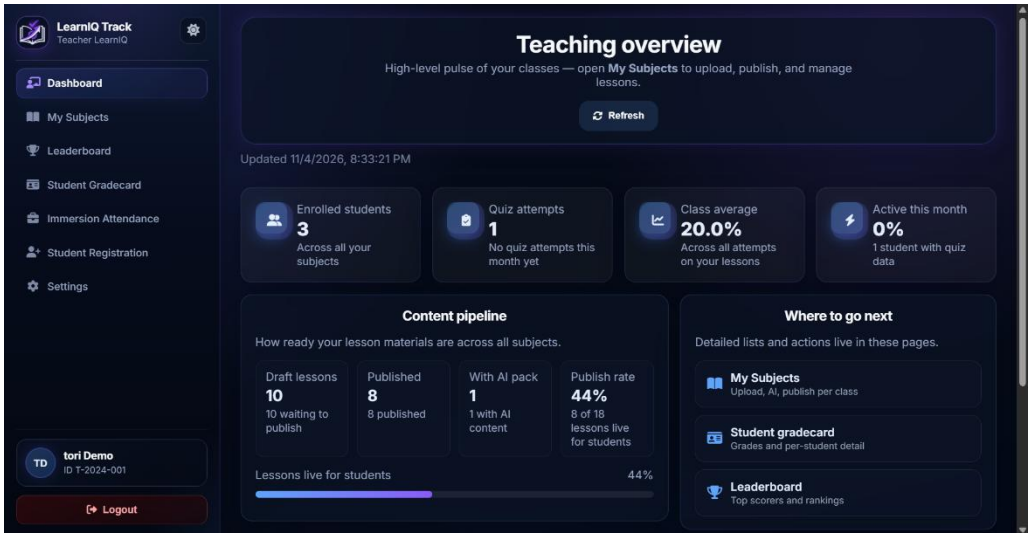


Figure A.7. Teacher account login and access to the teacher dashboard.

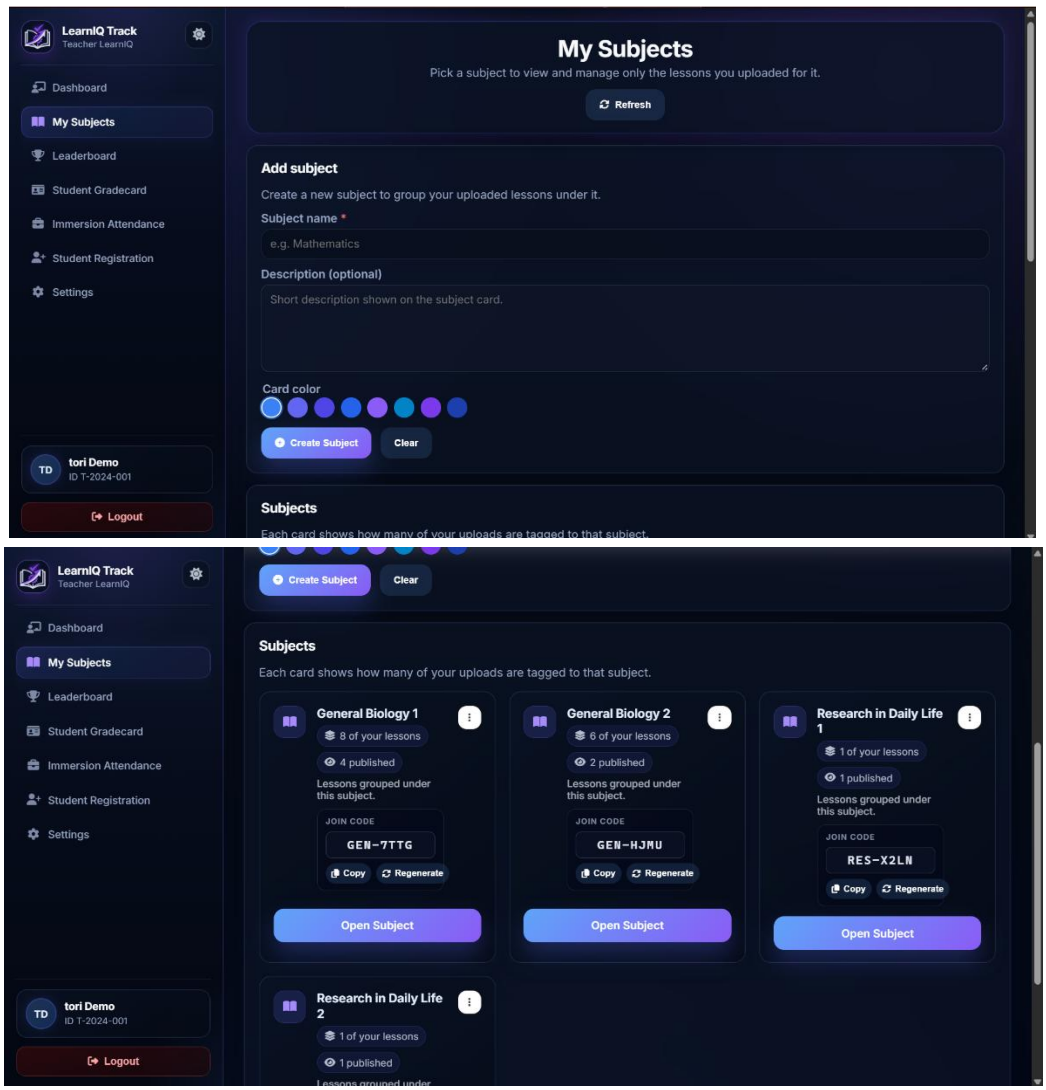


Figure A.8. Managing learning materials.

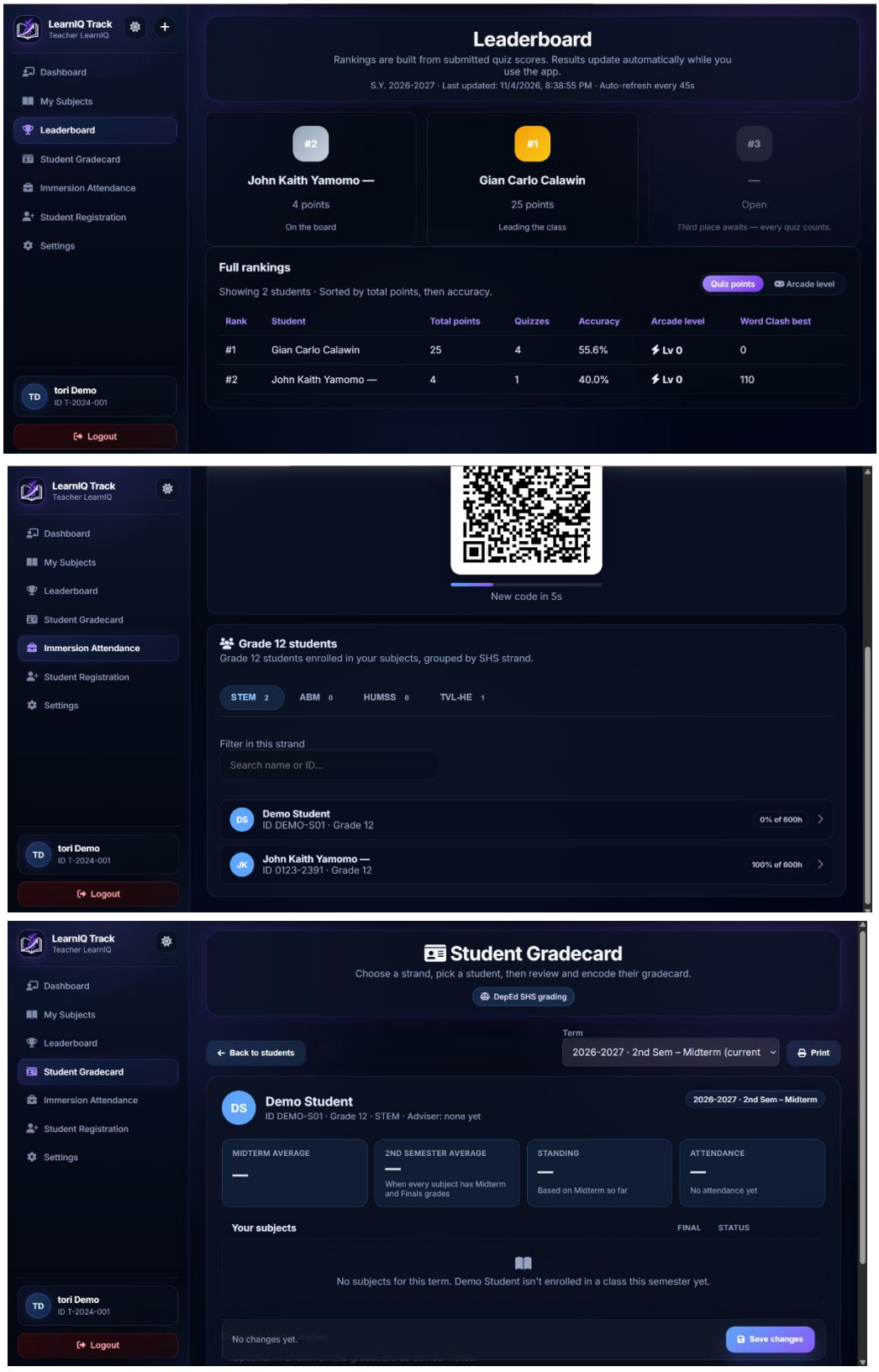


Figure A.9. Monitoring student performance, attendance records, grade card progress.

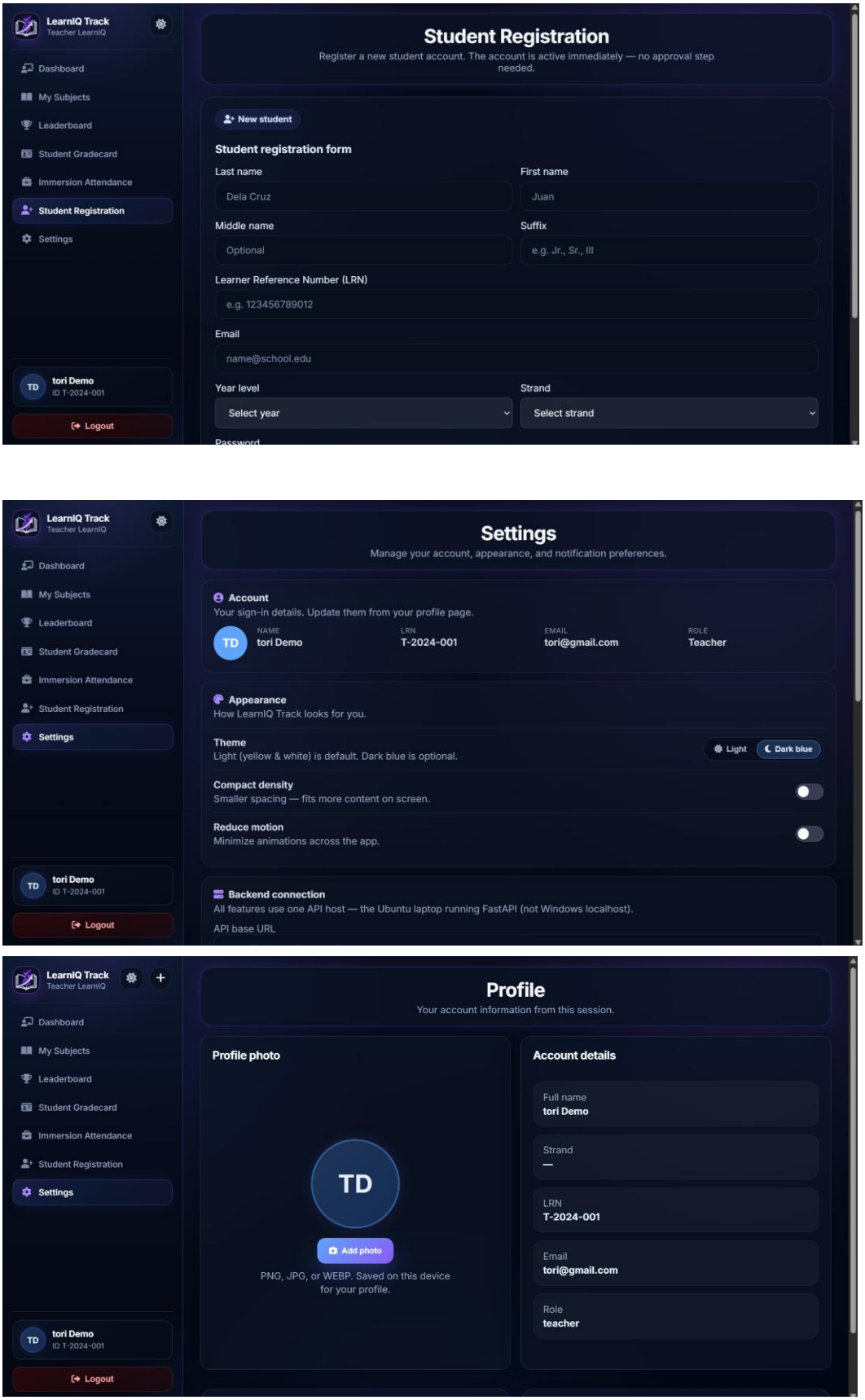


Figure A.10. Student Registration, Settings, Profile Pages.

ADMINISTRATOR JOURNEY

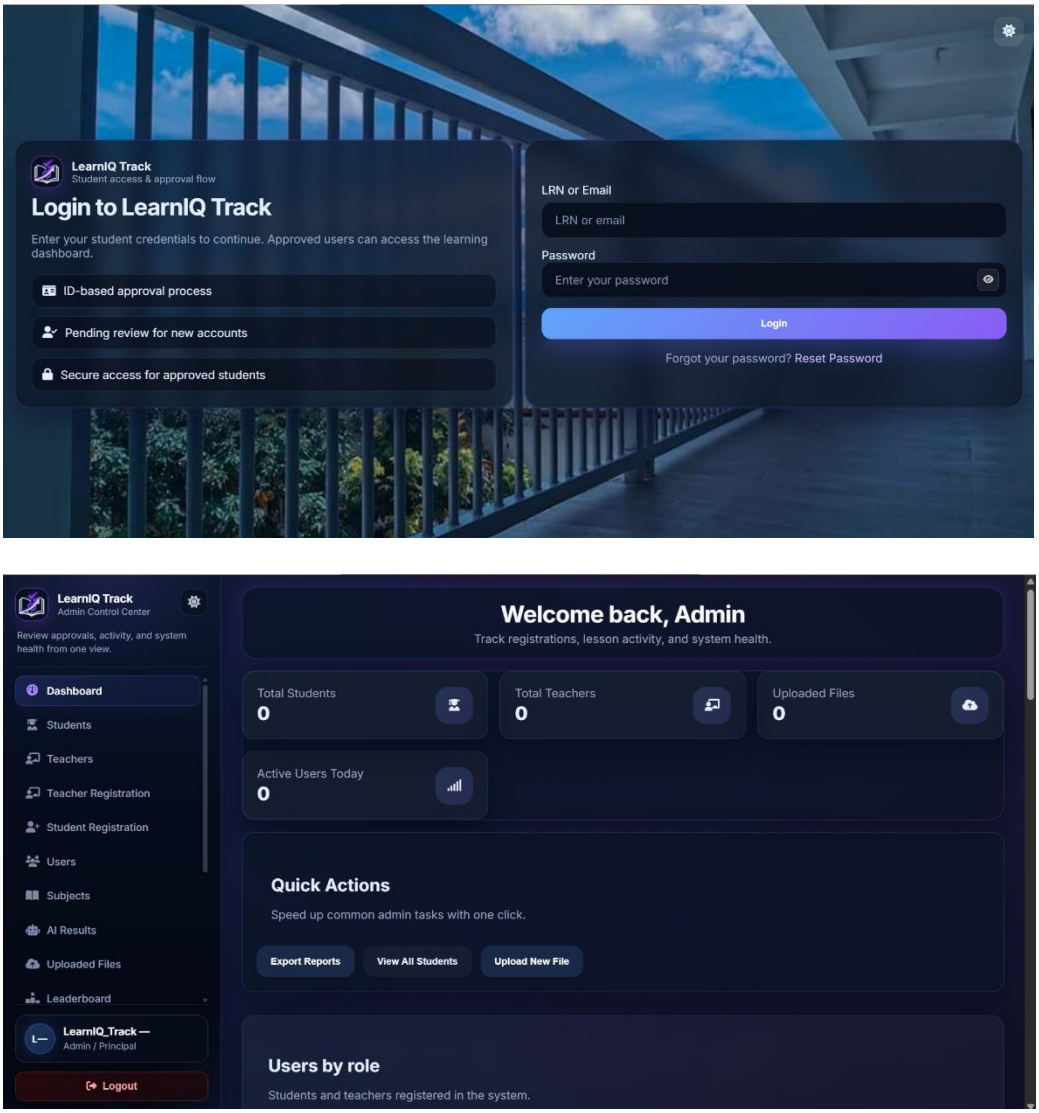
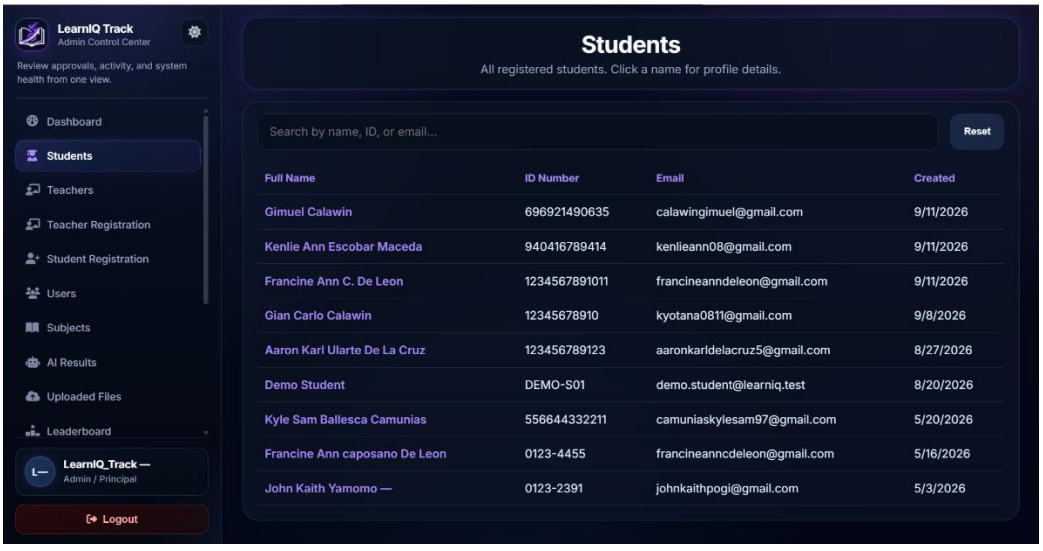


Figure A.11. Administrator account login and access to the system dashboard.



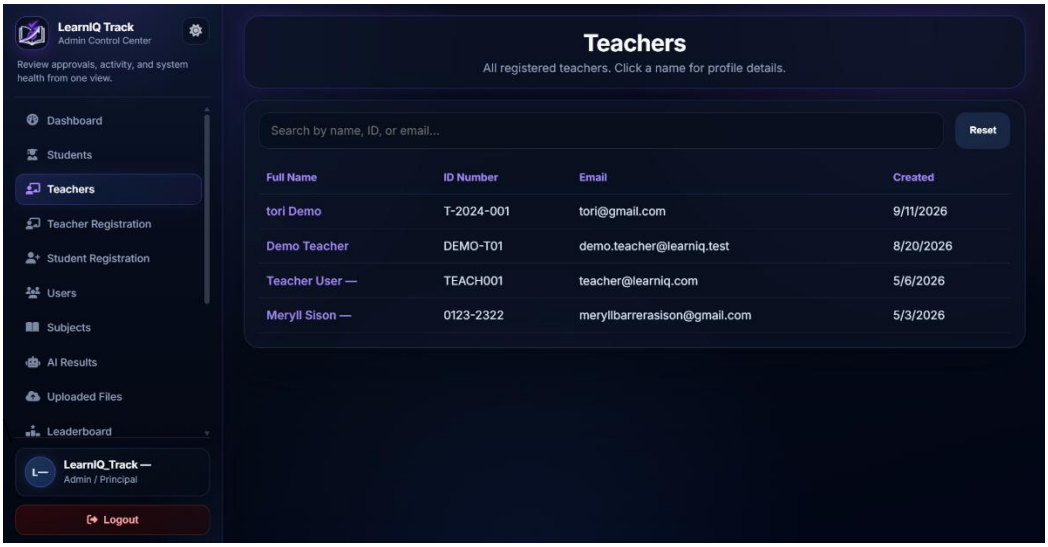


Figure A.12. Managing student and teacher accounts and maintaining user records.

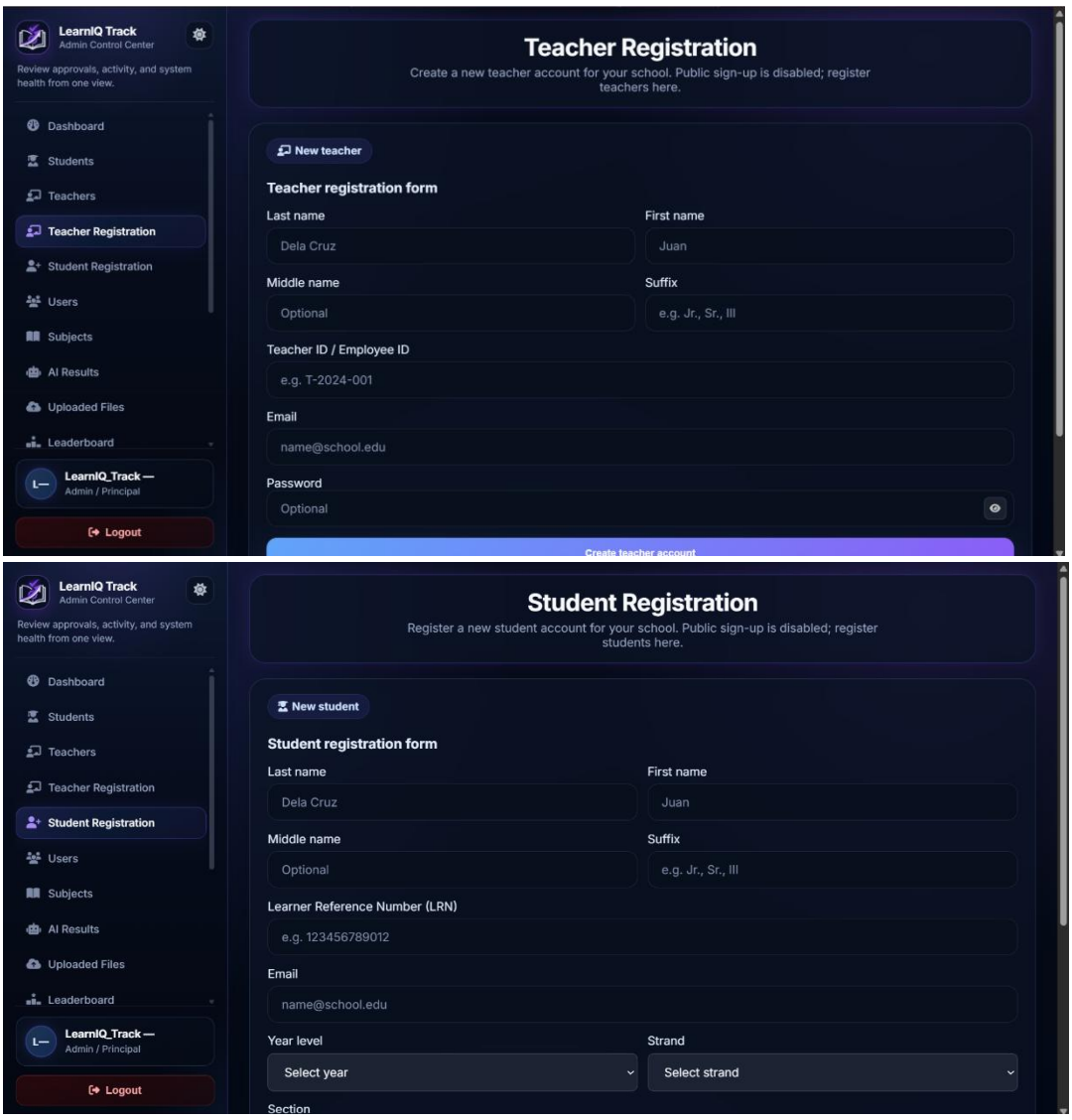


Figure A.13. Teacher Registration and Student Registration pages.

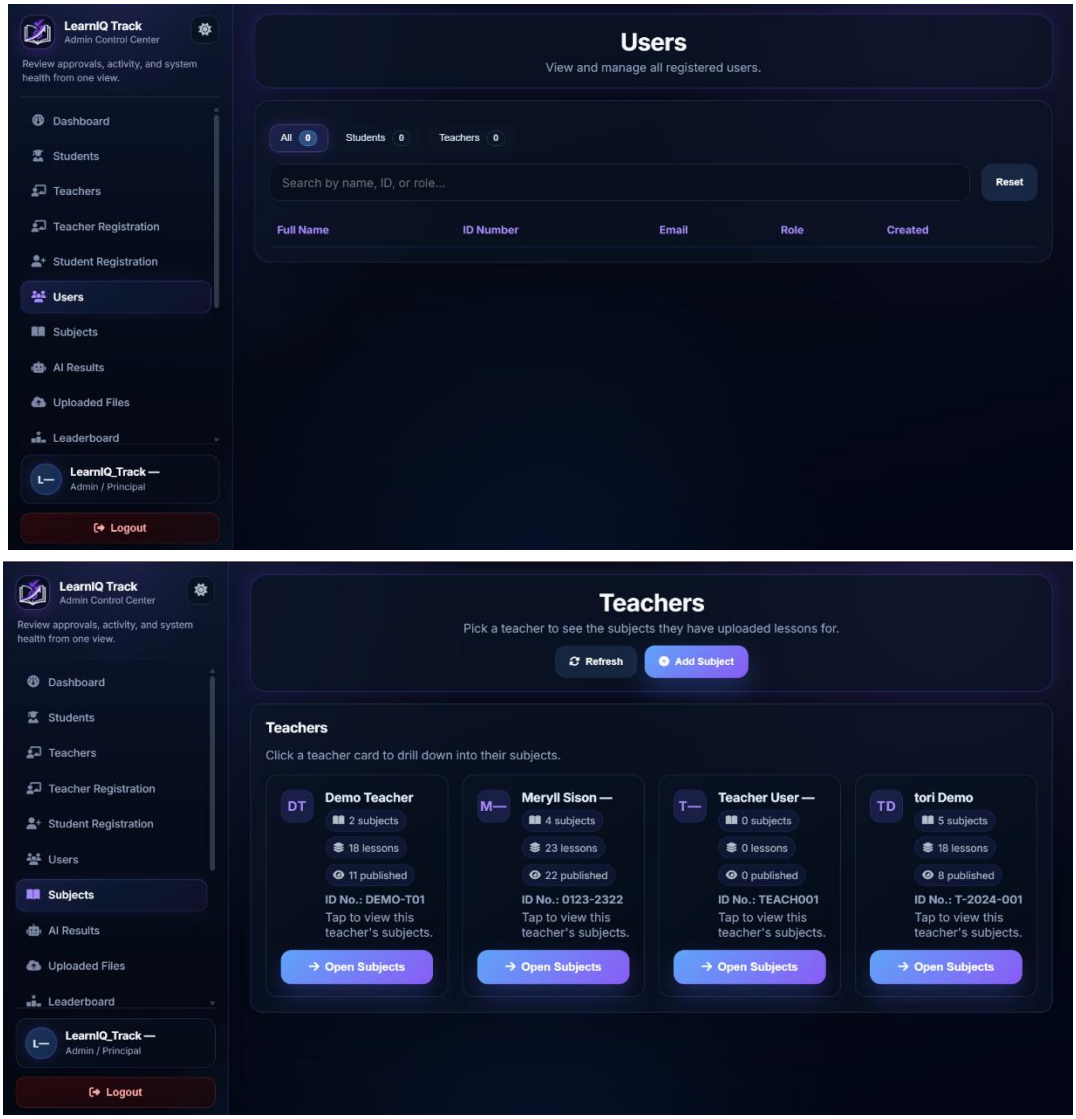
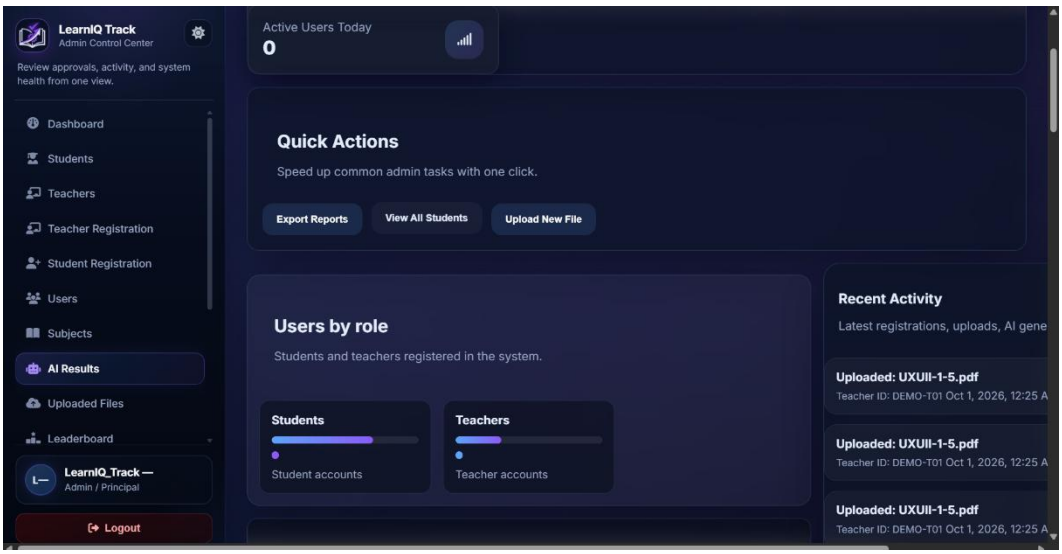


Figure A.14. Users and Subjects pages.



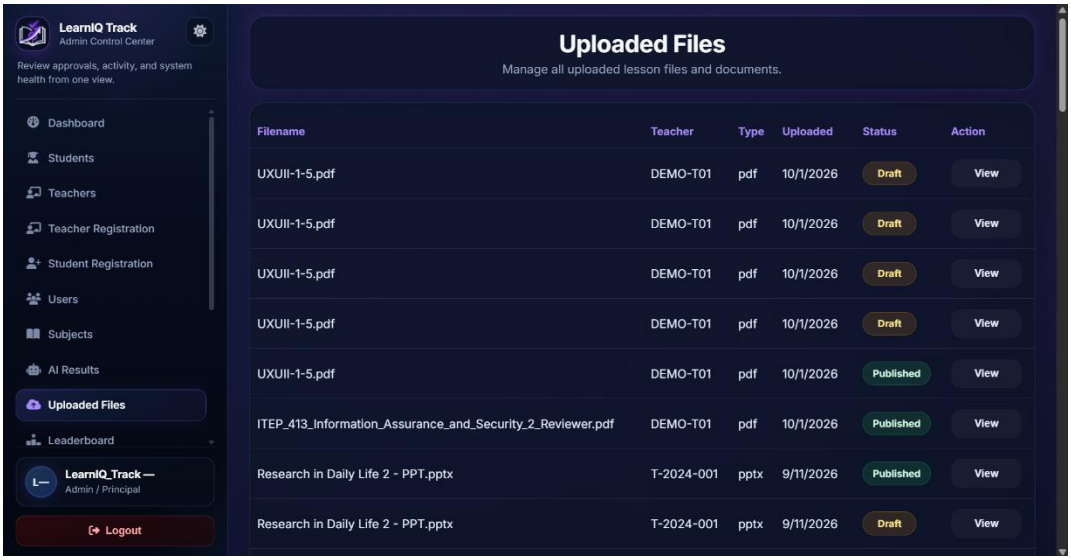


Figure A.15. AI Results and Uploaded files pages.

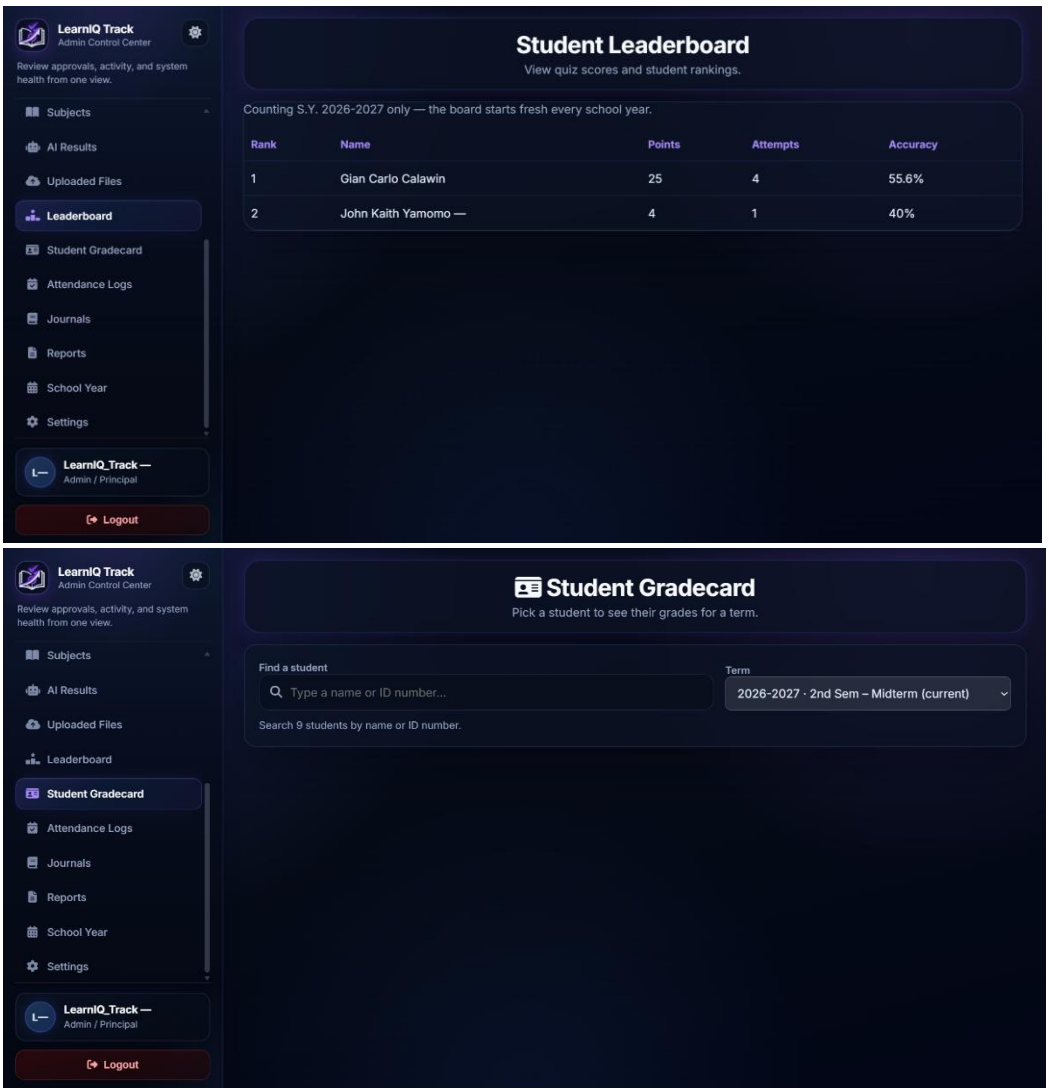


Figure A.16. Leaderboard and Student Gradecard pages.

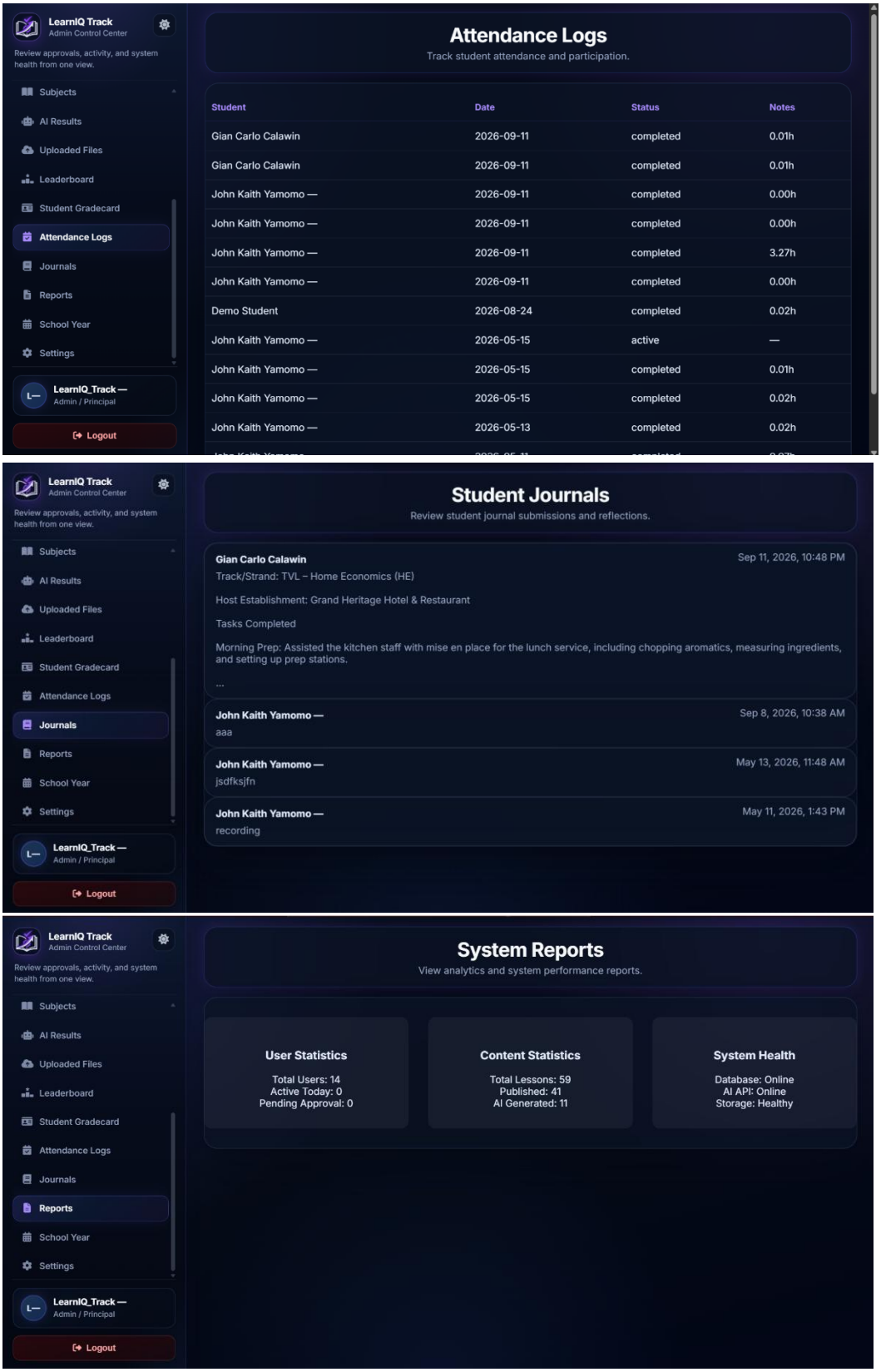


Figure A.17. Attendance Logs, Journals, and Reports pages.

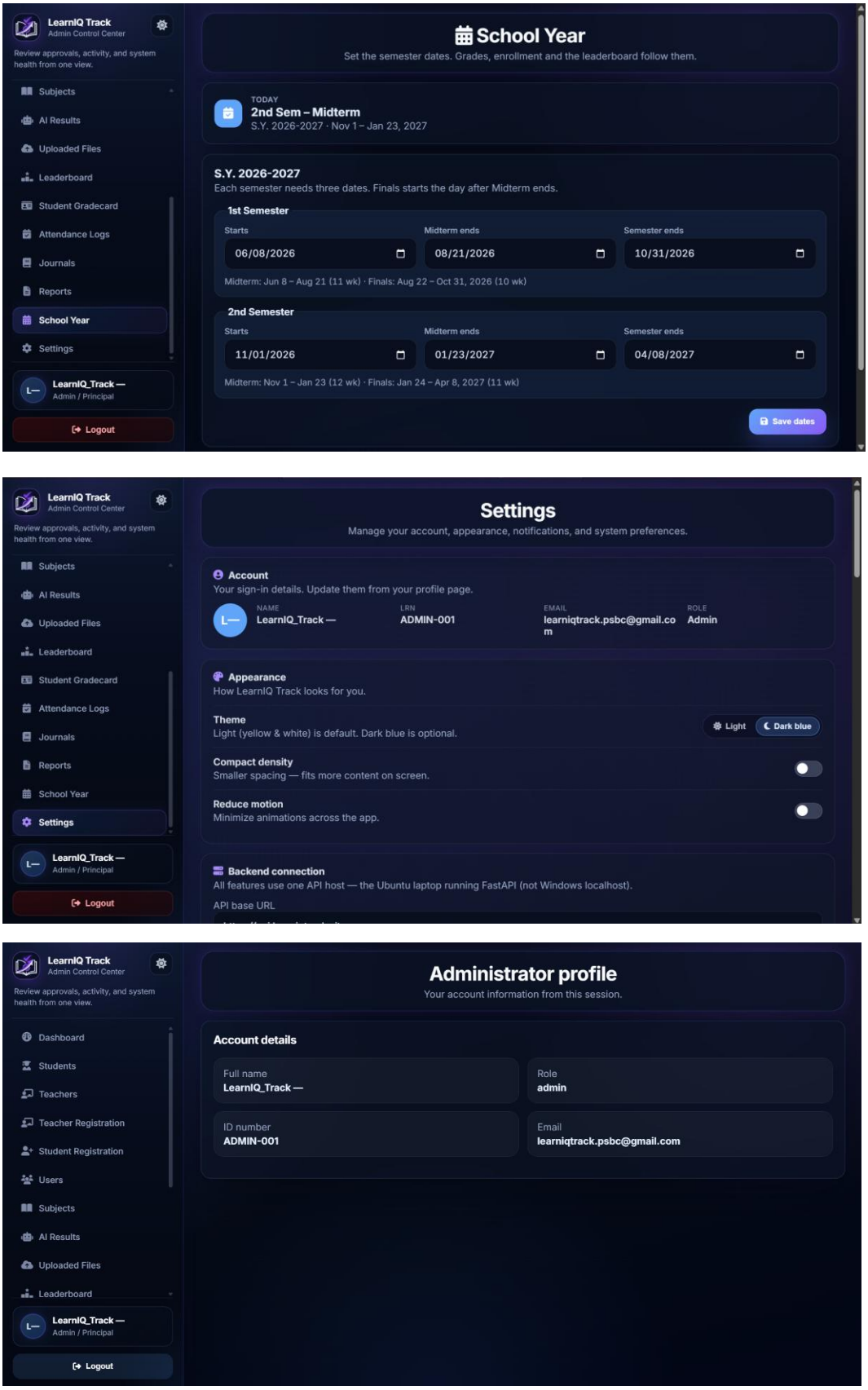


Figure A.18. School Year, Settings. And Profile pages.

Current Process Flowchart

A. Current Academic Learning Process

This shows the current process followed by Paete Science and Business College Inc. in conducting academic lessons and providing learning materials to Senior High School students. Teachers provide lessons and modules, which students use as references for studying and completing their learning activities.

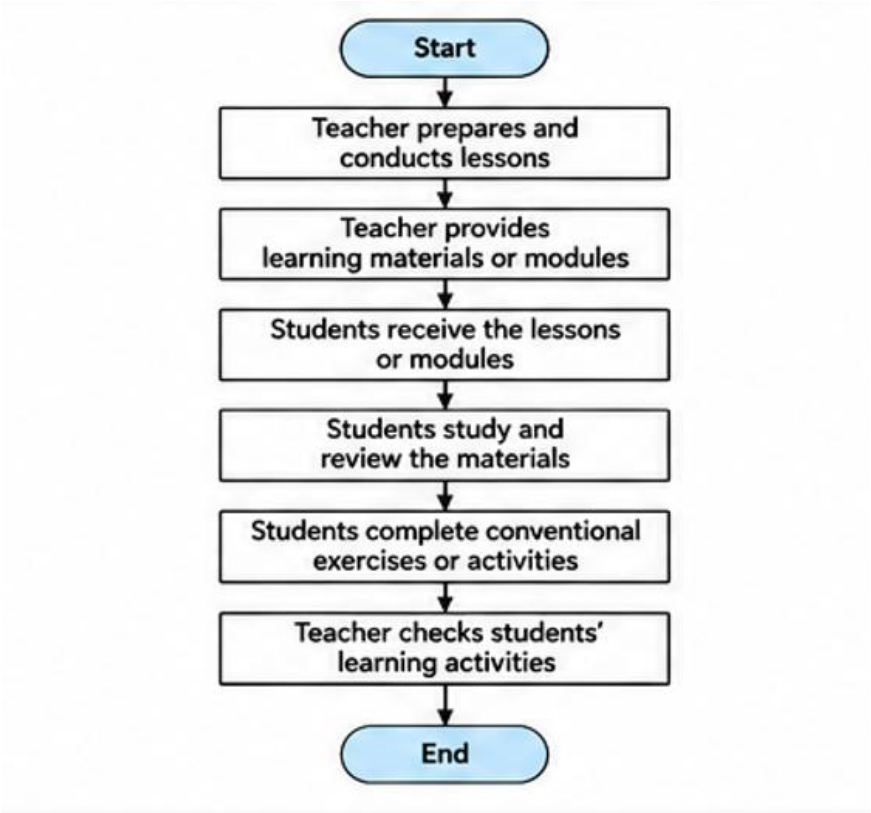


Figure A.19. This figure shows the current process flowchart for academic learning and lesson review at Paete Science and Business College Inc.

B. Current Work Immersion Monitoring Process

This shows the current process followed by the school in monitoring the work immersion activities of Grade 12 students. Students record their attendance using paper-

based attendance sheets and submit their daily journals through handwritten or printed documents. The teacher assigned to work immersion checks and monitors the submitted records.

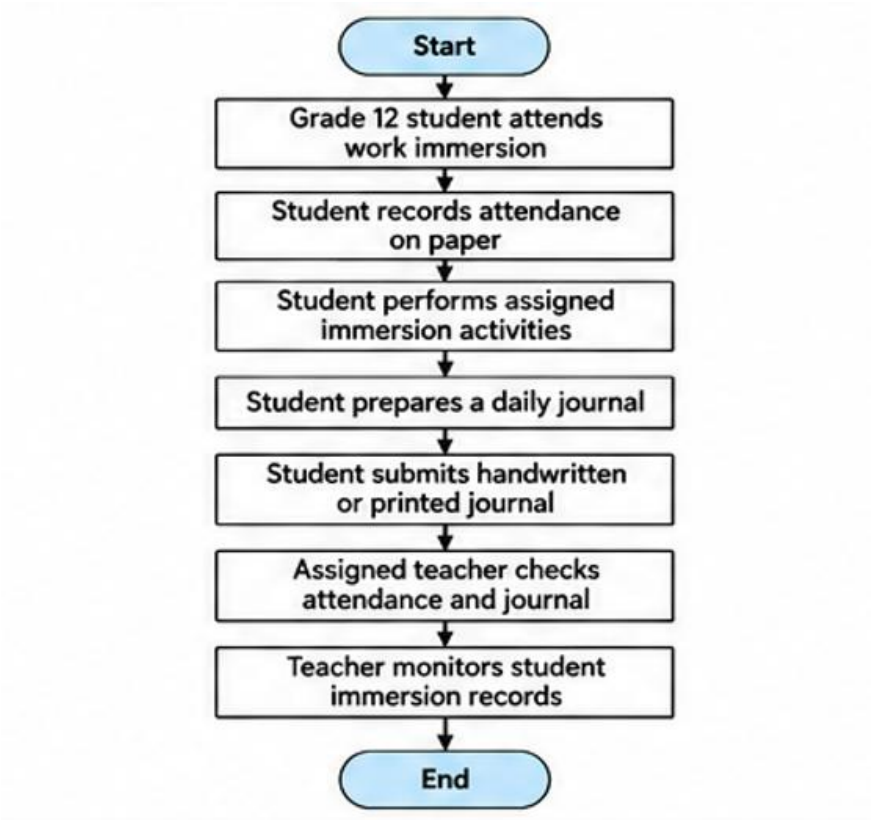


Figure A.20. This figure shows the current process flowchart for work immersion attendance and daily journal monitoring at Paete Science and Business College Inc.

Proposed Process Flowchart

A. Proposed Process Flowchart – Student

This shows the proposed process flowchart for students using LearnIQ Track. Students access the system using their approved accounts and select the module they wish to use. Through the LearnIQ module, they can access learning materials, answer AI-generated quizzes and activities, and view their scores and leaderboard rankings. Grade 12 students can also use the Immersion Tracker module to record their attendance through QR code scanning, submit daily journals, and access their immersion records.

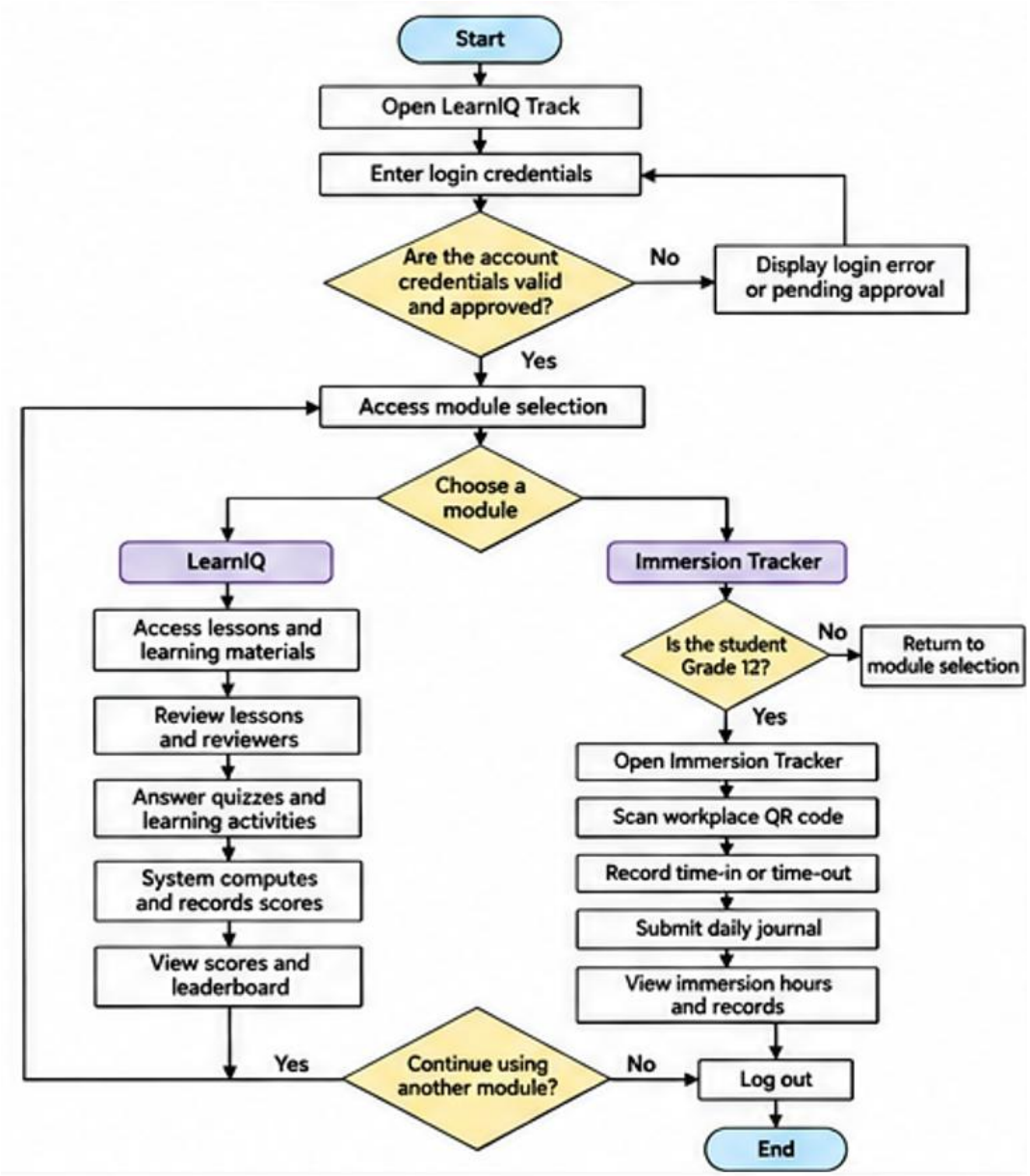


Figure A.21. This figure shows the proposed process flowchart for students using the LearnIQ Track web application.

B. Proposed Process Flowchart – Teacher

This shows the proposed process flowchart for teachers using LearnIQ Track. Teachers log in to their accounts to access the dashboard, upload learning materials, and generate AI-assisted quizzes, activities, and reviewers using Google Gemini AI.

The generated learning content is reviewed and validated by teachers before being made available to students. Teachers can also monitor student quiz performance, attendance records, and daily journal submissions through the system.

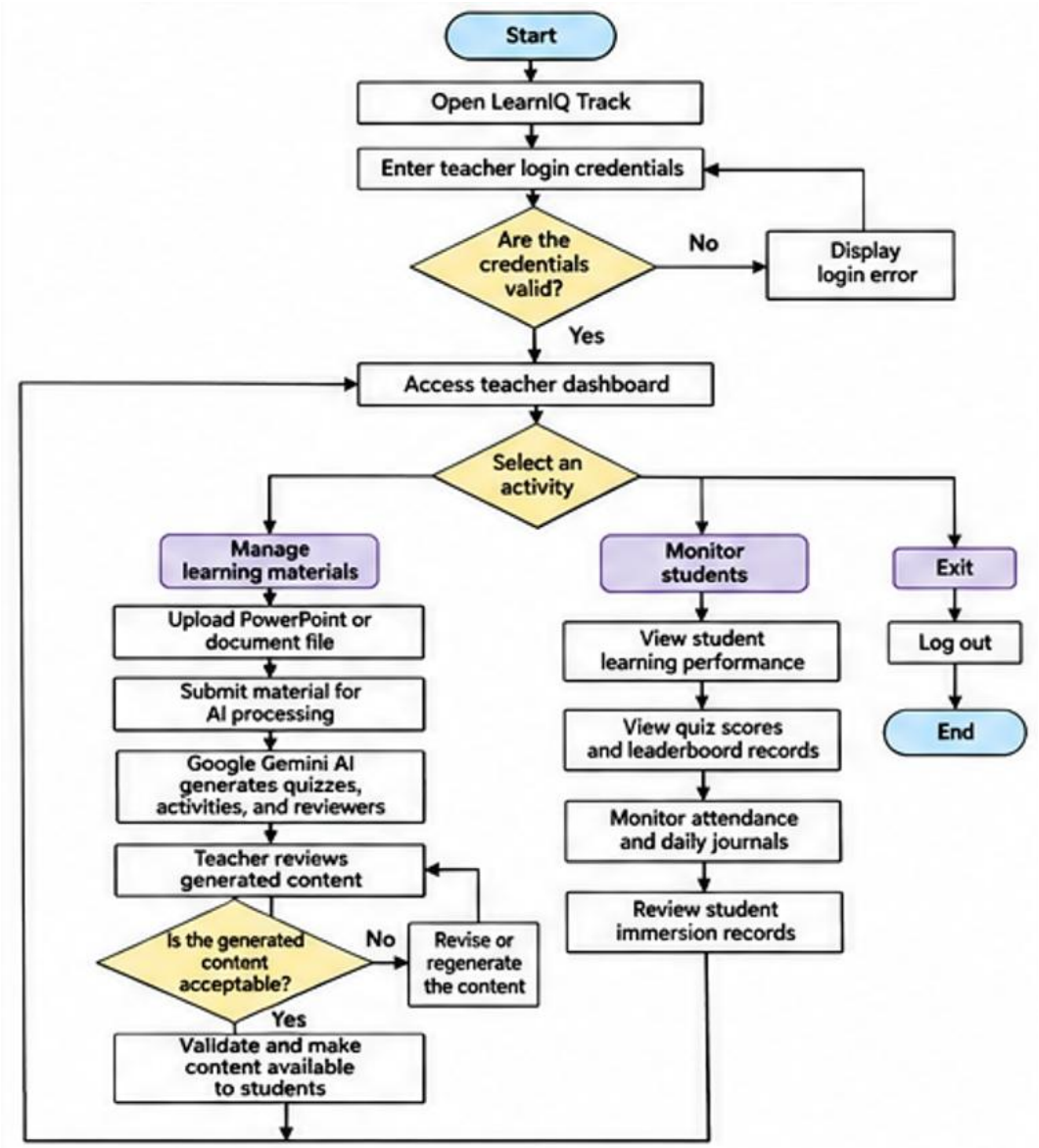


Figure A.22. This figure shows the proposed process flowchart for teachers using the LearnIQ Track web application.

C. Proposed Process Flowchart – Admin

This shows the proposed process flowchart for the Administrator of LearnIQ Track. The Administrator, represented by the school principal, accesses the system through an authorized account to manage student and teacher accounts, view system records, monitor available reports, and oversee the overall operation of the platform.

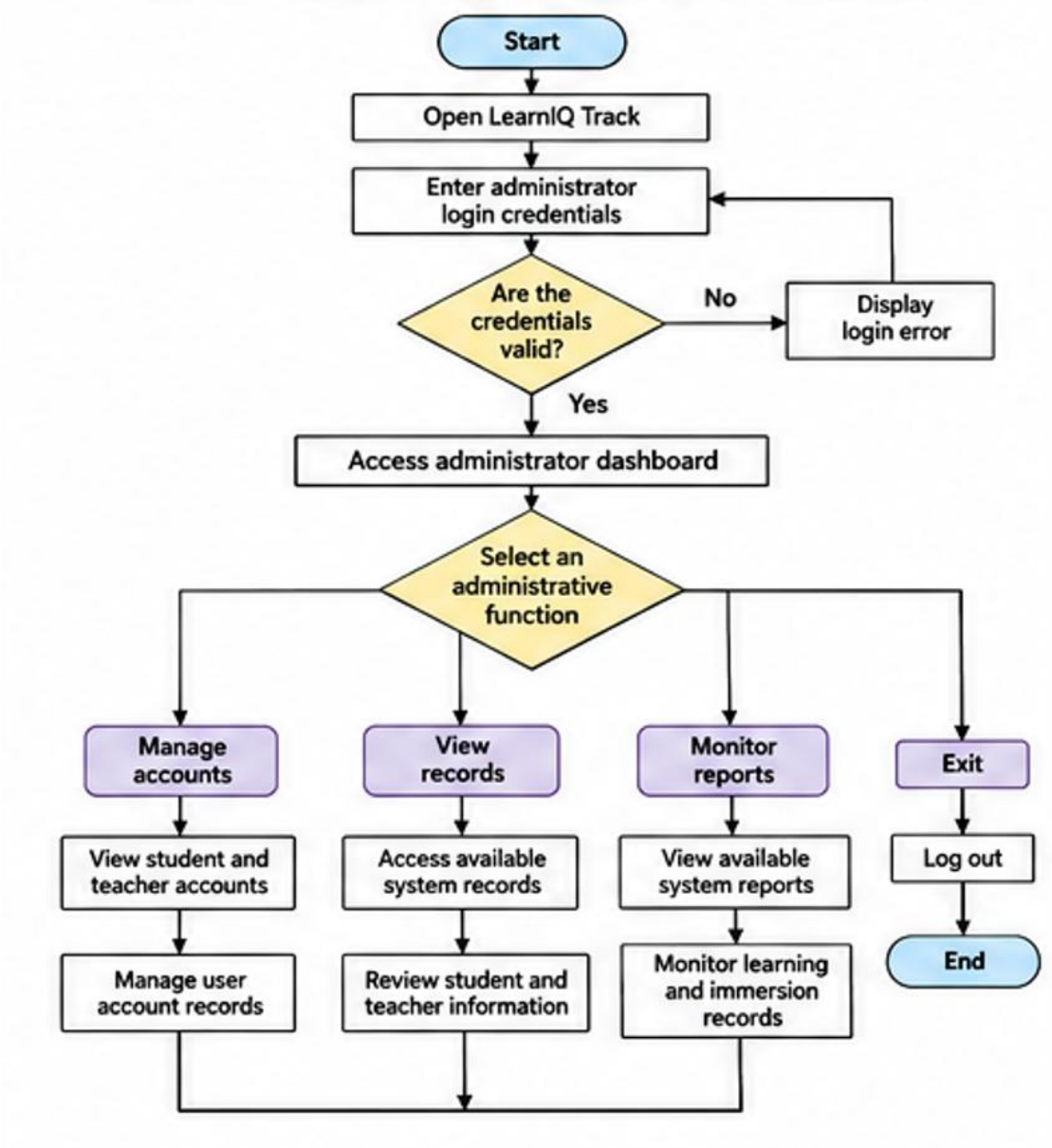


Figure A.23. This figure shows the proposed process flowchart for the Administrator of the LearnIQ Track web application.

PROJECT SCHEDULE

This section displays the research study's project schedule. In order to complete this research project, the researchers went through several stages.

Task	Start	End	Jan	Feb	Mar	Apr	May	Jun
Formulation of title	**1/27/26**	**2/16/26**	■	■				
Initial Client Interview	**2/17/26**	**2/27/26**		■				
Formulation of Proposal	**3/1/26**	**3/14/26**			■			
Title Oral Defense	**3/16/26**	**3/16/26**			■			
System Development (60-80%)	**3/17/26**	**5/16/26**			■	■	■	
Writing of Manuscript (Chapters 1-3)	**3/17/26**	**5/16/26**			■	■	■	
Research Colloquium	**5/20/26**	**5/20/26**					■	
			Jul	**Aug**	**Sept**	**Oct**	**Nov**	**Dec**
System Development (100%)	**8/2/26**	**10/29/26**		■	■	■		
Writing of Manuscript (Chapters 4-5)	**8/2/26**	**10/29/26**		■	■	■		
Final Oral Defense	**11/16/26**	**11/20/26**					■	
Book Binding of Final Manuscript	**11/23/26**	**11/27/26**					■	

Figure A.24. Project Schedule Diagram

APPENDICES B

(COMMUNICATION LETTERS AND FORMS)

REQUEST LETTER

(INITIAL INTERVIEW WITH CLIENT)

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna
COLLEGE OF COMPUTER STUDIES

MARIA GEMMA B. CALMA
PSBC Principal
Paete Science and Business College INC
Brgy. Poblacion Dos, Pagsanjan, Laguna

SUBJECT: Request for Initial Interview for Capstone Project System Development

Dear Maria Gemma B. Calma,

Warm greetings!

We are third-year students of the Bachelor of Science in Information Technology under the College of Computer Studies. As part of the academic requirements for our Capstone Project, we respectfully request permission from your good office to conduct an initial interview and requirements gathering with your organization.

The purpose of this interview is to understand your current system or processes and to identify possible areas where an information system may be designed and developed to help improve operational efficiency. The information gathered will be used strictly for academic purposes only and will be treated with utmost confidentiality.

Should you grant our request, the interview may be conducted at your most convenient date and time, either face-to-face or through an online platform.

We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time and cooperation.

Respectfully yours,

De La Cruz, Aaron Karl U.
BS Information Technology - 3E

De Leon, Francine Ann C.
BS Information Technology - 3E

Yamomo, John Kaith L.
BS Information Technology - 3E

Maria Gemma B. Calma
PSBC Principal II

Noted by:

Maria Laureen B. Miranda, LPT, MIT
Capstone Project Subject Instructor
College of Computer Studies

TITLE PROPOSAL

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Proposal

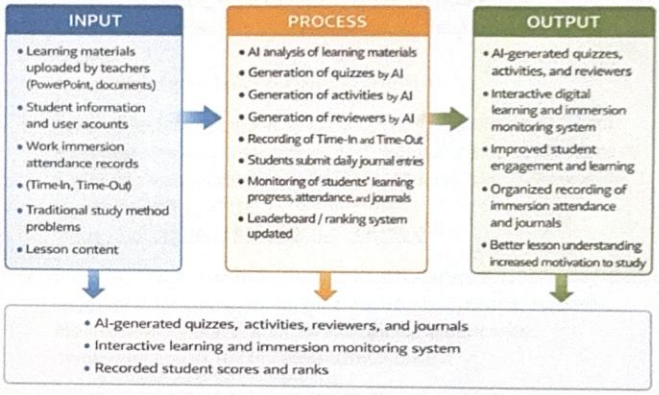
Proponent/Researcher:	**EW10**
Project Title: LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc.	
Introduction	Learning and work immersion are important parts of the Senior High School (SHS) program because they help students develop knowledge, skills, and real-world experience. However, many students still experience difficulties in studying their lessons using traditional methods, which can sometimes make learning boring and less engaging. At the same time, some schools still use manual methods such as paper in recording attendance and daily journals during work immersion, which can sometimes lead to lost records or difficulty in monitoring student activities. With the development of modern technology, schools can now use digital systems to improve both learning and student monitoring. Artificial Intelligence (AI) is now being used in educational systems to help students study more effectively by generating quizzes, activities, and reviewers from learning materials. Digital systems can also help teachers and immersion coordinators monitor students' attendance and activities more efficiently. Because of this, the researchers propose LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The system will allow teachers to upload learning materials such as PowerPoint and document files, which will then be analyzed by AI to generate quizzes, activities, and reviewers that students can use for studying. The system will also include a ranking or leaderboard feature to motivate students to improve their performance. In addition, the system will include a digital platform where students can record their Time-In and Time-Out during their work immersion and submit their daily journal entries. Teachers and immersion coordinators will be able to monitor student attendance, journals, and learning progress through a web-based dashboard. The main goal of the system is to make learning more interactive and engaging while also making the monitoring of work immersion activities easier, more organized, and more efficient for both students and teachers.
Statement of the Problem	This study aims to develop LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. Many students still rely on traditional study methods, which can sometimes make learning less interactive and engaging. In addition, some schools still use manual methods such as paper in

Form 2

	recording attendance and daily journals during work immersion, which may lead to lost or incomplete records. Specifically, it seeks to answer the following questions: 1. How can LearnIQ Track be designed and developed to help students review their lessons using AI-generated quizzes, activities, and reviewers? 2. How can the system allow teachers to upload learning materials that can be analyzed by AI to generate learning content? 3. How can the system record students' Time-In and Time-Out during their work immersion? 4. How can the system allow students to submit their daily journals or reflections about their immersion activities? 5. How can the system help teachers and immersion coordinators monitor students' learning progress, attendance, and journal entries? 6. How can the system make learning and immersion monitoring more organized, interactive, and efficient?
Objective of the Study	The main objective of this study is to develop LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. that will help students review their lessons and record their work immersion activities using a digital platform. Specifically, this study aims to: 1. To design and develop an AI-based learning system that can generate quizzes, activities, and reviewers from uploaded learning materials. 2. To allow teachers to upload lesson materials such as PowerPoint and document files into the system. 3. To develop a leaderboard or ranking system that records student scores and performance. 4. To create a digital attendance system that records students' Time-In and Time-Out during work immersion. 5. To allow students to submit their daily journals or reflections about their immersion activities. 6. To provide a platform where teachers and immersion coordinators can monitor students' learning progress, attendance, and journal submissions.
Scope and Limitation of the Study	This study focuses on the development of LearnIQ Track, an AI-based learning game and digital work immersion monitoring system for Senior High School students of Paete Science and Business College Inc. The system will allow teachers to upload learning materials such as PowerPoint presentations and document files. The AI will analyze these materials and automatically generate quizzes, activities, and reviewers that students can use for studying. The system will also include features for recording students' Time-In and Time-Out during their work immersion and for submitting daily journal entries about their immersion activities. Teachers and immersion coordinators will be

	able to monitor student attendance, journal entries, and learning progress through a web-based dashboard. However, the system has some limitations. The AI-generated questions and reviewers will depend on the quality of the uploaded learning materials and may still need checking by teachers. The system also requires an internet connection to access its features. In addition, the system will only be used by Senior High School students and authorized personnel of PSBC Pagsanjan Branch and will only support selected file formats such as PowerPoint and document files.
Review of Literatures and Related Systems	The researchers collected the information that are greatly related to the study. These related studies and literature are compiled in this section as a basis in the development of the system. Related studies and projects are listed below in thematic order: **Theme 1: Artificial Intelligence in Education** Artificial Intelligence (AI) is becoming an important technology in education because it can help improve how students learn. AI systems can analyze learning materials and student performance to provide personalized learning experiences. According to Holmes et al. (2022), AI technologies can support teaching and learning by providing automated feedback, intelligent tutoring systems, and adaptive learning tools. These technologies help students understand lessons more effectively and allow teachers to manage learning activities more efficiently. AI can also assist in creating learning materials automatically. Zawacki-Richter et al. (2020) explained that AI applications in education can analyze data from learning systems to improve instructional design and support students' academic performance. **Theme 2: Automated Quiz Generation** Automated question generation is one of the useful applications of AI in educational technology. It allows systems to automatically create quizzes or test questions from learning materials. According to Kurdi et al. (2021), automatic question generation uses natural language processing techniques to produce questions from text-based materials. This helps reduce the workload of teachers and allows students to practice answering questions more frequently. Similarly, Liu et al. (2023) explained that deep learning techniques can be used to generate multiple-choice questions from educational texts. These systems can analyze lesson content and create relevant questions that help students review their knowledge. **Theme 3: Game-Based Learning and Student Engagement** Game-based learning is widely used to make learning more engaging and interactive for students. Educational systems that include game elements such as points, leaderboards, and rewards can motivate students to participate more actively in learning activities. According to Dichev and Dicheva (2020), gamification in education helps increase student motivation and engagement when game elements are integrated into learning platforms.

	In addition, research by Al-Adwan et al. (2022) showed that gamified learning systems improve students' participation and help them develop better learning outcomes. These systems encourage students to study while enjoying the learning process. **Theme 4: Web-Based Learning Systems (Online Learning Platforms)** Web-based learning systems allow students to access educational materials anytime and anywhere using the internet. These systems support flexible learning and allow students to continue studying even outside the classroom. According to Dhawan (2020), online learning platforms became very important during emergencies such as school closures because they allowed students and teachers to continue education through digital tools. Web-based platforms also support independent learning by giving students access to lessons, quizzes, and other educational resources. These systems help improve learning accessibility and provide opportunities for students to study at their own pace. **Theme 5: Work Immersion Programs in Senior High School** Work immersion programs are designed to help students gain real-world work experience and develop practical skills related to their field of study. These programs allow students to apply the knowledge they learned in school to actual workplace situations. According to Meriam M. Alawin-Ligwang (2025), work immersion helps improve students' technical skills, workplace behavior, and readiness for employment. The study found that students who participate in immersion programs become more confident in performing real work tasks and communicating in professional environments. Similarly, Democrito C. Cabile Jr. (2024) evaluated the implementation of work immersion programs and found that proper supervision and monitoring of students during immersion help improve their performance and learning experience. The study emphasized that structured immersion activities allow students to gain meaningful workplace knowledge and improve their motivation to complete tasks effectively. In another study, Jocelyn J. Pestijo (2023) explained that work immersion helps students develop responsibility, discipline, and communication skills in real workplace settings. The research also highlighted that proper monitoring and documentation of student activities during immersion are important to ensure that students complete their required training hours and learning objectives. **Theme 6: Digital Internship and Work Immersion Monitoring Systems** With the help of modern technology, many schools now use digital systems to monitor internship and work immersion activities. These systems allow students to record their attendance and submit reports online while teachers and supervisors can easily track their progress. According to Kelfin Nurfaizi and Djarot Hindarto (2023), web-based internship attendance systems improve the accuracy of attendance records and reduce manual errors compared to traditional paper-based methods. The study

	also explained that digital monitoring platforms help administrators manage student activities more efficiently. Similarly, Rica A. Barrocan et al. (2025) developed a web-based internship monitoring system that allows supervisors to track student attendance and progress in real time. Their findings showed that digital monitoring systems help improve organization, transparency, and efficiency in internship programs. Another study by Thanakorn Sangkudluo (2023) also emphasized that online monitoring platforms help educators easily supervise student internship activities and improve communication between students and supervisors.
Conceptual Framework	 **Figure 1.** Conceptual Framework of **LearnIQ Track:** An AI-Based Learning Game and Digital Work Immersion Monitoring System **Figure 1.** Conceptual Framework of LearnIQ Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Paete Science and Business College Inc. The conceptual framework of the study illustrates how the research will be conducted. The framework reflects the research objectives and activities to be conducted
SWOT ANALYSIS	**Strengths** • Uses Artificial Intelligence (AI) to generate quizzes, activities, and reviewers. • Helps students study in a **Weaknesses** • The system depends on internet connection. • AI-generated content may still need checking by teachers. **Opportunities** • The system can be expanded and used by other schools. • Additional features such as reports and **Threats** • Technical issues such as system bugs or server downtime may affect the system. • Internet connectivit

	more interactive and engaging way. • Allows digital recording of work immersion attendance and journals. • Helps teachers and coordinators easily monitor students.	• Some users may not be familiar with digital systems.	analytics can be added in the future. • The system can help schools improve their digital learning methods.	y problems may limit system usage. • Some users may prefer traditional methods instead of using technology.
References	1. Holmes, W., Bialik, M., & Fadel, C. (2022). Artificial intelligence in education: Promise and implications for teaching and learning. *AI & Society*. https://doi.org/10.1007/s00146-021-01205-4 2. Zawacki-Richter, O., Marín, V. I., Bond, M., & Gouverneur, F. (2020). Systematic review of research on artificial intelligence applications in higher education. *International Journal of Educational Technology in Higher Education, 17*(1). https://doi.org/10.1186/s41239-019-0171-0 3. Kurdi, G., Leo, J., Parsia, B., Sattler, U., & Al-Emari, S. (2021). A systematic review of automatic question generation for educational purposes. *International Journal of Artificial Intelligence in Education*. https://doi.org/10.1007/s40593-020-00224-0 4. Liu, M., Huang, S., & Zhang, Y. (2023). Deep learning for automatic question generation in educational systems. *Computers & Education: Artificial Intelligence, 4*. https://doi.org/10.1016/j.caeai.2022.100115 5. Dichev, C., & Dicheva, D. (2020). Gamifying education: What is known and what is next? *Educational Technology & Society, 23*(3), 75–88. https://www.jstor.org/stable/26981756 6. Al-Adwan, A. S., Al-Adwan, A., & Smedley, J. (2022). Exploring students' acceptance of e-learning systems in higher education. *Education and Information Technologies*. https://doi.org/10.1007/s10639-021-10761-7 7. Dhawan, S. (2020). Online learning: A panacea in the time of COVID-19 crisis. *Educational Technology Research and Development, 68*(5), 2385–2402. https://doi.org/10.1007/s11423-020-09806-z 8. Alawin-Ligwang, M. M. (2025). Work immersion performance of senior high school students: A foundation for enhancing implementation. *Journal of Education, Society and Behavioural Science*. https://doi.org/10.9734/jesbs/2025/v38i41403			

9. Anajao, L. J. L., Intano, J. A., & Sumaoy, R. C. (2025). Industry feedback on work immersion: A qualitative exploration. *International Journal of Research and Innovation in Social Science*. https://rsisinternational.org/journals/ijriss/articles/industry-feedback-on-work-immersion-a-qualitative-exploration/

10. Barrocan, R. A., Calzo, M. J. R., Carambas, C. B., Tugade, M. J. E., & Bartolome, M. B. (2025). An architectural approach of a web-based monitoring system for efficient internship time tracking. *International Journal of Research and Innovation in Applied Science*. https://rsisinternational.org/journals/ijrias/articles/an-architectural-approach-of-a-web-based-monitoring-system-for-efficient-internship-time-tracking/

11. Cabile Jr., D. C. (2024). Implementation of work immersion in the new normal: Basis for students' quality performance and satisfaction. *International Journal of Multidisciplinary: Applied Business and Education Research*. https://www.babmrjournal.org/index.php/ijmaber/article/view/1587

12. Nurfaizi, K., & Hindarto, D. (2023). Web-based student internship attendance application system for effective student attendance monitoring. *International Journal Software Engineering and Computer Science*. https://journal.lembagakita.org/index.php/ijsecs/article/view/1760

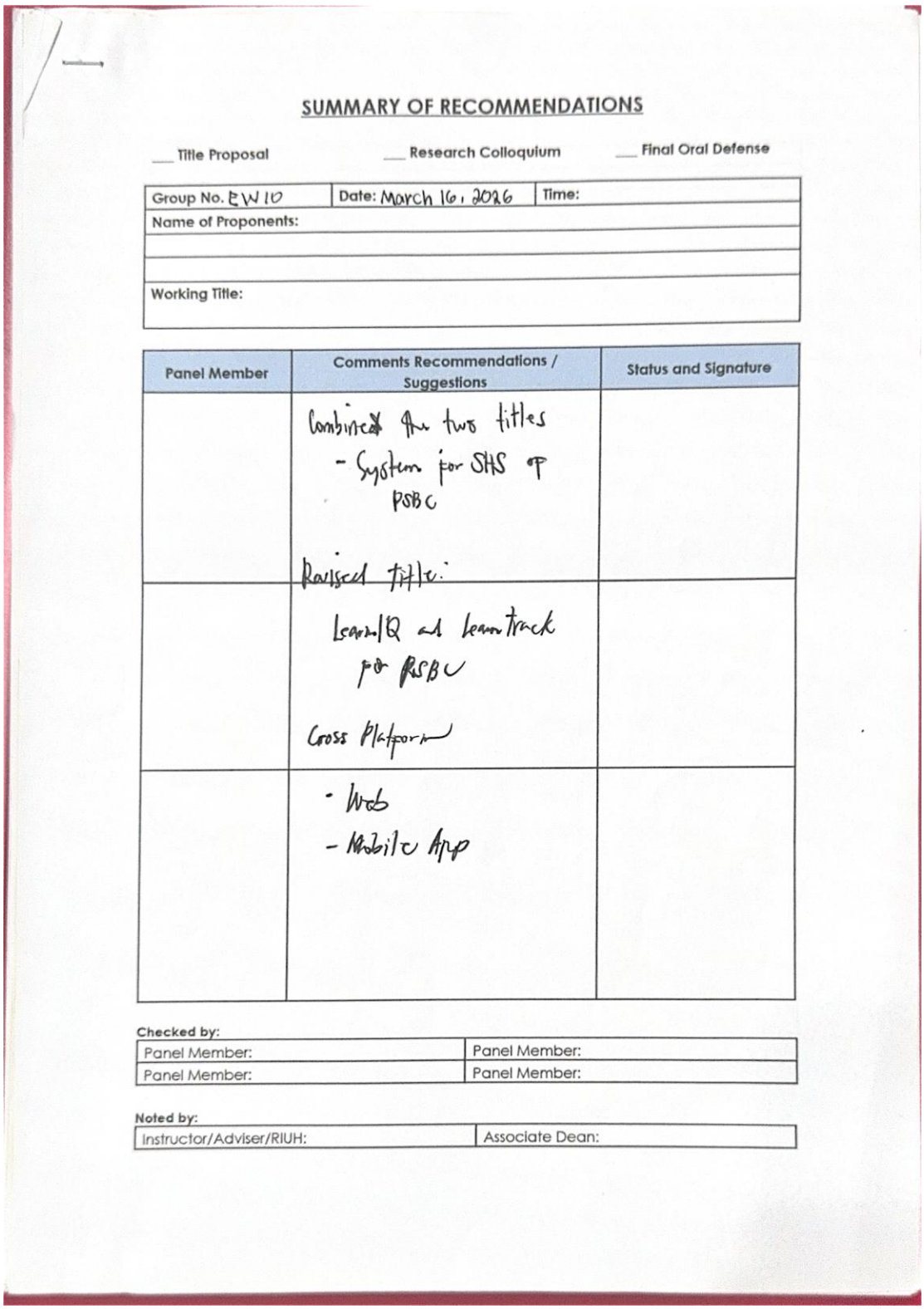
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14. Sangkudluo, T. (2023). Development of supervision monitoring internship system in educational institutions. *Journal of Library and Information Science Srinakharinwirot University*. https://so02.tci-thaijo.org/index.php/JLIS_SWU/article/view/265901

Prepared by:
De La Cruz, Aaron Karl U.
De Leon, Francine Ann C.
Yamomo, John Kaith L.

SUMMARY OF RECOMMENDATIONS

(TITLE PROPOSAL)



SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. BW 10	Date: March 16, 2026	Time:
Name of Proponents:		
Working Title:		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	Combined the two titles - System for SHS of PSBC Revised title: Learn IQ ad Learn track for PSBC Cross Platform	
	- Web - Mobile App	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EW 10	Date: March 16, 2026	Time:
Name of Proponents:		
Working Title: Learn IQ & Immersion Track PSBC		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	PDF/PPT → 1) Quiz generator (all types) 2) Game based Activities 3) Flash Cards 4) Feed back & explanation after 7 Multi-part uploads	
	AI - Powered Course Structuring - Adaptive & Personalized learning - LMS	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

MIA V. ULLA, Ph.D.

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EW10	Date: March 16, 2026	Time:
Name of Proponents:		
Working Title: PSBC LEARNIQ W/ OJT TRACKING.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	– COVERAGE: GRADE 11, ALL STRANDS – CONSIDER IF APPLICABLE, THE SCHOOL YEAR. – SHOULD BE APPLICABLE TO ALL BRANCHES OF THE SCHOOL (PSBC) – INCLUDE THE OJT IMMERSION MONITORING.	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

RATING SHEET

(TITLE DEFENSE)

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW 10	Date: March 16, 2026	Time:
Name of Proponents:		
Research Title:		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	87

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW10	Date: March 16, 2026 Time:
Name of Proponents:	
Research Title:	

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	89
		Total	

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW 10	Date: March 16, 2026	Time:
Name of Proponents:		
Research Title:		

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	82

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Form 3

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Title Defense Rating Sheet

Group No. EW10	Date: March 16, 2026 Time:
Name of Proponents:	
Research Title:	

CRITERIA	DESCRIPTION	WEIGHT	SCORE
TOPIC ALIGNMENT	The proposed title is timely and aligned to LSPU/ CCS Research Agenda Thrust and Priority	20%	
NOVELTY	Had checked the novelty of the proposed topic, identified the gap in the existing literature and /or introduced new ideas	20%	
PURPOSE	The purpose of the proposed capstone/thesis topic is clearly stated and properly identified	20%	
REVIEW OF RELATED LITERATURES AND SYSTEM	Provided sufficient reviews of related studies and system relative to the proposed topic	20%	
FEASIBILITY	The proposed topic is reliable, practical and feasible	20%	
		Total	83

VERDICT

[ ] **APPROVED.** Minor revisions are necessary, but they do not have to be presented in front of and checked by all panelists. **86 – 100**

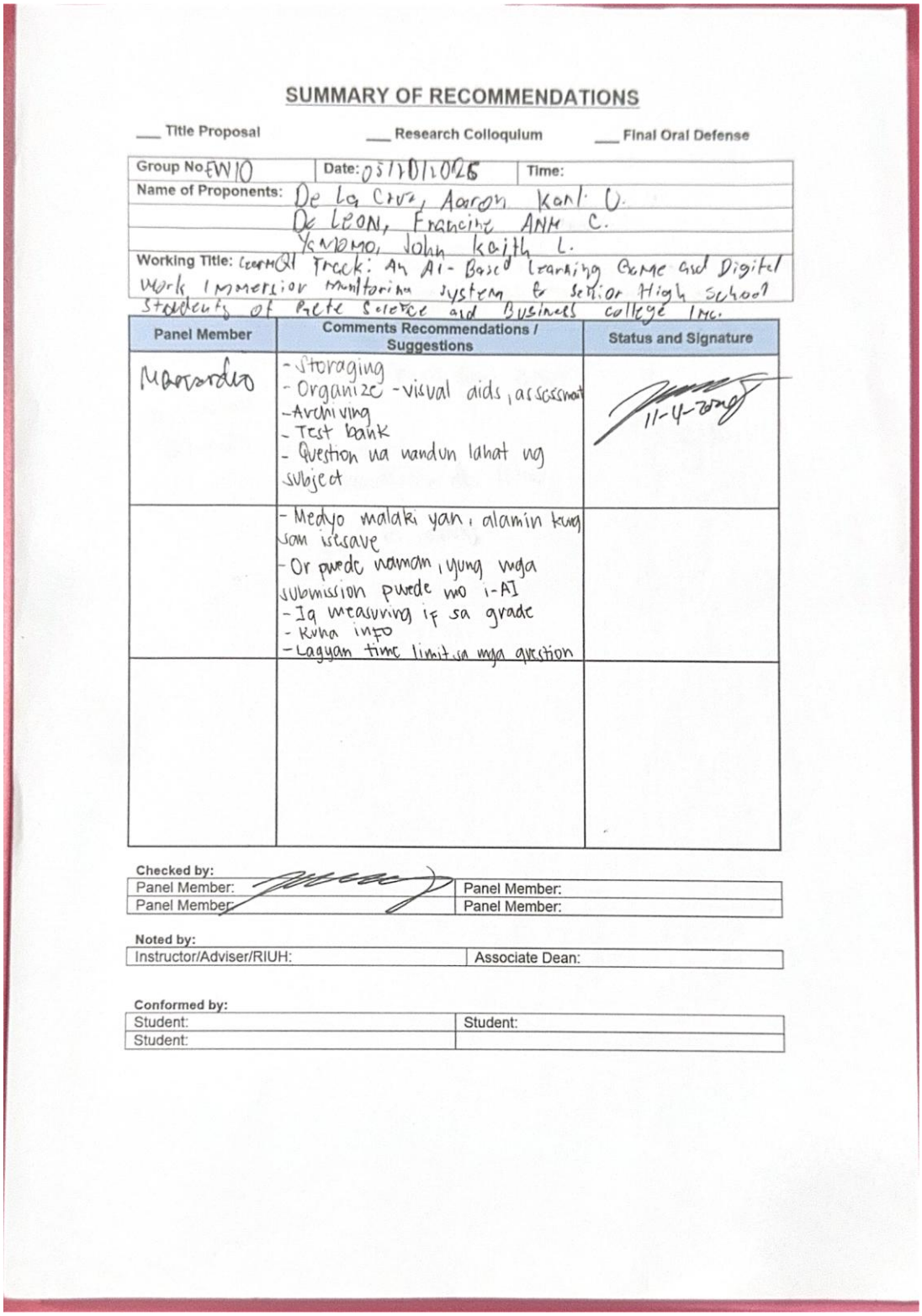
[ ] **APPROVED WITH REVISIONS.** Major revisions shall be incorporated in the final copy of the revised Project Proposal summary. These must be checked by the panelists. **75 – 85**

[ ] **DISAPPROVED.** The Proponents/Researchers failed to propose a researchable or scholarly Thesis / Capstone Project. **Below 70**

Form 3

SUMMARY OF RECOMMENDATIONS

(RESEARCH COLLOQUIUM)



SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EW10	Date: 05/10/2025	Time:
Name of Proponents:	De La Cruz, Aaron Karl U. De LEON, Francine ANN C. Ysmano, John Keith L.	
Working Title: Learning Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Pete Science and Business college Inc.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Maorsdus	- storaging - Organize - visual aids, assessment - Archiving - Test bank - Question na nandun lahat ng subject	11-4-2025
	- Medyo malaki yan, alamin kung san isesave - Or pwede naman, yung mga submission pwede mo i-AI - Iq measuring if sa grade - Kuha info - Lagyan time limit sa mga question	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EW10	Date: 05/10/2025	Time:
Name of Proponents:	De La Cruz, Aaron Karl U. De Leon, Francine Anh C. Ysarmo, John Kaith L.	
Working Title: Capstone Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Pete Science and Business College Inc.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Maam Sadia	- storaging - Organize - visual aids, assessment - Archiving - Test bank - Question na nandun lahat ng subject	11-4-2025
	- Medyo malaki yan, alamin kung san isesave - Or pwede naman, yung mga submission pwede mo i-AI - Ig measuring if sa grade - Kuha info - Lagyan time limit sa mga question	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EWP	Date: 06/20/2026	Time:
Name of Proponents:	De la Cruz, Aaron Karl U. De Leon, Francine Ann C. Ysmaine, John Keith L.	
Working Title:	Learning Track: An AI-Based Learning and Digital Work Immersion Monitoring System for ~~Senior~~ High School Students of Pasig and Business College Inc.	

Panel Member	Comments Recommendations / Suggestions	Status and Signature
Reynaldo Ignacio (Chairman)	1. Improve the AI Game Based learning 2. Accuracy 3. All materials from the school	
	should be included.	

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

SUMMARY OF RECOMMENDATIONS

___ Title Proposal ___ Research Colloquium ___ Final Oral Defense

Group No. EW10	Date: 05/20/2026	Time:
Name of Proponents:	De La Cruz, Aaron Karl V. De Leon, Francine Ann C. Yamoma, John Isgith L.	
Working Title: Learn/Q Treats: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students at Pete Science and Business College Inc.		

Panel Member	Comments Recommendations / Suggestions	Status and Signature
	- Copyright - materials	LLM

Checked by:

Panel Member:	Panel Member:
Panel Member:	Panel Member:

Noted by:

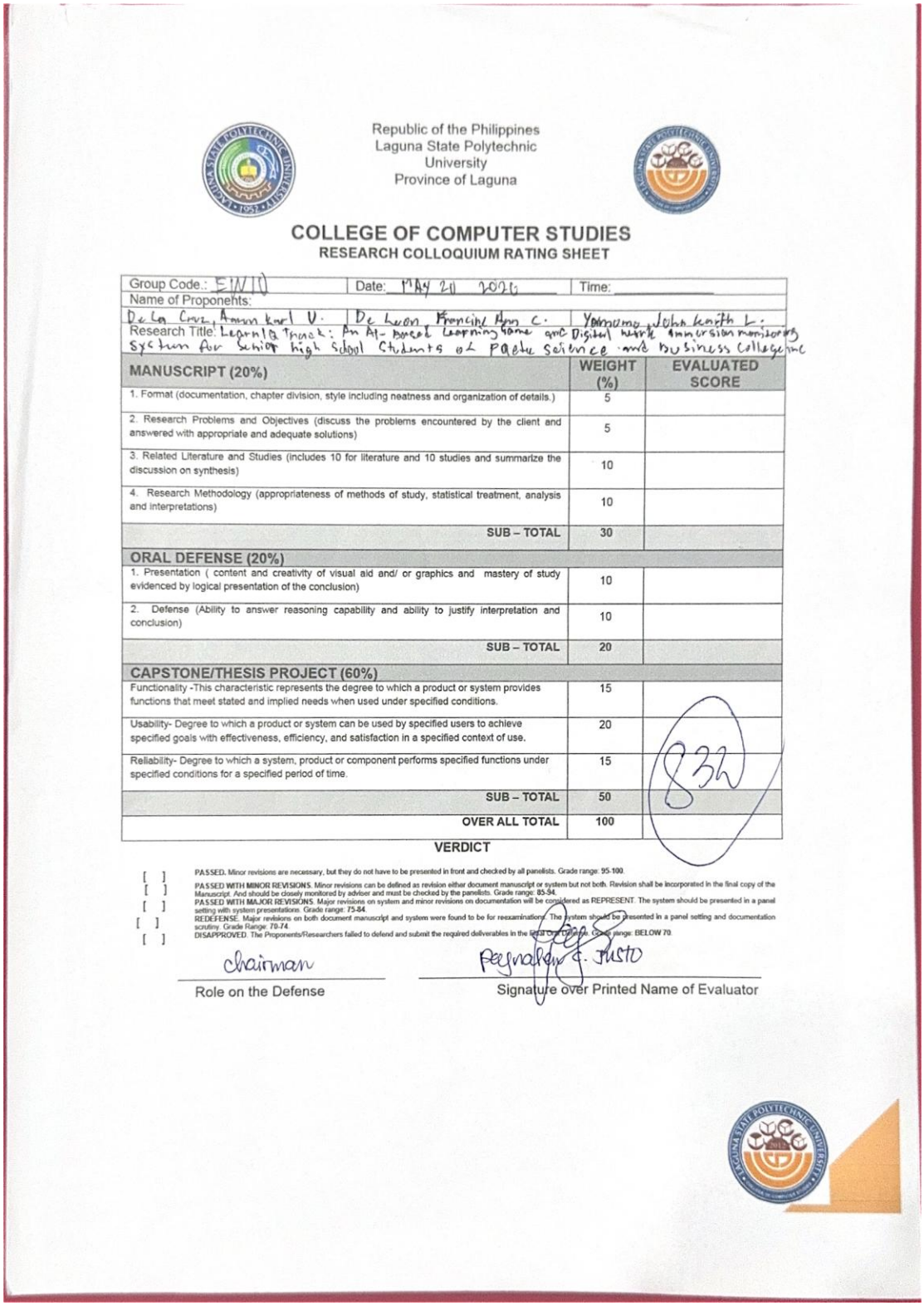
Instructor/Adviser/RIUH:	Associate Dean:

Conformed by:

Student:	Student:
Student:	

RATING SHEET

(RESEARCH COLLOQUIUM)



Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EWIO	Date: MAY 21 2026	Time:
Name of Proponents:		
De La Cruz, Aaron Karl V.	De Leon, Francine Ann C.	Yamsuan, John Keith L.

Research Title: Learning Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System for Senior High School Students of Pasig Science and Business Colleges Inc

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	83
SUB – TOTAL	50	
OVER ALL TOTAL	100	

VERDICT

[ ] PASSED. Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: 95-100.

[ ] PASSED WITH MINOR REVISIONS. Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: 85-94.

[ ] PASSED WITH MAJOR REVISIONS. Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: 75-84.

[ ] REDEFENSE. Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: 70-74.

[ ] DISAPPROVED. The Proponents/Researchers failed to defend and submit the required deliverables in the [illegible]. Grade range: BELOW 70.

Chairman
Role on the Defense

Reynaldo C. Justo
Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: EWD	Date: MAY 20 2026	Time:
Name of Proponents: De La Cruz, Aaron Lemuel U. De Leon, Francine Ann C. Yamoma, John Keith		
Research Title: Learn|B Track: An AI-Based Learning Game and Digital Work Immersion Monitoring System		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	93

VERDICT

[ ] PASSED. Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: 95-100.

[ ] PASSED WITH MINOR REVISIONS. Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: 85-94.

[ ] PASSED WITH MAJOR REVISIONS. Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: 75-84.

[ ] REDEFENSE. Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: 70-74.

[ ] DISAPPROVED. The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: BELOW 70.

Role on the Defense

Signature over Printed Name of Evaluator

Republic of the Philippines
Laguna State Polytechnic
University
Province of Laguna

COLLEGE OF COMPUTER STUDIES
RESEARCH COLLOQUIUM RATING SHEET

Group Code.: BWD	Date: MAY 20 2026	Time:
Name of Proponents:		
De La Cruz, Aaron Kyle U.	De Leon, Francine Ann C.	Yamamoto, John Keith L.
Research Title: Learn It Track: An AI-Based Learning Course and Distant Monitoring System for Senior High School Students of Pantu Science and Business College Inc		

MANUSCRIPT (20%)	WEIGHT (%)	EVALUATED SCORE
1. Format (documentation, chapter division, style including neatness and organization of details.)	5	
2. Research Problems and Objectives (discuss the problems encountered by the client and answered with appropriate and adequate solutions)	5	
3. Related Literature and Studies (includes 10 for literature and 10 studies and summarize the discussion on synthesis)	10	
4. Research Methodology (appropriateness of methods of study, statistical treatment, analysis and interpretations)	10	
SUB – TOTAL	30	
ORAL DEFENSE (20%)		
1. Presentation (content and creativity of visual aid and/ or graphics and mastery of study evidenced by logical presentation of the conclusion)	10	
2. Defense (Ability to answer reasoning capability and ability to justify interpretation and conclusion)	10	
SUB – TOTAL	20	
CAPSTONE/THESIS PROJECT (60%)		
Functionality -This characteristic represents the degree to which a product or system provides functions that meet stated and implied needs when used under specified conditions.	15	
Usability- Degree to which a product or system can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use.	20	
Reliability- Degree to which a system, product or component performs specified functions under specified conditions for a specified period of time.	15	
SUB – TOTAL	50	
OVER ALL TOTAL	100	88

VERDICT

[ ] **PASSED.** Minor revisions are necessary, but they do not have to be presented in front and checked by all panelists. Grade range: 95-100.

[ ] **PASSED WITH MINOR REVISIONS.** Minor revisions can be defined as revision either document manuscript or system but not both. Revision shall be incorporated in the final copy of the Manuscript. And should be closely monitored by adviser and must be checked by the panelists. Grade range: 85-94.

[ ] **PASSED WITH MAJOR REVISIONS.** Major revisions on system and minor revisions on documentation will be considered as REPRESENT. The system should be presented in a panel setting with system presentations. Grade range: 75-84.

[ ] **REDEFENSE.** Major revisions on both document manuscript and system were found to be for reexaminations. The system should be presented in a panel setting and documentation scrutiny. Grade Range: 70-74.

[ ] **DISAPPROVED.** The Proponents/Researchers failed to defend and submit the required deliverables in the Final Oral Defense. Grade range: BELOW 70.

Role on the Defense

Signature over Printed Name of Evaluator

REQUEST LETTER

(IT EXPERTS TESTER)

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Vic Marco Orbista
Mega Cat Studios
Spec Ops

Dear Mr. Orbista

Warm greetings!

We are fourth-year students of the Bachelor of Science in Information Technology, College of Computer Studies of Laguna State Polytechnic University. As part of the academic requirements for our Capstone Project, currently conducting our Capstone Project entitled **"LEARNIQ TRACK: AN AI-BASED LEARNING GAME AND DIGITAL WORK IMMERSION MONITORING SYSTEM FOR SENIOR HIGH SCHOOL STUDENTS OF PAETE SCIENCE AND BUSINESS COLLEGE INC."**

The purpose of this evaluation is to obtain professional feedback, opinions, and suggestions from IT experts. The testing will use Black Box Testing, where you will execute predefined test cases, record result that will help us determine whether the system is sufficiently appropriate and ready for system demonstration to our intended client.
For documentation of tester qualification, we also kindly request a copy of your updated Curriculum Vitae/Resume and any relevant certification or professional credentials, if available.

We assure you that all information and feedback gathered during the evaluation will be used strictly for academic purposes and will be treated with utmost confidentiality. We would be grateful for the opportunity to you at your most convenient date and time, either face-to-face or through an online platform.
We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time, assistance, and cooperation.
Sincerely,

De La, Cruz Aaron Karl U.
Researcher BSIT 4E

De Leon, Francine Ann C.
Researcher BSIT 4E

Yamomo, John Kaith L.
Researcher BSIT 4E

Noted by:

MIA V. VILLARICA, DIT
Thesis Adviser
College of Computer Studies

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Mike Angelo Matrido
Director of Technical Services
One Depot, Makati, Philippines

Dear Mr. Matrido

Warm greetings!

We are fourth-year students of the Bachelor of Science in Information Technology, College of Computer Studies of Laguna State Polytechnic University. As part of the academic requirements for our Capstone Project, currently conducting our Capstone Project entitled **"LEARNIQ TRACK: AN AI-BASED LEARNING GAME AND DIGITAL WORK IMMERSION MONITORING SYSTEM FOR SENIOR HIGH SCHOOL STUDENTS OF PAETE SCIENCE AND BUSINESS COLLEGE INC."**

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Sincerely,

De La, Cruz Aaron Karl U.
Researcher BSIT 4E

De Leon, Francine Ann C.
Researcher BSIT 4E

Yamomo, John Kaith L.
Researcher BSIT 4E

Noted by:

MIA M. VILLARICA, DIT
Thesis Adviser
College of Computer Studies

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Ferdinand T. Celis
Website Developer / Designer

Dear Mr. Celis

Warm greetings!

We are fourth-year students of the Bachelor of Science in Information Technology, College of Computer Studies of Laguna State Polytechnic University. As part of the academic requirements for our Capstone Project, currently conducting our Capstone Project entitled **"LEARNIQ TRACK: AN AI-BASED LEARNING GAME AND DIGITAL WORK IMMERSION MONITORING SYSTEM FOR SENIOR HIGH SCHOOL STUDENTS OF PAETE SCIENCE AND BUSINESS COLLEGE INC."**

The purpose of this evaluation is to obtain professional feedback, opinions, and suggestions from IT experts. The testing will use Black Box Testing, where you will execute predefined test cases, record result that will help us determine whether the system is sufficiently appropriate and ready for system demonstration to our intended client.
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We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time, assistance, and cooperation.
Sincerely,

De La, Cruz Aaron Karl U.
Researcher BSIT 4E

De Leon, Francine Ann C.
Researcher BSIT 4E

Yamomo, John Kaith L.
Researcher BSIT 4E

Noted by:

MIA V. VILLARICA, DIT
Thesis Adviser
College of Computer Studies

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

COLLEGE OF COMPUTER STUDIES

Edmar A Dela Cruz
Business Support Specialist

Dear Mr. Dela Cruz

Warm greetings!

We are fourth-year students of the Bachelor of Science in Information Technology, College of Computer Studies of Laguna State Polytechnic University. As part of the academic requirements for our Capstone Project, currently conducting our Capstone Project entitled **"LEARNIQ TRACK: AN AI-BASED LEARNING GAME AND DIGITAL WORK IMMERSION MONITORING SYSTEM FOR SENIOR HIGH SCHOOL STUDENTS OF PAETE SCIENCE AND BUSINESS COLLEGE INC."**

The purpose of this evaluation is to obtain professional feedback, opinions, and suggestions from IT experts. The testing will use Black Box Testing, where you will execute predefined test cases, record result that will help us determine whether the system is sufficiently appropriate and ready for system demonstration to our intended client.
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We sincerely hope for your kind consideration and support in the completion of our Capstone Project. Thank you very much for your time, assistance, and cooperation.
Sincerely,

De La, Cruz Aaron Karl U.
Researcher BSIT 4E

De Leon, Francine Ann C.
Researcher BSIT 4E

Yamomo, John Kaith L.
Researcher BSIT 4E

Noted by:

MIA V. VILLARICA, DIT
Thesis Adviser
College of Computer Studies

ISO FORMS

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS ADVISER NOMINATION

Student/s Name and Signature : De La Cruz, Aaron Karl U. -
: De Leon, Francine Ann C. -
: Yamomo, John Keith L. -
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Thesis Adviser

Name of Faculty : Mia V. Villarica, DIT
College : College of Computer Studies
Field of Specialization : Computer Science / Information Technology

I, hereby indicate my willingness to serve as the Thesis Adviser.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PHD, FFP
R&D Director/Chairperson

LSPU-RDO-SF-011 Rev. 0 8 August 2018

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : De La Cruz, Aaron Karl V. -
: De Leon, Francine Ann C. -
: Yamomo, John Kaith L. -
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Statistician)

Name of Faculty : Reynalen C. Justo, LPT, DIT
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : _____ Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PHD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : De La Cruz, Aaron Karl V. -
: De Leon, Francine Ann C. -
: Yamano, John Kaith L. -
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Language Critic)

Name of Faculty : Ma. Cezanne D. Dimaculangan
College : College of Arts and Science
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : 09/09/2020
Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mya V. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : De La Cruz, Aaron Karl U.
: De Leon, Francine Ann C.
: Yamano, John Faith L.
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Technical Editor)

Name of Faculty : Edward S. Flores
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : 09/09/2026
Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia M. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FFP
R&D Director/Chairperson

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

THESIS PANEL MEMBER NOMINATION

Student/s Name and Signature : De La Cruz, Aaron Karl V.
: De Leon, Francine Ann C.
: Yamomo, John Kaith L.
:

Degree Sought : Bachelor of Science in Information Technology
Major Field (if applicable) :

I/we, hereby nominate as Panel Member (Subject Expert)

Name of Faculty : Mark P. Bernardino, MSCS
College : College of Computer Studies
Field of Specialization :

I, hereby indicate my willingness to serve as a Panel Member.

Signature of Faculty : 9/9/2022
Date Signed

Endorsed By:

Mark P. Bernardino, MSCS
Research Coordinator

Recommending Approval:

Mia M. Villarica, DIT
Dean/Associate Dean

APPROVED/DISAPPROVED:

Rosario G. Catapang, PhD, FFP
R&D Director/Chairperson

APPENDICES C

(CURRICULUM VITAE OF SYSTEM TESTERS)



Kevin Adlaon

Analyst Programmer

Summary

Analyst Programmer with 10+ years of experience in analyzing and developing software solutions. Adept in working with other highly competent colleagues.

Experience

2022-11 - 2024-07

Data Scientist

Kasatria **Technologies** Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Built pipelines using Google Cloud Platform and **Amazon Web Services**
- Analyzed data and developed software solutions

2017-10 - 2022-11

Analyst Programmer

iTAc MSC Outsourcing Sdn Bhd, Kuala Lumpur, Malaysia

Responsibilities:

- Developed systems as innovative visualization tools for different data representation and analysis
- Created Google Ads using Google's platform

2016-07 - 2017-10

Systems Developer and Administrator

Management Information Systems Office, Provincial Government of Laguna, Philippines

2012-04 - 2016-07

Computer Programmer I

Management Information Systems Office, Provincial Government of Laguna, Philippines

2011-04 - 2011-05

On-the-job Training

Management Information Systems Office, Provincial Government of Laguna, Philippines

Personal Info

Address
3773 P. Zamora Street, Brgy. Santo Angel Sur, Santa Cruz, Laguna

Phone
+63 965 427 1050

E-mail
adlaon.kevin@gmail.com

Place of birth
Calamba City, Laguna, Philippines

Date of birth
1992-07-30

LinkedIn
https://linkedin.com/in/kevin-adlaon

Facebook
https://facebook.com/adlaon.kevin

Marital status
Widowed

Citizenship
Filipino

Skills

PHP

MySQL

CSS

jQuery

Ajax

JavaScript

Java

TypeScript

Python

Git

Google Analytics

Google Cloud Platform

Amazon Web Services

Salesforce Marketing Cloud

Education

2008 - 2012
ACTS Computer College
Sta. Cruz, Laguna, BS in Computer Science

2004 - 2008
Masico National High School
Pila, Laguna, Salutatorian

2004 - 1998
Mangumit Elementary School
Calamba City, Laguna

Certificates

- Google Cloud Certified Professional Data Engineer
- Salesforce Marketing Cloud Developer
- Salesforce Marketing Cloud Email Specialist

Languages

Filipino
English

References

Adoree Ramos
Professor
ACTS Computer College
Sta. Cruz, Laguna
(049) 501 1484

Shing Nien Fong
Director
Research & Development
Kasatria Technologies Sdn Bhd
+60 16 261 6060



ASPIRING QA ENGINEER |
WEB DEVELOPER |
IT SUPPORT TECHNICIAN

Contact

+63 976 238 5693

jhayjustinlopez@gmail.com

Santa Cruz, Laguna, 4009, Phillipines

Skills

- Leadership & Team Collaboration
- Computer Literacy (MS Office, Email, Digital Filing)
- Visual Design & Editing
- Photo & Video Editing
- Basic UI/UX & Web Content Management

Language

- English
- Filipino

Justine Jhay Lopez

Education

(2022 - 2026)
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bachelor of Science in Information Technology
1st - 4th Year College

(2020 - 2022)
LAGUNA SENIOR HIGH SCHOOL
Science, Technology, Engineering, and Mathematics strand
Grade 11 - Grade 12

(2016 - 2019)
PEDRO GUEVARA MEMORIAL NATIONAL HIGH SCHOOL
Grade 7 - Grade 10

Experience

(2025 - Present)
MANAGEMENT INFORMATION SYSTEMS OFFICE
- MIS Programmer

(2024 - 2026)
MUNICIPAL SOCIAL WORKER AND DEVELOPMENT OFFICE
- Data Entry Clerk

Projects

- **IKNOWLGRC - KNOWLEDGE MANAGEMENT SYSTEM**
  -2026 (Full Stack Developer)
- **EXELENCIA ACADEMY PORTAL**
  -2024 (Full Stack Developer)
- **SHE WEAR (E-COMMERCE WEB)**
  -2024 (Programmer)
- **BARANAGAY RECORDING SYSTEM**
  -2024 (Full Stack Developer)
- **JAMAICA RECORDING SYSTEM**
  -2023 (Full Stack Developer)



MHARJUN V. TOBIAS

Graphic Artist / Technical Support

EDUCATION

2008 - 2013
BS INFORMATION TECHNOLOGY
Laguna State Polytechnic University

2004 - 2008
SECONDARY EDUCATION
Pedro Guevara Memorial National High School

SKILLS

- Networking Fundamentals
- Hardware Diagnostics
- Graphic Arts
- Microsoft Office Savvy
- GIS Mapping
- Technical Support

CONTACT

09171302395
tobiasmharjun666@gmail.com
Patimbao, Santa Cruz, Laguna
www.reallygreatsite.com

PROFILE INFO

Highly skilled professional with extensive experience in graphic design, technical support, and administrative operations. Proven ability to handle complex networking tasks and digital asset creation with precision and efficiency. Dedicated to delivering high-quality visual solutions and reliable technical troubleshooting.

EXPERIENCE

2025 - Present
CASUAL CLERK
MIS Office - MGO Santa Cruz, Laguna
Handled management information systems and administrative support for the local government unit.

2023 - 2025
CASUAL CLERK
Assessor's Office - MGO Santa Cruz, Laguna
Assisted in data management and office operations for tax assessment and land records.

2020 - 2022
GRAPHIC ARTIST / TECH SUPPORT
Mayor's Office - MGO Santa Cruz, Laguna
Provided digital creative solutions and technical maintenance for municipal operations.

2019 - 2019
ADMINISTRATIVE AIDE
PIO - MGO Santa Cruz, Laguna
Supported public information dissemination and media relations.

ACHIEVEMENTS

2022
CIVIL SERVICE PROFESSIONAL
Passed with a rating of 80.56.

2019
CIVIL SERVICE SUB-PROFESSIONAL
Passed with a rating of 82.26.



JONAS T. SANTOS

Barangay. Jasaan, Santa Cruz, Laguna, Philippines, 4009
Jonassantos122113@gmail.com | 09675977572

- IT professional with three years of experience in technical support, system administration, and IT operations in the Municipal Government of Santa Cruz, Laguna. Skilled in troubleshooting hardware, software, printers, and network issues, providing IT support across multiple municipal offices, and maintaining systems to ensure efficient operations. Known for strong problem-solving abilities, adaptability, and delivering reliable technical solutions that improve productivity and service delivery.

Work Experience

- **MUNICIPAL GOVERNMENT OF SANTA CRUZ**

  Management Information Systems office | MIS IT Staff — **2025 – Present**
  - Provide technical support for hardware, software, printers, and network issues across municipal offices.
  - Install, configure, and maintain computer systems, peripherals, and user accounts.
  - Perform system administration, preventive maintenance, and software updates.
  - Troubleshoot network connectivity issues and support IT projects and system implementations.
  - Maintain IT asset inventory, service records, and technical documentation.

  Municipal Administrator's Office / Municipal ICT Office | Administrative Aide — **2024 – 2025**
  - Performed administrative and clerical functions to support daily office operations.
  - Prepared and maintained records, reports, and official correspondence.
  - Assisted with document processing, office coordination, and administrative support.

  Municipal Government of Santa Cruz | DOLE - Government Intern — **2023 – 2024**
  - Municipal Administrator's Office / Municipal ICT Office

- **PROVINCIAL GOVERNMENT OF LAGUNA | ON-THE-JOB TRAINING** — **2022**
  - Laguna Parole and Probation Office - R4A

Key responsibilities

- Provide technical and systems support for multiple LGU information systems, including the *General Services Office Management System, Motorized Tricycle Operators Permit System*, and *EBOSS Queueing System.*
- Troubleshoot and resolve system issues, perform database record updates and corrections, manage user accounts, and generate operational reports.
- Perform source code modifications, including layout adjustments (margins and page breaks), to improve system-generated reports and document formatting.
- Administer and maintain EBOSS Queueing System kiosks, ensuring reliable operation and resolving system-related concerns.
- Perform daily backups of LGU system databases to ensure data integrity, security, and disaster recovery readiness.
- Provide technical support for municipal offices by installing, maintaining, and troubleshooting computers, printers, network connectivity, and other IT equipment.
- Prepare and process Purchase Requests (PRs) for MIS office supplies, IT equipment, maintenance requirements, and other operational needs.
- Manage inventory and monitoring of office supplies, computer equipment, and IT tools to ensure availability and accountability.

Technical Skills & Abilities

- IT Technical Support
- Hardware Installation, Configuration & Troubleshooting
- Backup and recovery deleted files
- Software Installation & Maintenance
- Printer Installation & Troubleshooting
- Network Troubleshooting
- System Administration
- System User Account Creation and Management
- Operating System Installation & Configuration (Windows)
- Technical Documentation
- Computer System Maintenance
- Problem Solving & Troubleshooting
- Analytical Thinking
- Customer Service & End-User Support
- Communication Skills & Team Collaboration
- Time Management
- Attention to Detail
- Adaptability & Continuous Learning
- Organizational Skills
- Ability to Work Under Pressure

Education

- **TERTIARY LEVEL:** **2018 – 2022**
  Bachelor of Science in Information Technology
  Major in Service Management Program
  Laguna State Polytechnic University – Santa Cruz Campus
- **SENIOR HIGH SCHOOL:** **2016 – 2018**
  Science, Technology, Engineering and Mathematics – STEM
  AMA Computer Collage – Santa Cruz Campus

Civil Service Eligibility

- **Career Service Subprofessional Eligibility**
  Passed
  March 2026

Personal Information

- Age: 26 years old
- Birthdate: May 11, 2000
- Birthplace: Santa Cruz, Laguna
- Sex: Male
- Height: 4'11"
- Weight: 40kg
- Citizenship: Filipino
- Civil status: Single
- Religion: Roman Catholic
- Languages: English/Tagalog

Seminars & Training

- **Introduction to Networking**
  Provincial Government of Laguna - Miso Training Room
  April 30, 2026
- **E-LGU User's Training**
  DICT- CALABARZON
  April 23, 2026
- **Web Development (Laravel & MySQL) & Server Administration**
  Provincial Government of Laguna - Miso Training Room
  January 5 – 16, 2026
- **Data Analytics Essentials**
  December 5 – 11, 2025
  DICT-ITU DTC (Net. Acad. Program)
- **Gender Sensitivity and Safe Space Act of the Philippines**
  LGU Santa Cruz
  December 2, 2025
- **E-LGU Admin Training and System Set-Up**
  DICT- CALABARZON
  November 25 – 28, 2025
- **Google Workspace for Educators/LGU Employees**
  DICT – CAR
  October 27, 2025
- **Cybersecurity and PNPKI Awareness at LGU - Santa Cruz**
  DICT - Calabarzon
  August 11, 2025
- **Basic & Intermediate Geographic Information System (GIS) Training**
  Uplink Integrated Solution Inc.
  January 30 – February 1, 2025
- **City/Municipal-Level Training Coordination, and**
  Field Visit of LGU on the Adoption of the Philippines Standard Industrial Classification
  PHILLIPPINES STATISTICS AUTHORITY
  July 6, 2024
- **Computer System Servicing Training**
  Provincial Government of Laguna - Miso Training Room
  October 16 – 20, 2023
- **Installing and Configuring Computer System & Introduction to Computer System Servicing**
  TESDA Online Program
  July 2022

Character Reference

Reference available upon request.

I hereby certify that the statement above are true and correct.



Jonas T. Santos

John Marvin A. Tabo

3192 Biya Street
Santo Angel Norte,
09754068759
itsotavs0099@gmail.com

Birthdate

May 29, 2001

EDUCATION

Santo Angel Norte Elementary School

June 2006 - April 2013

Academic Achiever (Grade 6)

Pedro Guevarra Memorial National High School (P.Gueverra Street, Sta Cruz Laguna. Junior High School)

June 2013- April 2017

Academic Achiever

Saint Therese Martin of Lisieux School and Business High School. - Sto Angel Sur, Santa Cruz, Laguna.(Information and Communication Technology)

June 2013- April 2017

Supreme Student Government President

With Honor

Laguna State Polytechnic University - Santa Cruz Campus (Bachelor of Science in Information Technology)

June 2019- July 2023

Dean`s Lister Second Semester AY 2022-2023

CSE PROFESSIONAL PASSER

SKILLS

Computer Literate

Knowledge in Microsoft Office

Hardware Configuration

Installation and Configuration

Computer Troubleshooting

AWARDS

Academic Achiever - Pedro Guevara Memorial National High School

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2017)

With Honor - Saint Therese Martin Of Lisieux School and Business High School (2018)

Dean`s Lister - Laguna State Polytechnic University (2022)

LANGUAGES



WILMAR T. MOROTA

09202603909
Lynville 8, Bagumbayan, Sta Cruz, Laguna

EDUCATION

College

AMA Computer College
Computer Programming – Under Graduate
2007-2012

Secondary

Pedro Guevarra Memorial High School
Sta Cruz, Laguna
2003-2007

Primary

Our Lady of Maulawin
1998-2003

REFERENCE

RYAN R. DUÑGO
09166738313
Bagumbayan, Sta Cruz, Laguna

JUNE DAE L. MORGA
09686058086
Bubukal, Sta Cruz, Laguna

PROFILE

Birthdate : December 8, 1990
Place of Birth : Sta Cruz, Laguna, Philippines
Civil Status : Married
Religion : Roman Catholic

WORK EXPERIENCE

Management Information Systems Office — **2025 - Present**
Job-Order

Economic Enterprice Management Office — **2022 - 2025**
Job-Order

- Daily submission of Collection Report of Market Collectors
- Daily Printout of Market Collector Reciepts

Freelance Encoder — **2017-2020**

OTHER ATTAINMENTS

CAREER SERVICE EXAMINATION PROFESSIONAL PASSER
MAY 2024



ROBIN PAOLO BONA

Bachelor of Science in Information Technology

To utilize experience and qualification in different and challenging position that professionally growth and enhance my future opportunities.

EDUCATION

2018

UNION COLLEGE OF LAGUNA

- Bachelor of Science in Information Technology

CERTIFICATE

2022

NATIONAL CERTIFICATION II

- Computer System Servicing NC II

SKILL

- Troubleshoot 100%
- Technical Support 90%
- Microsoft Office 100%
- Creative 90%
- Team Work 95%

EXPERIENCE

2019 - PRESENT

Municipal Government Santa Cruz, Laguna

Administrative Aide

- Information Technology

2018 - 2019

DEP-ED LAGUNA

Administrative Aide

- Information Technology

2018

SM Calamba

Stock Clerk

- Warehouse Stock Clerk

CONTACT

09650499411

Hushbona@gmail.com

229 Jamaica Subdivision, Santo Angel Central, Santa Cruz, Laguna

REFERENCE

BRYAN BOUVIER BONZA
Municipal Government Santa Cruz, Laguna
HRMO V
09177059405

MELVIN O. BONZA
Municipal Government Santa Cruz, Laguna
ADMINISTRATOR
09967079115

Ferdinand T. Celis

Website Developer / Designer

Lilian st. Brgy. Bulihan, Famy, Laguna Philippines 4021
+63 (992) 301 9995 / +63 (938) 038 6385
celisinan@gmail.com
https://inancelis.com/



ABOUT ME

I'm a professional website developer and designer from the Philippines with over 5 years of experience building clean, fast, and easy-to-use websites. I specialize in WordPress, WooCommerce, PHP, Laravel, React, and Vue, with hands-on experience in client management, strategic planning, and full-cycle project delivery.

SKILLS & ENDORSEMENT

- Website Development
- Responsive Web Design
- Front-End Development
- Back-End Development
- WordPress Development
- WooCommerce Development
- System Development
- API Integration
- PSD to HTML
- Database Management
- AI Website Vibe Coding
- AI Automation Workflow

TOOLS & TECHNOLOGIES

BACKEND & FRAMEWORKS

- Laravel
- PHP
- NodeJS
- Python
- Java

FRONTEND

- ReactJS
- Vue / Nuxt
- JavaScript
- jQuery
- HTML5
- CSS / SASS
- Bootstrap

CMS & TOOLS

- WordPress
- Elementor Pro
- Divi Builder
- Wix
- GoHighLevel
- n8n
- Git
- AWS

GOHIGHLEVEL (GHL)

- GHL Funnel & Website Builder
- Automation
- AI Studio

HOSTING & SERVER

- Hosting Management
- SSL Installation
- Domain & DNS Setup
- Pointing Domains

DATABASE

- MySQL

AI TOOLS

- ChatGPT
- Claude AI
- Vibe Coding

EXPERIENCE

Freelancer · Part time
2019 – Present

Flexible role — Multiple Location

As a website developer based in the Philippines, I handle multiple projects at once and enjoy learning something new from each one. I work with clients on web development and design, creating simple, clean designs and building full websites that fit exactly what they need. I communicate clearly, solve problems fast, and always focus on delivering high-quality results that help businesses grow.

Web Development | Converting PSD to HTML | Graphic Designing | T-shirt Design

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Web Designer and Developer · Part-time to Full-time
Aug 2024 – July 2026

Blue Apple Launchpad — Sacramento, California, United States - Remote

Responsibilities

- Design and develop responsive WordPress websites using Elementor Pro
- Build and maintain custom themes and page layouts for client projects
- Collaborate with the team on site structure, UX, and content updates
- Optimize site performance, responsiveness, and cross-browser compatibility
- Troubleshoot and resolve website issues as they arise
- Build and manage funnels, websites, and automations in GoHighLevel (GHL)
- Create workflow automations using n8n to streamline business processes
- Handle hosting management for client sites and servers
- Configure DNS records and domain settings for hosting and email setup
- Deploy websites to live servers and manage release updates
- Develop custom WordPress plugins tailored to client requirements

WordPress | Elementor Pro | Responsive Web Design | PHP | JavaScript | GoHighLevel (GHL) | n8n | Hosting Management | DNS Configuration | Website Deployment | Custom Plugin Creation

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Full Stack Web Developer · Full-time
March 2025 – December 2025

International Property Alerts — MCS Tower, Chino Roces Ave, Makati City, Philippines

Responsibilities

- Creating custom WordPress plugins and themes
- Building and maintaining Elementor pages as needed
- Creating and maintaining API endpoints
- Managing and maintaining databases
- Maintaining scraper systems
- Managing IPA listings and CRM system
- Troubleshooting and addressing issues as reported
- Creating web applications as needed

PHP | WordPress | Elementor Pro | JavaScript | GHL | Namecheap | Kinsta

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Software Engineer · Full-time
March 2023 – December 2024

Accenture — Uptown Taguig, Philippines

Responsibilities

- Develop, test, and deploy scalable software solutions for enterprise-level clients.
- Collaborate with cross-functional teams, including analysts, designers, and other developers,to deliver client-centric solutions.
- Identify and resolve software defects through rigorous testing and debugging processes.
- Provide technical support and maintenance for existing applications to ensure optimal performance.
- Ensure compliance with coding standards and best practices, including security and data protection protocols.

Asp.net | C# | Oracle Database

Front-End Web Developer · Full-time — May 2022 – March 2023

Asticom Group of Companies — Bonifacio Global City, Taguig City, Philippines

Responsibilities

- Develop and maintain front end component of the ERP
- Collaborating with back-end developers and B2B endpoints on the integration of elements.
- Running diagnostic tests, repairing defects, and providing technical support.
- Recommending and implementing improvements to processes and solutions.
- Implement and apply Continuous Integration and Continuous Development practices.

Php | Laravel | WordPress | Elementor Pro | AWS | React Native | JavaScript

---

Web Developer · Full-time — Jan 2021 – April 2022

Valens Research Asia — Pasig City, Philippines

Responsibilities

- Helps the Digital Team design, develop, deploy, and maintain web applications, scripts,and data systems for both internal use and our clients.
- Ensures that all work done by the team is bug-free via Test-Driven Development methodologies.
- Administers live websites as necessary.

Php | Laravel | WordPress | Elementor Pro | VueJs | Node

EDUCATION

Bachelor of Science in Information Technology — 2017 – 2021

Laguna State Polytechnic University — Sta. Cruz, Laguna, Philippines

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Bachelor of Science in Computer — 2015 – 2017

Colegio De San Lorenzo — Congressional Avenue, Barangay Bahay Toro, Project 8, Quezon City, Philippines

Contact

vicmarcoorbista@gmail.com

www.linkedin.com/in/vicmarcoorbista (LinkedIn)

Top Skills

Unity
Adobe Photoshop
Microsoft Office

Vic Marco Orbista

Spec Ops at Mega Cat Studios
Laguna, Calabarzon, Philippines

Summary

Keep adapting.

Experience

Mega Cat Studios
Spec Ops
December 2020 - Present (5 years 10 months)

Finally had my dream job! :)

NGGI
Indie Game Developer
July 2020 - Present (6 years 3 months)
Philippines

Already created and published my first mobile game in apkpure titled as deadspot. I'm planning to make some more but I first need to have a stable job and grind my graphic designing skills to create better art for the games that I will make.

Freelance
Freelance Graphic Designer
June 2020 - Present (6 years 4 months)
Philippines

The Graphic Artist
Assistant Graphic Designer
October 2019 - June 2020 (9 months)
Laguna, Philippines

I mainly use Adobe Photoshop and Illustrator on my work. I do different invitations, Ads, and presentations.

K Business
System Quality Assurance
October 2019 - October 2019 (1 month)

Evolutionworld
Freelance Proofreader
September 2019 - October 2019 (2 months)

kgb
Quality Assurance Specialist
July 2019 - August 2019 (2 months)
NCR - National Capital Region, Philippines

2 months Project

CLA Central Mall Inc.
Purchasing Staff
June 2016 - June 2019 (3 years 1 month)
Philippines

Education

AMA Computer College
Bachelor's degree, Computer Science · (2007 - 2012)



EDMAR A. DELA CRUZ

Brgy. Duhat, Santa Cruz, Laguna
Contact No: 09494620078
Email Address: edmardelacruz101688@gmail.com

OBJECTIVE

To find a responsible and challenging position that will use my education, work experience, background, expand my knowledge and offer opportunities for personal and professional growth.

EXPERIENCES

WILLIS TOWERS WATSON PHILIPPINES, INC.
Business Support Specialist / Senior Associate
23rd/F, W City Center, 7th Avenue corner 30th St., Bonifacio Global City, Taguig, Metro Manila, Philippines

April 17, 2023 – Present

Accenture, Inc.
Business Process Delivery Associate / Transaction Processing New Associate
1880 Eastwood Avenue, Corner Richmonde Street 1110, Quezon City, Philippines
April 06, 2015 – May 31, 2019

SPi Global
Indexer
SPi Bldg. Carmelrey Park 2, Calamba City, Laguna, Philippines
July 09, 2012 – February 10, 2014

SPi Global
Data Technician
SPi Bldg. Carmelrey Park 2, Calamba City, Laguna, Philippines
May. 21, 2012 – June 30, 2012

SPi Global
Data Technician
SPi Bldg. Carmelrey Park 2, Calamba City, Laguna, Philippines
November 15, 2010 – March 03, 2012

DILG Office
On-The-Job-Training
Municipality of Magdalena, Magdalena, Laguna
April 2009 – June 2009

EDUCATIONAL BACKGROUND

Tertiary	Laguna State Polytechnic University Bachelor of Science in Information Technology Santa Cruz, Laguna 2006 -2010
Secondary	Pedro Guevara Memorial National High School Santa Cruz, Laguna 2001-2005
Primary	Duhat Elementary School Duhat, Santa Cruz, Laguna 1995- 2001

SKILLS

Literacy in Microsoft Office and Word Excel Applications,Internet Surfing, Turbo C, Microsoft Visual Basic, Multimedia (Adobe Photoshop CS3, Flash Macromedia)

SEMINARS ATTENDED

Basic in PHP 5
Laguna State Polytechnic University – Main Campus, Santa Cruz, Laguna
Santa Cruz, Laguna

Business Process Outsourcing for Technical Support
Laguna State Polytechnic University – Main Campus, Santa Cruz, Laguna
Santa Cruz, Laguna

Philippine Youth Congress in Information Technology 2009
University of the Philippines – Diliman
Quezon City

Edge of an IT Graduates in BPO Industry
Laguna State Polytechnic University – Main Campus, Santa Cruz, Laguna
Santa Cruz, Laguna

REFERENCE

Name	:	Ana Carmelia Ramiscal
Position	:	Team Leader
Company	:	Accenture, Inc.
Contact #	:	09175213585

PERSONAL DATA

Age	:	36 years old	**Date of Birth**	:	October 16, 1988
Nationality	:	Filipino	**Place of Birth**	:	Santa Cruz, Laguna
Civil Status	:	Single	**Weight**	:	64 kg
Gender	:	Male	**Height**	:	5'7"

I hereby certify that the above information are true and correct to the best of my knowledge.

APPENDICES D

(CURRICULUM VITAE OF THE RESEARCHERS)



DE LA CRUZ, AARON KARL U.

STUDENT

+09072590557 | aaronkarldelacruz5@gmail.com | 01011 Sitio Matahimik Brgy Duhat Sta Cruz Laguna

PROFILE

Motivated Information Technology student with an interest in web and mobile application development, networking, and emerging technologies. Skilled in developing user-focused digital solutions and learning modern tools and technologies. A responsible and adaptable individual who enjoys solving technical problems, working on projects, and continuously improving technical and creative skills.

EXPERIENCE

Networking **Present**

- Built and configured network topologies using Cisco Packet Tracer.
- Configured VLANs, switches, wireless access points, and network connections.
- Tested network connectivity and troubleshot configuration issues.

Web Development

- Developed functional web applications using modern web development technologies.
- Created responsive and user-friendly web interfaces.
- Tested and debugged web applications to improve functionality.

EDUCATION

College: Laguna State Polytechnic University Santa Cruz Main Campus Bubukal, Santa Cruz, Laguna **Present**

SHS: Laguna State Polytechnic University Santa Cruz Main Campus Bubukal, Santa Cruz, Laguna **2019**

Secondary: Don Manuel Rivera Memorial National High School **2017**

Elementary: Duhat Elementary School – Duhat, Santa Cruz, Laguna **2016**

SKILLS

- Cisco
- Figma
- Python
- HTML
- CSS
- AI

ADDITIONAL INFORMATION

- **Languages:** English, Tagalog
- **Certifications:** Python Essentials 1 (01 Mar 2025)

DE LEON, FRANCINE ANN C.

Address: Greenfield Subd. Brgy. Sampaloc, Pagsanjan, Laguna
Mobile Number: 09177039406
E-mail Address: francineanncdeleon@gmail.com



CAREER OBJECTIVES

To gain valuable knowledge and skills to complement those that I've learned from school in an actual job environment. In return, I offer my service and determination as an asset to your company throughout the duration of my work immersion period.

PERSONAL INFORMATION

FULL NAME:	Francine Ann C. De Leon
BIRTHDATE:	September 9, 2004
AGE:	18
PLACE OF BIRTH:	Sta. Cruz, Laguna
SEX:	Female
CIVIL STATUS:	Single
NATIONALITY:	Filipino
RELIGION:	Roman Catholic

SKILLS: Reliable, Responsible, Proactive, Communication skills, Problem-Solving skills, People skills, Learning/adaptability skills, Collaboration skills

EDUCATIONAL ATTAINMENT

COLLEGE:	Laguna State Polytechnic University, Sta. Cruz Laguna S.Y 2023 - Present
SENIOR HIGH SCHOOL:	Philippine Women's University-CDCEC Sta. Cruz Laguna S.Y 2021 - 2023
STRAND:	Science, Technology, Engineering, and Mathematics (STEM)
JUNIOR HIGH SCHOOL:	St. Paul College of Makati S.Y 2017 - 2021
ELEMENTARY:	St. Paul College of Makati S.Y 2011 - 2017

SKILLS

- Web Development
- Front-End Development
- Web Design
- Mobile Design
- Back-End Development
- Mobile App Development
- Database Management
- Responsive Web Design

TOOLS & TECHNOLOGIES

- Python
- HTML
- CSS
- JavaScript
- Figma
- Canva
- MySQL
- Java

CHARACTER REFERENCE: I hereby certify that the information above is true and correct to the best of my knowledge and belief.

Francine Ann C. De Leon
Applicant's signature over printed name

John Kaith Yamomo

johnkaithyamomo22@gmail.com | +63-975-122-0721



PROFILE

I'm a web developer, UI/UX designer, and motion graphics editor from Laguna, Philippines, creating clean, responsive, and user-friendly web and app designs, as well as video content.

SKILLS

- Web Development
- AI Website Vibe Coding
- Motion Graphics Editing
- Cinematics Editing
- Front-End Development
- Web Design
- Mobile App Design
- Back-End Development
- Mobile App Development
- Database Management
- WordPress Development
- Responsive Web Design

TOOLS & TECHNOLOGIES

Technical & Design Stack

- Python
- PHP
- HTML
- CSS
- JavaScript
- Figma
- MySQL
- Supabase
- Git

Video & Media Editing

- Adobe After Effects
- Adobe Premiere Pro
- Adobe Media Encoder
- Alight Motion
- CapCut
- HandBrake

AI & Productivity Tools

- Claude Code
- Gemini
- ChatGPT

EDUCATION

Laguna State Polytechnic University — Santa Cruz, Laguna
Bachelor of Science in Information Technology — 2023 - 2027

AMA Computer College — Santa Cruz, Laguna
Accountancy Business Management (ABM) — 2021 - 2023

San Juan High School — Mexico, Pampanga
Secondary — 2017 - 2020

San Juan Elementary School — Mexico, Pampanga
Primary — 2011 - 2017

WORK EXPERIENCE

Atlas Home Products — Pagsanjan, Laguna
Puregold Merchandiser — June 2025 - August 2025

- Experienced working as a merchandiser managed daily inventory levels, ensured shelves were consistently stocked, and monitored product availability to prevent shortages